HKUSPACE
香港大學專業進修學院
HKU School of Professional and Continuing Education

Business Process Automation with VBA and Python

Mr. Eddie Chow / 17 October 2026

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Table Of Contents

Introduction to business process automation

Business Process automation with VBA

Business process automation with Python

Introduction to project management for business process automation

Development and implementation of business process automation

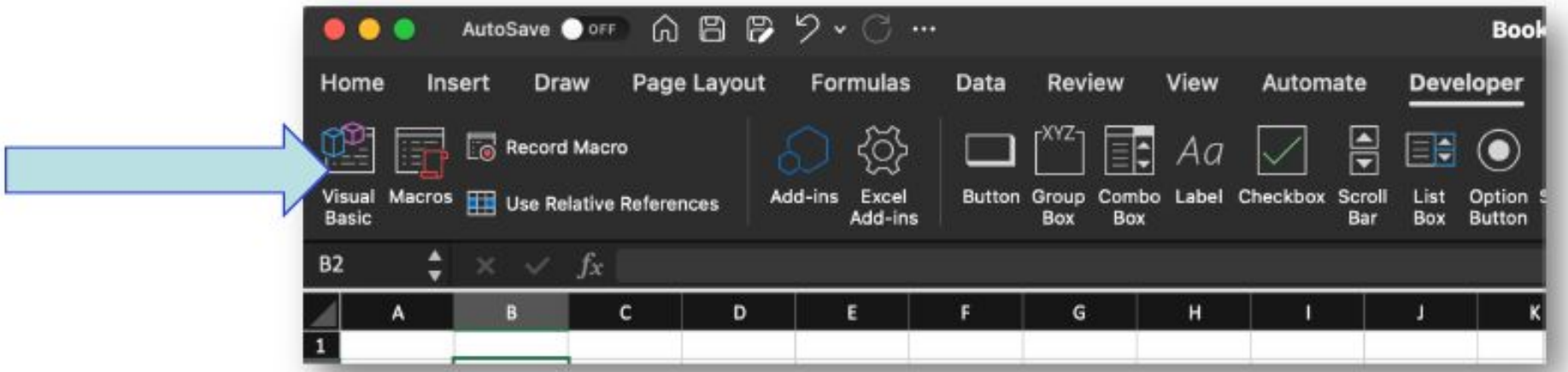
Final Group Presentation

Agenda

1. Excel VBA Advanced Exercise
2. Python Introduction, Basic Coding and Function
3. Python Process Automation Overview
4. Manage Your Budget with Simple Python Script
5. Business Analytics and Machine Learning
6. Applied Machine Learning Pipeline Overview
7. Data Collection - Hong Kong Open Data Collection
8. Data Collection - Reading Data from API
9. Data Collection - Processing Data using Pandas
10. (Optional) Data Collection - Reading Data from Google Sheet
11. (Optional) Data Collection - Reading Data from Gmail
12. (Optional) Automation Pipeline - Google Sheet Content Update
13. (Optional) Data Collection - Web Crawler
14. (Optional) Beautiful Soup for Web Scraping
15. Structured and Unstructured Data Collection Method

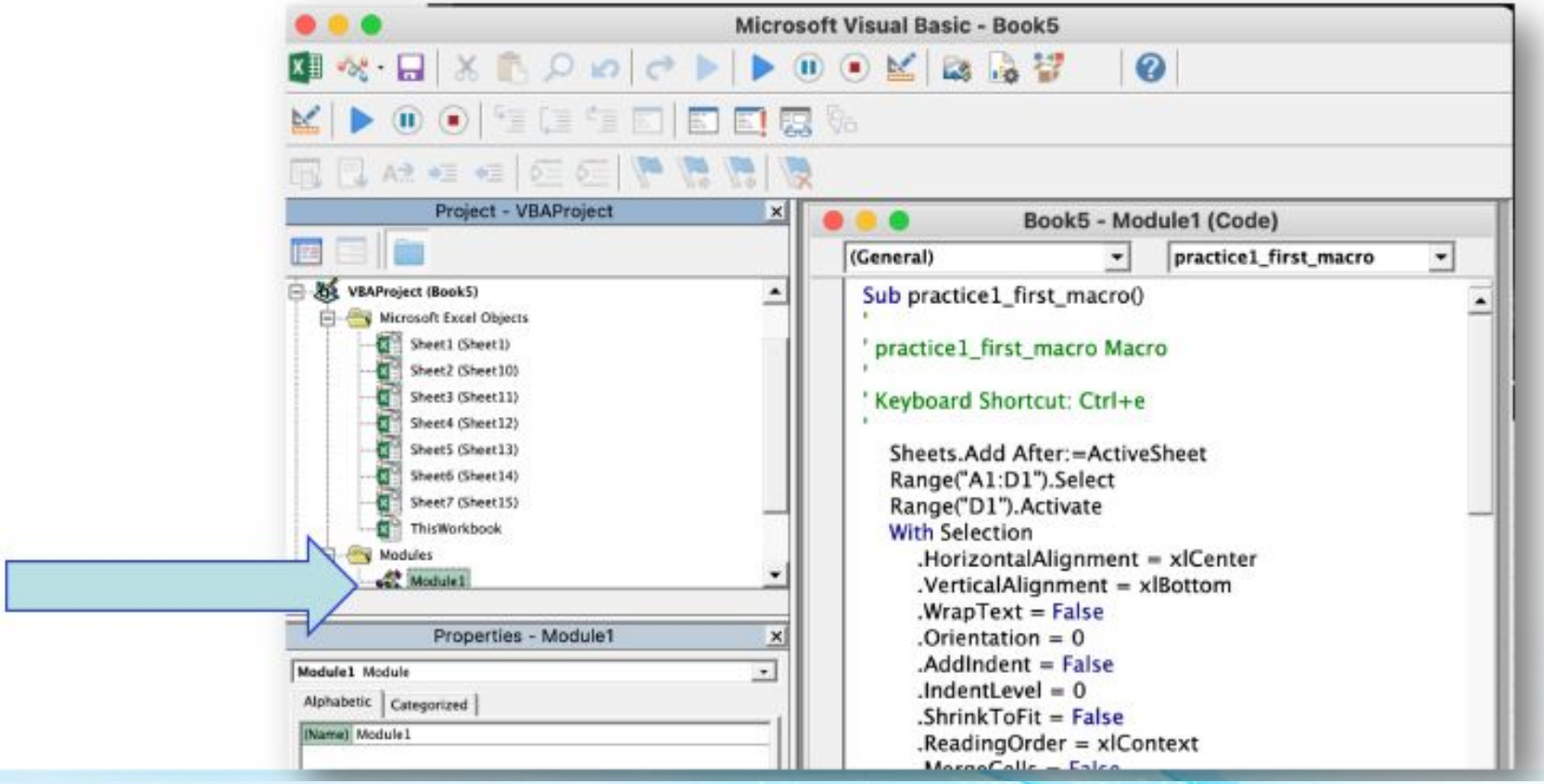
Entering Visual Basic Mode

- Developer tab ➔ Visual Basic



Entering Visual Basic Mode

- Integrated Development Environment (IDE)
- Modules = where codes are usually stored



Control flow – If, Then, Else

- Example: speed check



	A	B
1	Speed	40
2		

	A	B
1	Speed	50
2		

	A	B
1	Speed	70
2		

```
Sub HighlightCell()

    If Range("B1").Value > 50 Then
        Range("B1").Interior.Color = vbYellow
    Else
        Range("B1").Interior.Color = vbGreen
    End If

End Sub
```

Excel Function Definition - Syntax & Structure

Definition

A VBA Function is a procedure that performs operations and returns a single value to the calling code. Functions are used to create custom calculations that can be called from Excel worksheets or other VBA code.

Syntax

```
Function FunctionName(parameters) As ReturnType
    ' Comments: perform the task here
    FunctionName = returnValue
      'Assign the function result to the function name'
End Function
```

Key Points:

- ✓ The function returns a value by assigning it to the FunctionName
- ✓ Use the As keyword to specify the return data type (e.g. Integer, Double, String)
- ✓ Must end with End Function to close the function

Example — Calculate Discount

```
Function CalculateDiscount(price As Double,
discountPercent As Double) As Double

  If discountPercent < 0 Or discountPercent > 1 Then
    CalculateDiscount = -1 ' Return -1 as error for invalid
     input
Else
    CalculateDiscount = price * (1 - discountPercent)
     Apply the discount
End If
End Function
```

Usage in Excel:

=CalculateDiscount(100, 0.15)

→ Returns 85 (100 × 0.85)

- ✓ Reusable call from VBA or Excel formulas
- ✓ Can be used like built-in Excel functions
- ✓ Placed in a Standard Module via Insert > Module

Note: Functions return a value; Subs perform actions without returning a value

Control flow – If, Then, Else

- Mark ➔ Grade

Mark	*Letter Grade*
90+	A+
85-89	A
80-84	A-
77-79	B+
73-76	B
70-72	B-
67-69	C+
63-66	C
60-62	C-
57-59	D+
50-56	D
0-49	F

Control flow – Select, Case

- Range check

```
?GetGrade(90)
A

?GetGrade(80)
B
```

```
Function GetGrade(score As Integer) As String
        Select Case score
        Case Is >= 90
                GetGrade = "A"
        Case Is >= 80
                GetGrade = "B"
        Case Is >= 70
                GetGrade = "C"
        Case Is >= 60
                GetGrade = "D"
        Case Else
                GetGrade = "F"
        End Select
End Function
```

Control flow – Select, Case

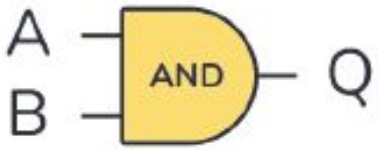
- Grade ➔ Marks

```
Dim letter_grade As String
Dim grade_point As Double

Select Case letter_grade
Case "A", "A+"
        grade_point = 4.0
Case "A-"
        grade_point = 3.7
Case "B+"
        grade_point = 3.3
...
Case Else
        grade_point = 0.0
End Select
```

Control flow – Conditions

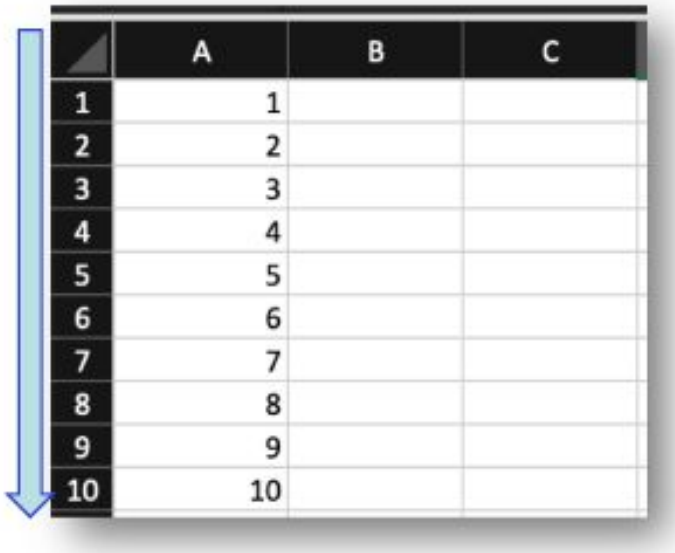
- AND
  - (True AND True) ➔ True
  - e.g., Month validity: 1 <= month AND month <= 12
- OR
  - (True OR False) ➔ True
  - e.g., Game Area: gold > 1000 OR level > 10
- NOT
  - NOT x=0 is same as x <> 0



A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

Loops – Using "For"

- Iterating through the rows
- What is changing?
  - A1 ➔ A2 ➔ A3 ➔ A4…

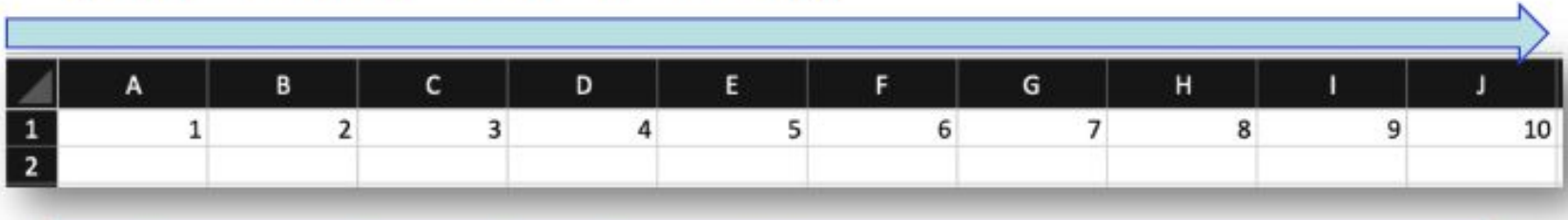


	A	B	C
1	1		
2	2		
3	3		
4	4		
5	5		
6	6		
7	7		
8	8		
9	9		
10	10		

```
Sub LoopThroughRows()
        For i = 1 To 10
                Range("A" & i).Value = i
        Next i
End Sub
```

Loops – Using "For"

- Iterating through the columns
- Again, track the change: A1 ➔ B1 ➔ C1 ➔ …
- Option 1: Make use of offset()



	A	B	C	D	E	F	G	H	I	J
1	1	2	3	4	5	6	7	8	9	10
2										

```
Sub LoopThroughColumns()
        For i = 1 To 10
                Range("A1").Offset(0, i - 1).Value = i
        Next i
End Sub
```

Loops – What is happening?

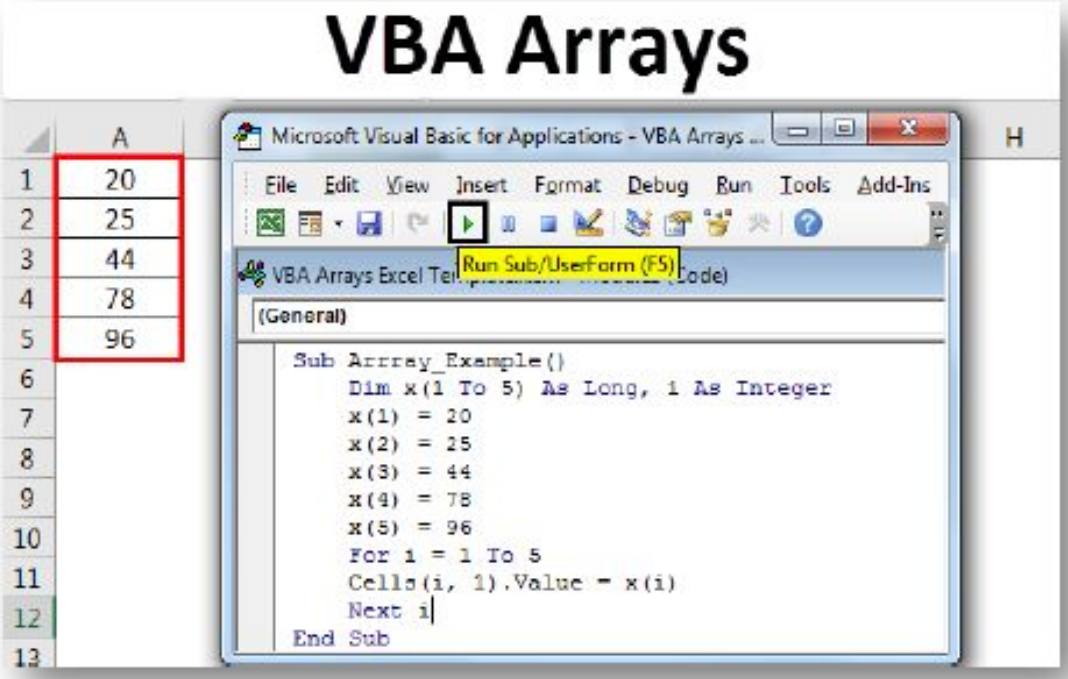
- Temporary variable + "Array"

```
Sub LoopThroughRows()
        For i = 1 To 10
                Range("A" & i).Value = i
        Next i
End Sub
```

```
Sub LoopThroughColumnsWithCells()
        Dim i As Integer: i = 1
        For Each cell In Range("A1:J1").Cells
                cell.Value = i
                i = i + 1
        Next cell
End Sub
```

Loops – What is happening?

- Array = many variables





Python – Origin

- Released in 1991 (same as Visual Basic)
- Free and open-source



Motivation – Soft Limits

- Protective measures not matching modern days

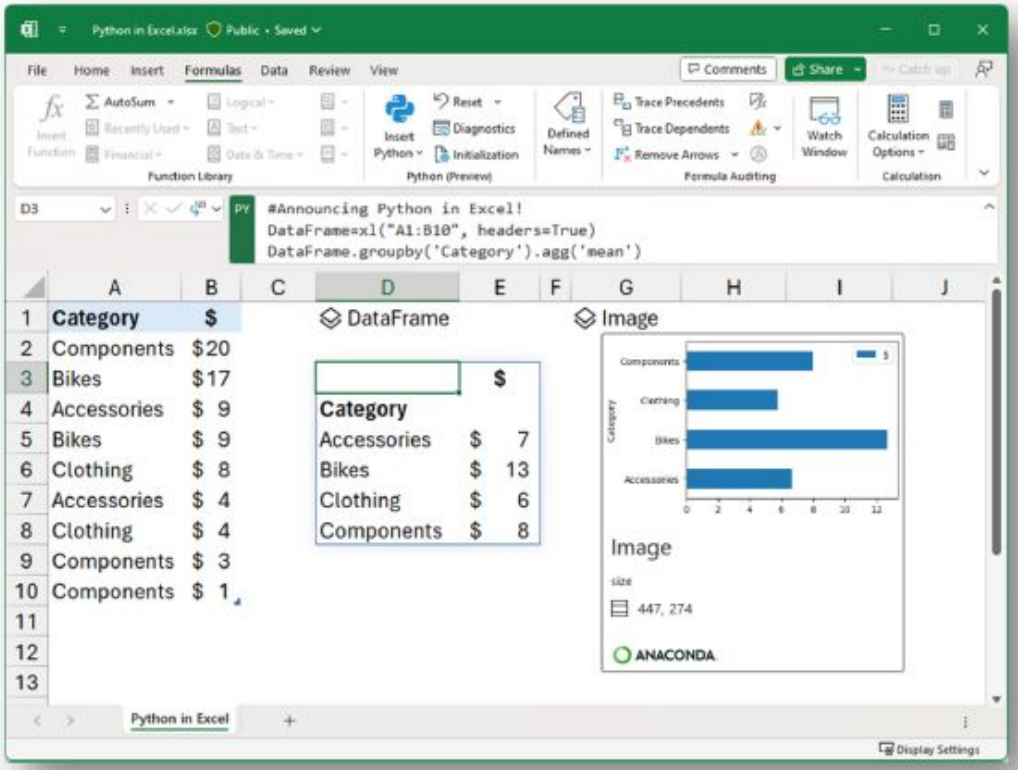
Feature	Maximum limit
Total number of rows and columns on a worksheet	1,048,576 rows by 16,384 columns
Column width	255 characters
Row height	409 points
Page breaks	1,026 horizontal and vertical

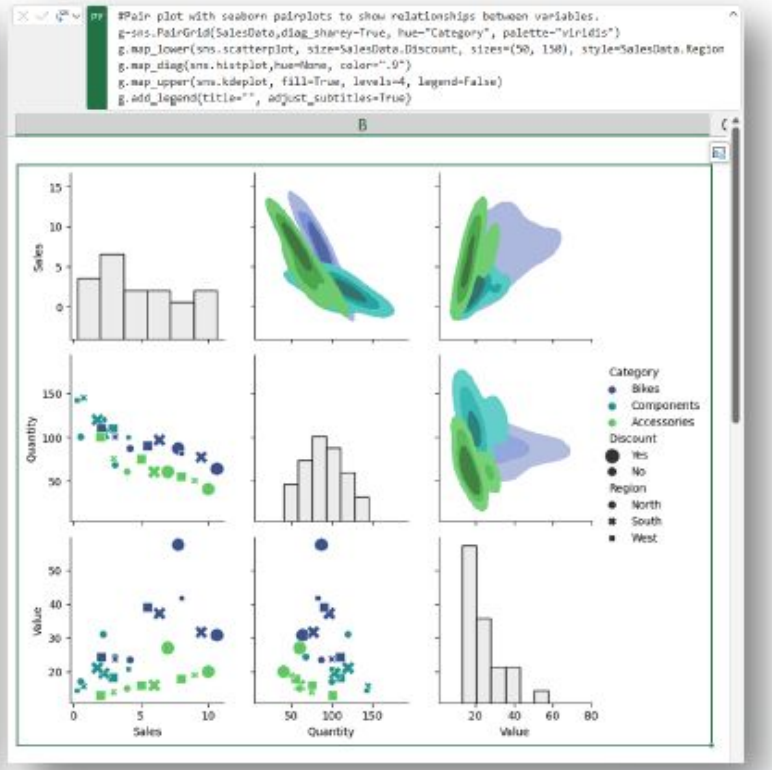
1. Microsoft. Excel Specifications and limits. https://support.microsoft.com/en-us/office/excel-specifications-and-limits-1672b34d-7043-467e-8e27-269d656771c3

Business Process Automation with Python

Motivation – Extensibility

- Python in Excel [1]



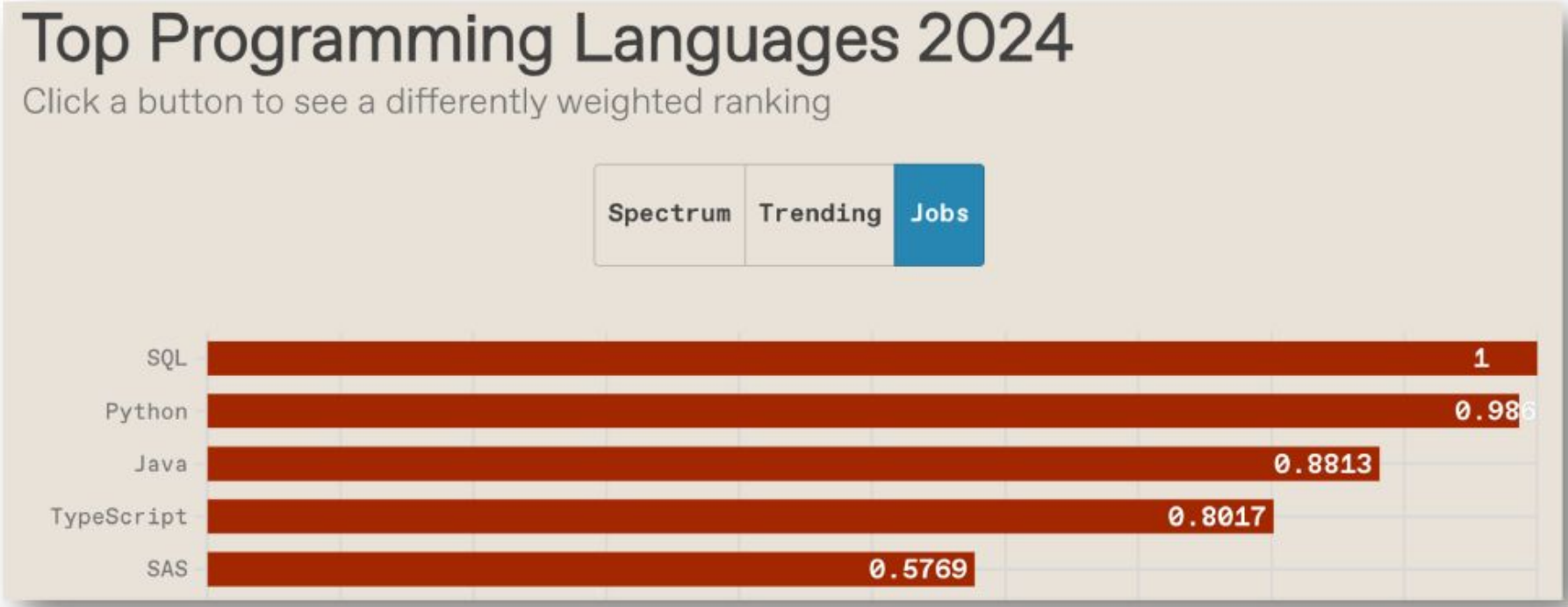


1. *Announcing Python in Excel: Combining the power of Python and the flexibility of Excel.* TECHCOMMUNITY.MICROSOFT.COM. https://techcommunity.microsoft.com/t5/excel-blog/announcing-python-in-excel-combining-the-power-of-python-and-the/ba-p/3893439

Business Process Automation with Python

Motivation – Jobs

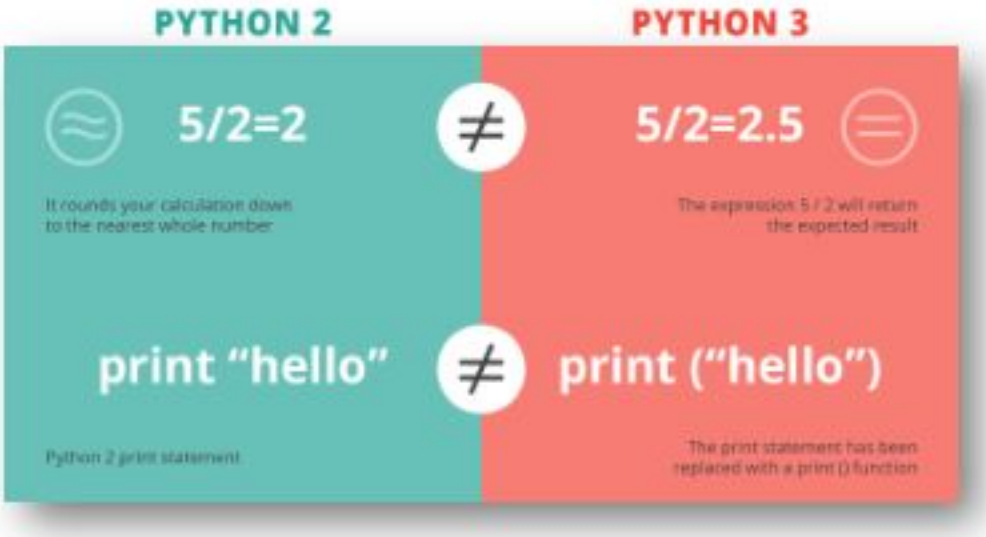
- Python ➔ 1 of the top languages for work



1. *The Top Programming Languages 2024*. (2024 August 22). IEEE Spectrum. https://spectrum.ieee.org/top-programming-languages-2024

Python runtime

- The language and basic runtime environment
- Just like Apple iOS, it keeps updating (e.g., Python 3.12.2)
  - Python 2 was no longer supported from 2020[1]

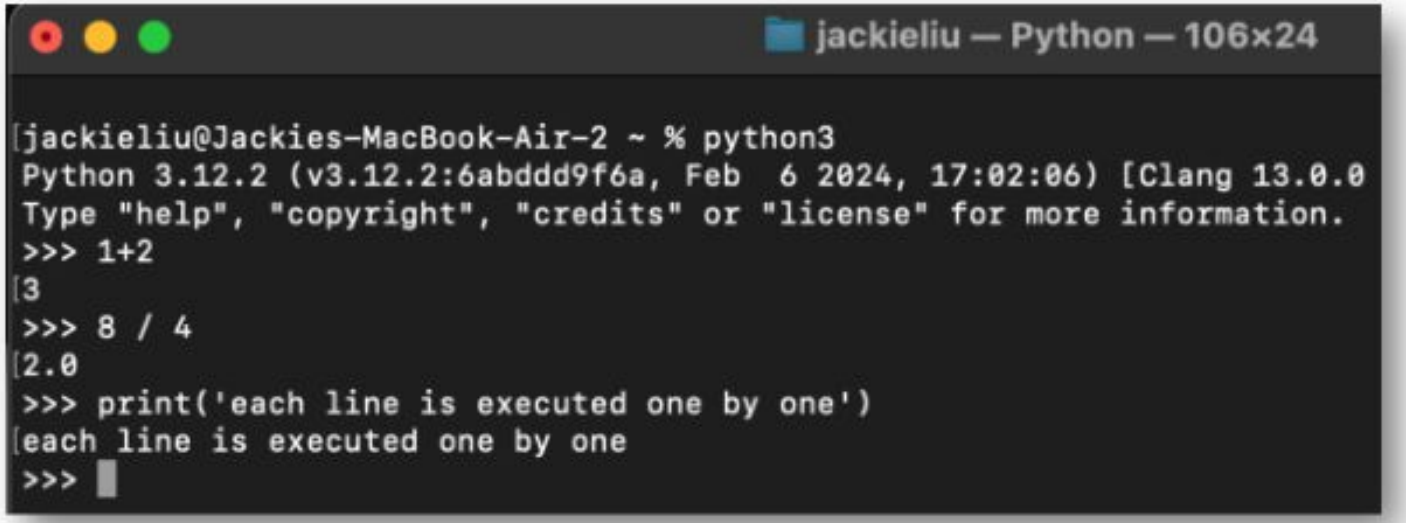


1. *Sunsetting Python 2*. Python.org. https://www.python.org/doc/sunset-python-2/

Business Process Automation with Python

Open Local Python Interpreter

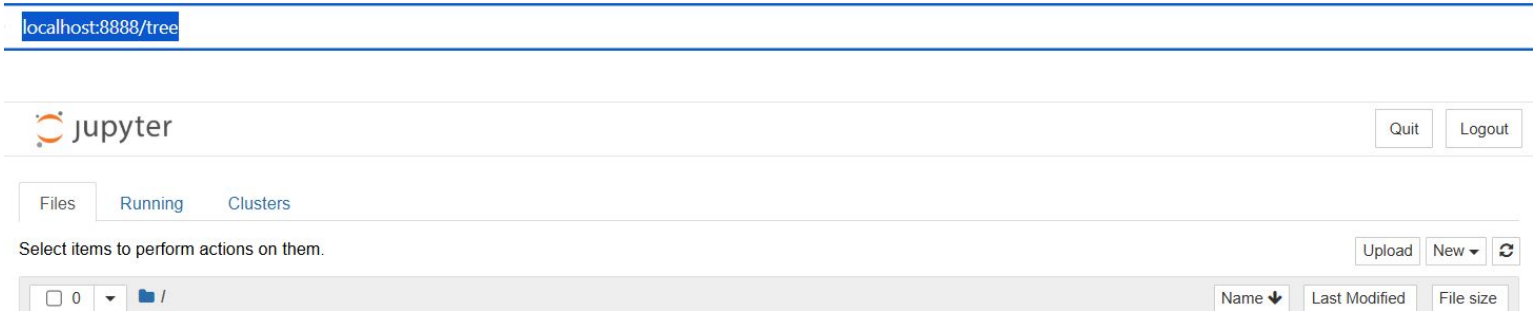
1. In Desktop, click ‘Startup’ button
2. Type “Anaconda Prompt (Anaconda3)“
3. Enter “python”
4. Input a line of codes -> Press Enter -> Run



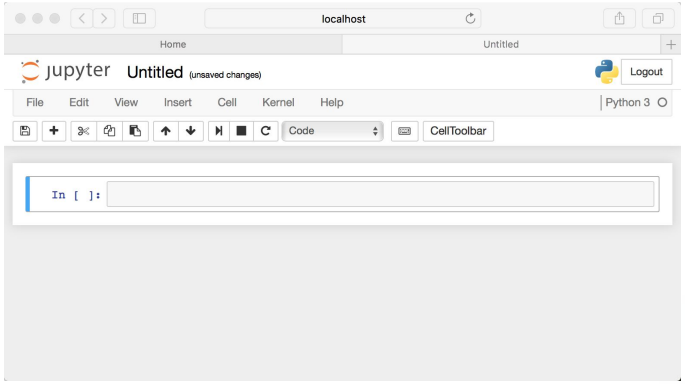
Python Coding Environment Practise - Jupyter notebook

Open Jupyter notebook

1. In Desktop, click ‘Startup’ button
2. Type “Anaconda Prompt (Anaconda3)“
3. Type “jupyter notebook“ to trigger your first notebook
4. http://localhost:8888/tree is opened

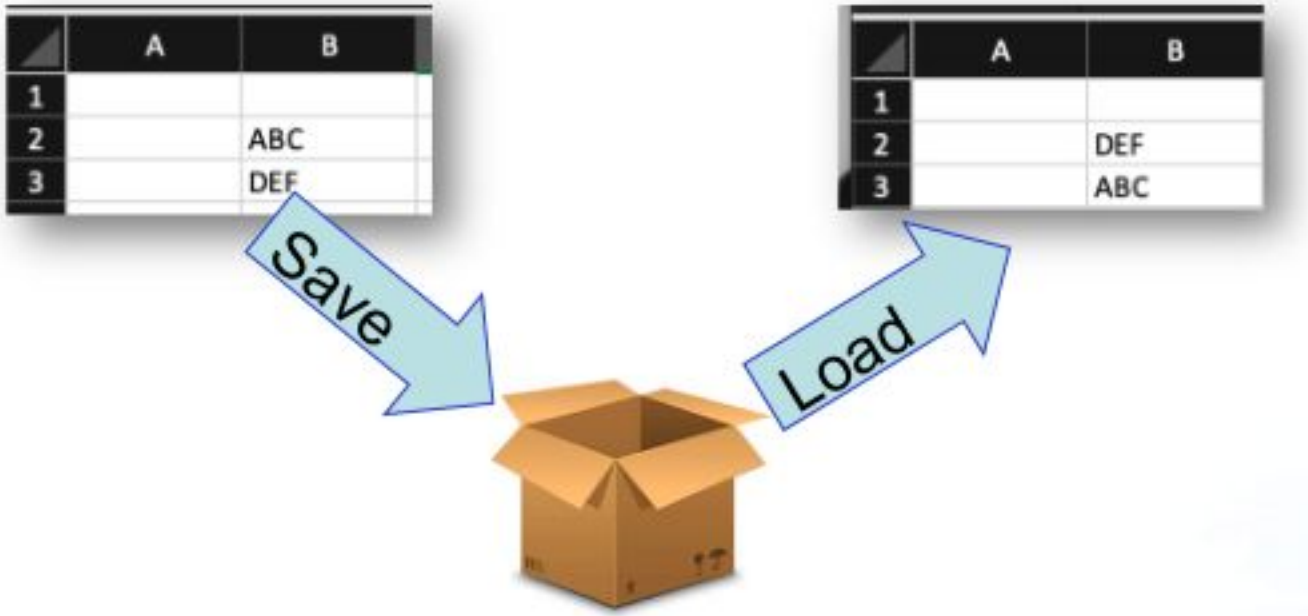


5. In right top side, click New -> Python 3
6. Renamed file name to “first_pythonlab”



Variables

- From the previous lecture
  - Variables ➔ Storage of values
  - Values ➔ carry data types



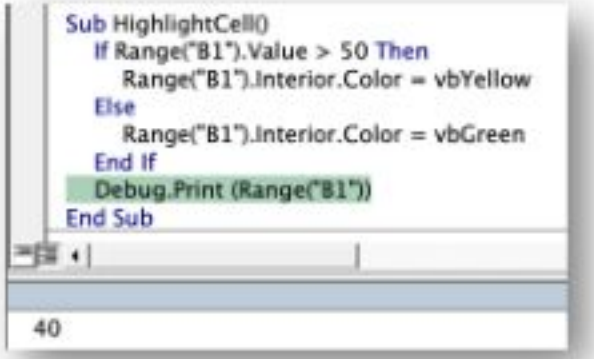
Variables

- Declaration
  - In VBA, declarations ("dim") are recommended
    - Dim year As **Integer**
    - year = 2024
  - In Python, variables are created during assignment
    - year = 2024
  - To assign without an actual value
    - year = None



Value Check

- VBA
  - Debug.Print()



- Python
  - print()

```
print(1 + 2)          # This gives 3
print(1 + 2 + 3)      # This gives 6
```

```
3
6
```

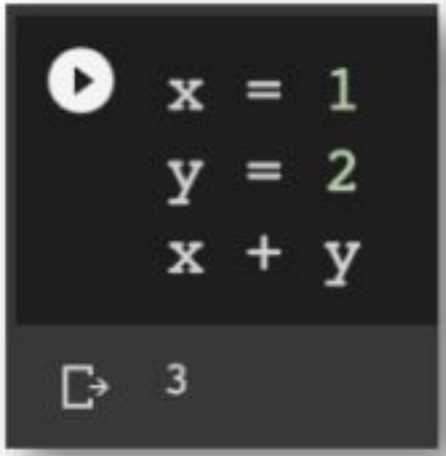
  - Last line in a block

```
1 + 2          # This is not shown
1 + 2 + 3      # This is shown
```

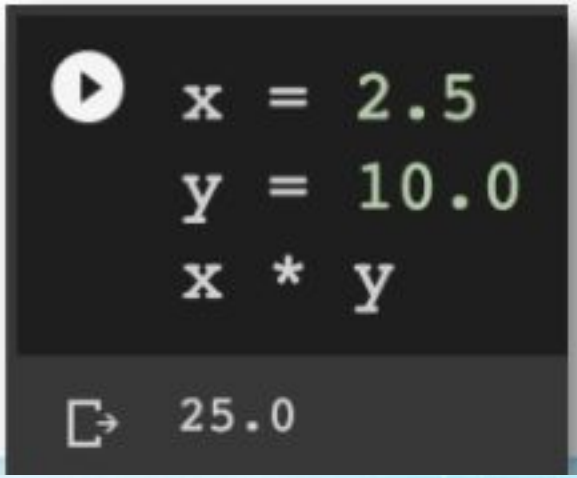
```
6
```

Primitive Data Types - Numeric

- Int

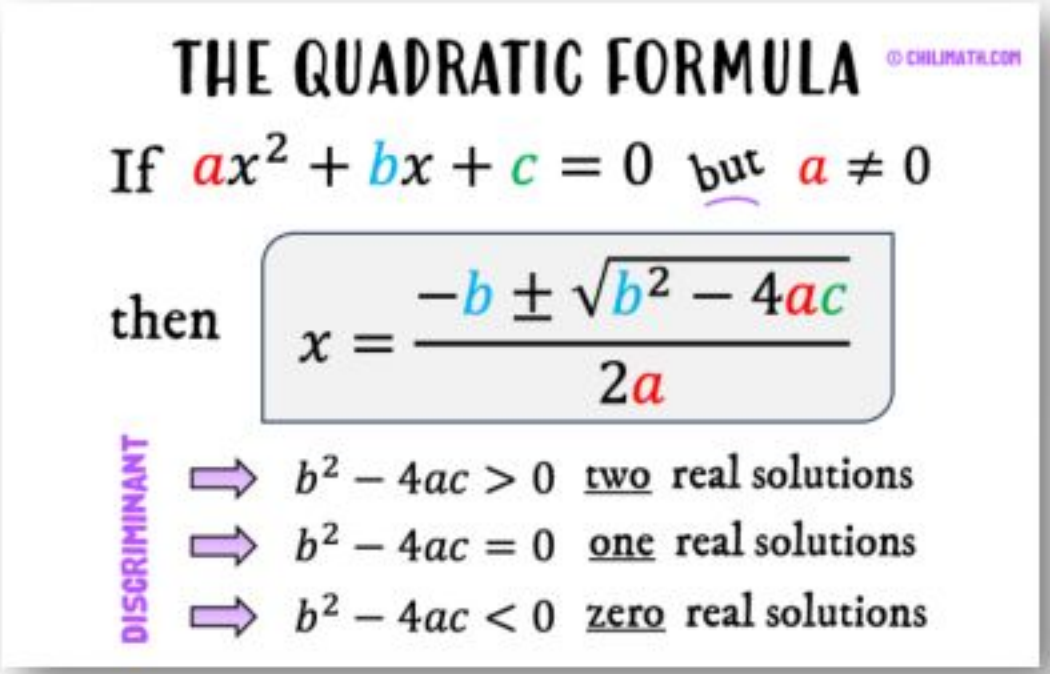


- Float



Primitive Data Types - Numeric

- Complex (For scientific calculations)




```
x = 3 + 4j
y = 2 + 2j
x - y
```

```
(1+2j)
```

$$i = \sqrt{-1}$$

Primitive Data Types - Numeric

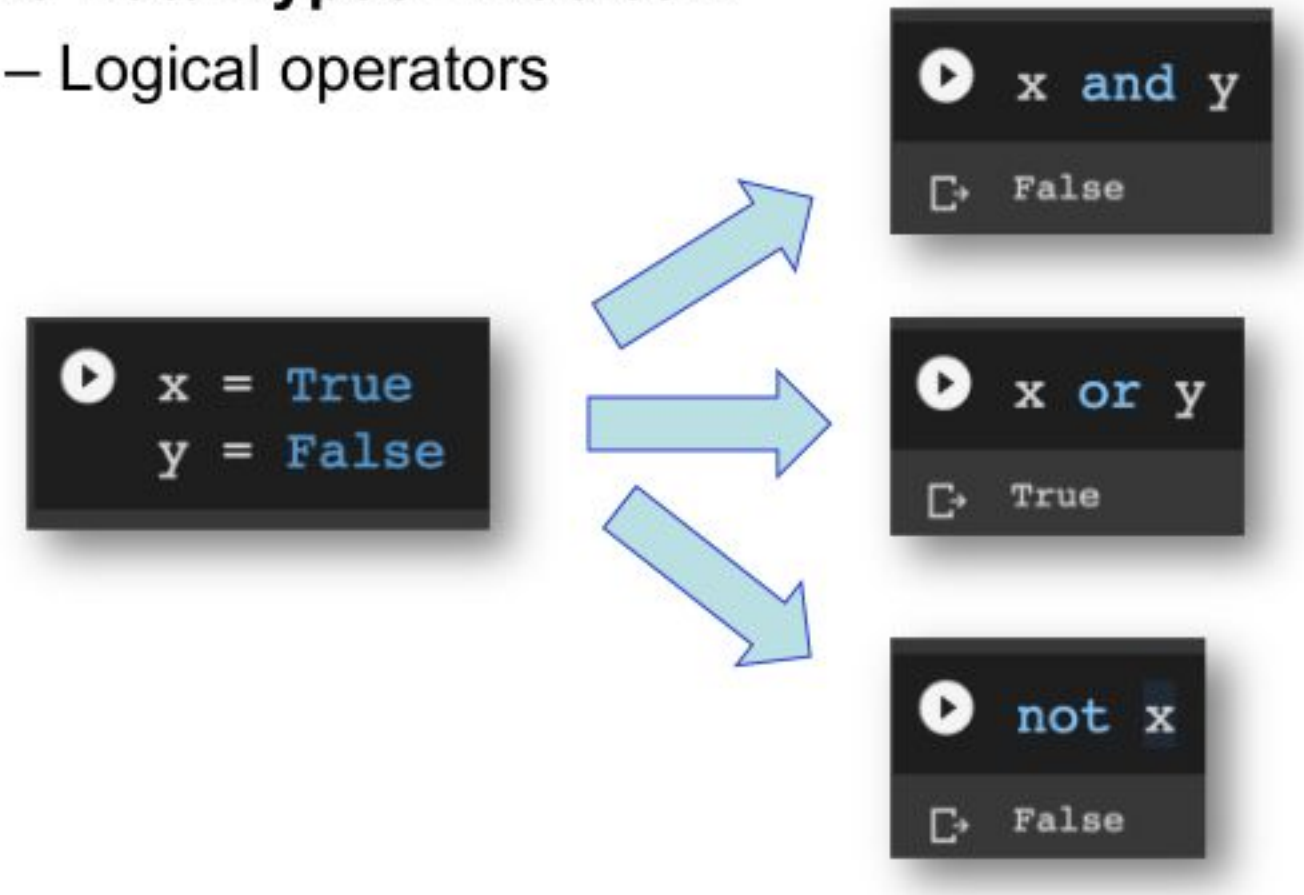
- Common Functions
  - round() / pow() / abs()

```
print(round(9876.54321, 2))    # Round to 2 decimal places
print(round(9876.54321, 0))    # Round to nearest integer
print(pow(2, 3))               # 2 * 2 * 2 = 8
print(abs(-5))                 # Negative numbers become positive
```

```
9876.54
9877.0
8
5
```

Primitive Data Types - Boolean

- Bool – Logical operators



Primitive Data Types - Boolean

- Bool – comparators
  - Generates True/ False

Operator	Name	Example
==	Equal	5 == 5
!=	Not equal	26 != 3
>	Greater than	100 > 67
<	Less than	89 < 216
>=	Greater than or equal to	90 >= 54
<=	Less than or equal to	23 <= 77

Primitive Data Types - Text

- String
  - Representation

```
x = 'This is a string'                    # Single quote
y = "This is also a string"               # Double quote
z = '''This is a very long string
spanning across more than 1 line'''       # Multi-line

print(x)
print(y)
print(z)
```

```
This is a string
This is also a string
This is a very long string
spanning across more than 1 line
```

Primitive Data Types - Text

- String
  - Concatenation ➔ "+" ➔ glue 2 strings together

```
x = 'This is a string'
y = "This is also a string"
```

```
print(x + ' and ' + y)  # Adding 2 strings together
```

```
This is a string and This is also a string
```

Primitive Data Types - Text

- String
  - Concatenation ➔ Number vs Text






```
lobster_price = 936
lobster_text = str(lobster_price)
print(2 * lobster_price)  # 1872
print(2 * lobster_text)   # 936936
```

```
1872
936936
```

```
print(10 * '=')
print('WELCOME')
print(10 * '=')
```

```
==========
WELCOME
==========
```

1. 星島日報. (2023, May 6). *荃灣中菜館驚現「天價龍蝦」兩隻竟索價90萬元 酒樓親解釋*. Singtaousa.com; 星島日報. https://www.singtaousa.com/2023-05-06/%e8%8d%83%e7%81%a3%e4%b8%ad%e8%8f%9c%e9%a4%a8%e9%a9%9a%e7%8f%be%e3%80%8c%e5%a4%a9%e5%83%b9%e9%be%8d%e8%9d%a6%e3%80%8d-%e5%85%a9%e9%9a%bb%e7%ab%9f%e7%b4%a2%e5%83%b990%e8%90%ac%e5%85%83-%e9%85%92/4488305

Primitive Data Types - String formatting

- F-String (Python 3.6 or newer)
  - A smart way to format strings

```
x = 4
y = 8.8
f'x is {x} and y is {y}; y divided by x is {y / x}'
```

```
'x is 4 and y is 8.8; y divided by x is 2.2'
```

1. *Input and Output*. Python Documentation. https://docs.python.org/3/tutorial/inputoutput.html

Primitive Data Types - String formatting

- F-String (Python 3.6 or newer)
  - A smart way to format strings (advanced)

```
z = 1234.56789
print(z)
print(f'{z:.2f}')
print(f'{z:,.2f}')
```

```
1234.56789
1234.57
1,234.57
```

1. *Input and Output*. Python Documentation. https://docs.python.org/3/tutorial/inputoutput.html

Primitive Data Types - String

- Common Functions
  - upper() / lower() / replace()

```
# Common functions
test_string = 'Hello Hong Kong!'
print(test_string.upper())
print(test_string.lower())
print(test_string.replace('Hello', 'Bello'))
```

```
HELLO HONG KONG!
hello hong kong!
Bello Hong Kong!
```

Primitive Data Types - Conversion

- **VBA**
  - CInt() / CDbl() / Format()
- **Python**
  - int() / float() / str() / bool()

```
int(4.5)
4
```

```
float(5)
5.0
```

```
str(5.55)
'5.55'
```



```
bool('Some values')
True
```

```
bool('')
False
```

Writing Python Function

Why Use Functions?

1. Avoid repeating the same code.
2. Break complex problems into smaller, manageable steps.
3. Make code easier to read and understand.
4. Reuse logic across multiple programs or modules.
5. Simplify testing and debugging.

What Is a Function?

1. A function is a named block of code that performs a specific task.
2. It can accept inputs, called **parameters** or **arguments**.
3. It can return an output using the `return` statement.
4. Functions are defined with the `def` keyword.

Methods

- VBA
  - Sub-routine: reusable code blocks
  - Function: sub-routine which gives an output
- Python
  - Methods: Use "return" if output is required

VBA

```
Function AddNumbers(x As Double, y As Double) As Double
    AddNumbers = x + y
End Function
```

```
?AddNumbers(1,2)
 3
```

Python

```
def add_numbers(x, y):
    return x + y
```

```
def add_numbers(x, y):
    return x + y

print(add_numbers(1,2))
```

Writing Python Function

Function Syntax

Python

```
def function_name(parameter1,
parameter2):

  """Optional docstring explaining the
    function.""")
  docstring
 # Function body

  return result
```

Key Points

- **def** starts the function definition
- **function_name** should use **lowercase** words separated by underscores
- Parameters are optional
- **return** sends a value back to the caller

Python Fundumentals • Function Definitions • Best Practices

Writing Python Function

Examples:

```
def greet():
    print("Hello, world!")

greet()
```

```
>>> def greet():
...     print("Hello, world!")
...
>>> greet()
Hello, world!
```

Parameters and Arguments

1. **Parameters** are variables listed in the function definition.
2. **Arguments** are the actual values passed when calling the function.

Examples:

```
def greet(name):
    print(f"Hello, {name}!")

greet("Eddie")
```

```
>>> def greet(name):
...     print(f"Hello, {name}!")
...
>>> greet("Eddie")
Hello, Eddie!
```

Writing Python Function

Return Values

- Use `return` to send a result back to the caller.
- A function can return one value, multiple values, or no value.
- If there is no `return`, Python returns `None` by default.

Examples(Add 2 Numbers):

```
def add(a, b):
    return a + b

result = add(3, 5)
print(result)
```

```
>>> def add(a, b):
...     return a + b
...
>>> result = add(3, 5)
>>> print(result)
8
```

Examples(2 Numbers Minus):

```
def minus(a, b):
    return a - b

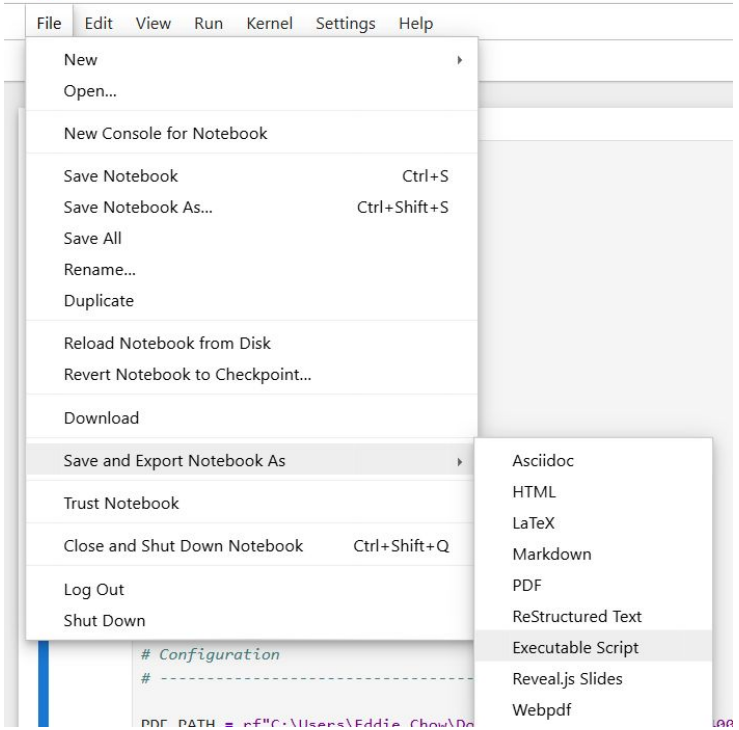
result = minus(5, 3)
print(result)
```

```
>>> def minus(a, b):
...     return a - b
...
>>> result = minus(5, 3)
>>> print(result)
2
```

Running python as a file

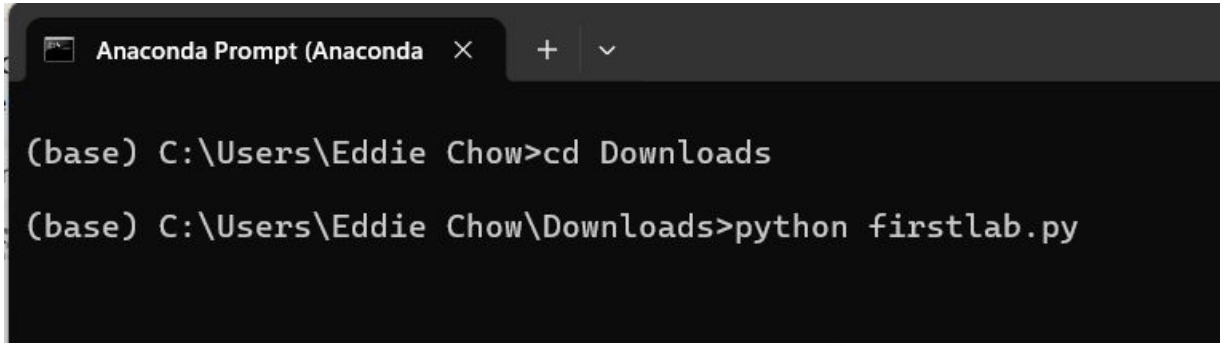
Download the .py file

1. Click “File” -> “Save and Export Notebook As”
2. Click “Executable Script” and “first_pythonlab.py” is download



Run the .py file

1. Open Anaconda Prompt and enter “Downloads”
2. Type command “first_pythonlab.py”



Practice 1 – Market Capitalization

- Compute market capitalization of 2 companies
- Show the difference between the two
- Format the result up to 2 decimal places



- **Starting notebook:**

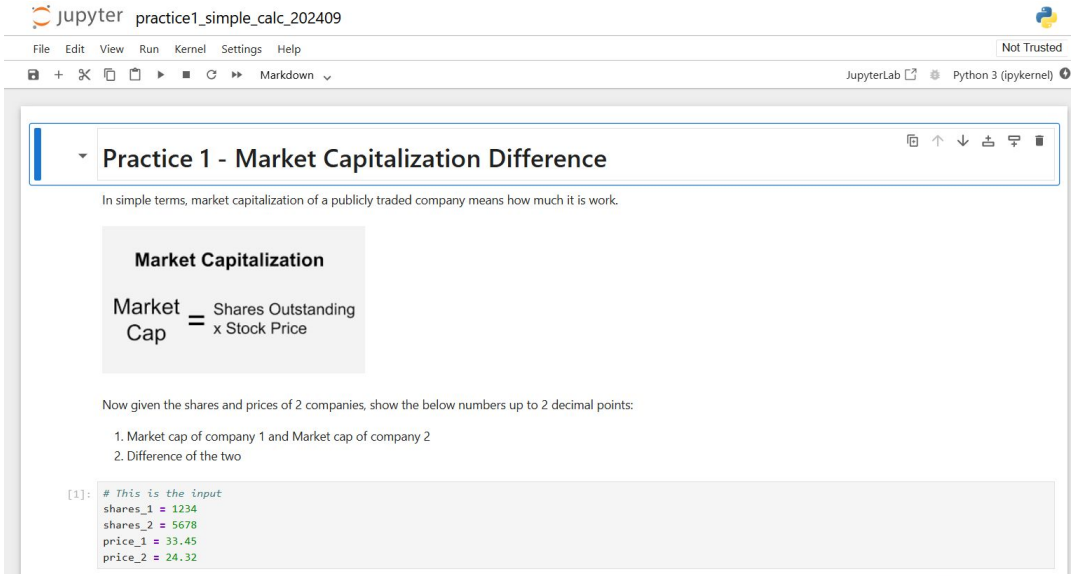
  https://github.com/innoviai/IPA_Courses-October2026/blob/main/practice1_simple_calc_202409.ipynb

How to download and edit jupyter notebook

1. Open https://github.com/innoviai/IPA_Courses-October2026/blob/main/practice1_simple_calc_202409.ipynb
2. Click button “Download raw file”
3. Don’t direct click the practice1_simple_calc_202409.ipynb
4. In jupyter notebook, click “Upload” and select practice1_simple_calc_202409.ipynb



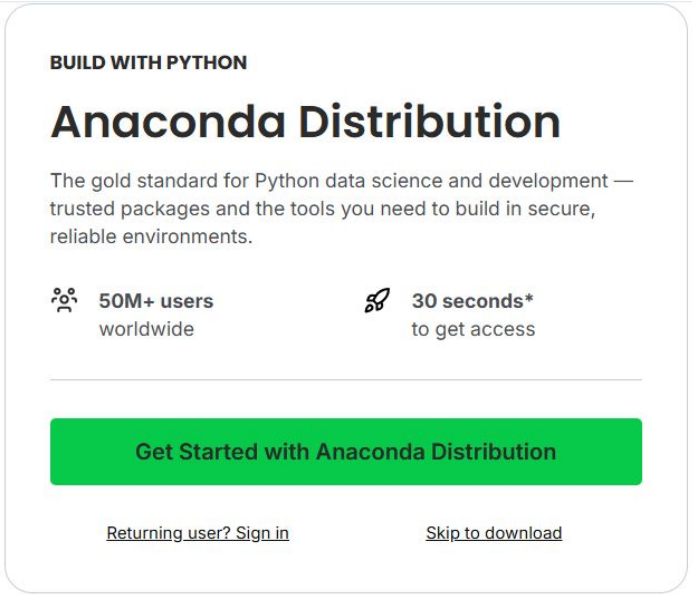
5. Then click practice1_simple_calc_202409.ipynb to edit the content



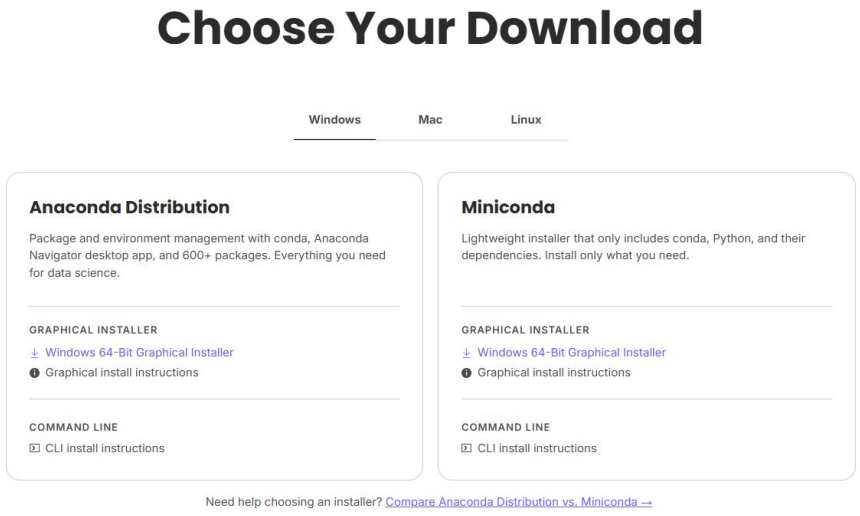
Business Process Automation with Python

Download and Install Local Python Interpreter

1. Download Anaconda Distribution at https://www.anaconda.com/download



2. Click “Skip to download”



3. Install in your own PC

How to run Python – Jupyter Notebook

- Google Colaboratory (Colab)
  - Jupyter Notebook powered by Google Cloud [1]
    - https://colab.research.google.com/
  - Benefits [2]
    - "Zero configuration required"
    - Easy sharing
    - Free access: CPU & GPU



1. *Google Colab FAQ*. https://research.google.com/colaboratory/faq.html
2. *Google Colaboratory*. https://colab.research.google.com

Cloud Software Introduction

Colaboratory, or Colab for short, is a Google Research product, which allows developers to write and execute Python code through their browser. Google Colab is an excellent tool for deep learning tasks. It is a hosted Jupyter notebook that requires no setup and has an excellent free version, which gives free access to Google computing resources such as GPUs and TPUs.

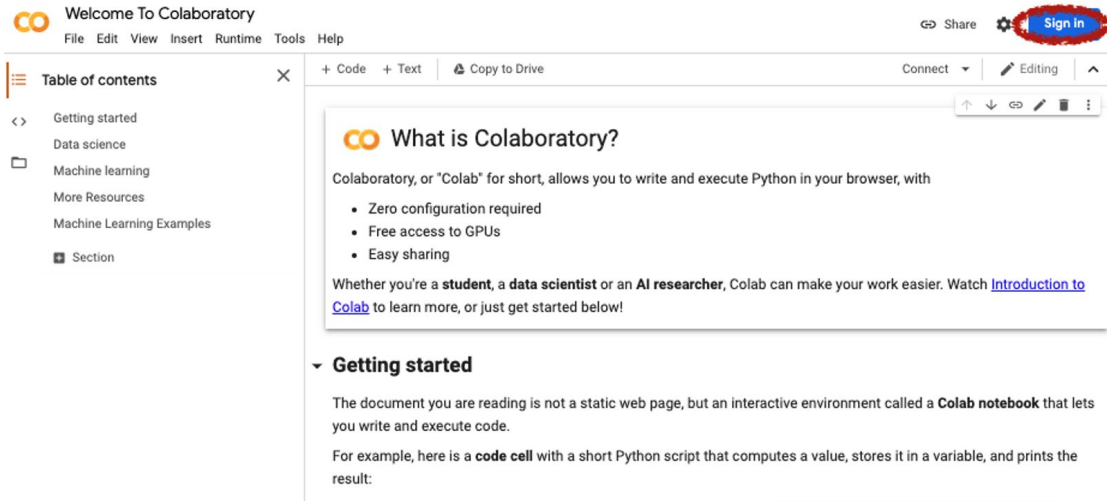
Why Should I Use Google Colab?

There are several reasons to opt to use Google Colab instead of a plain Jupyter Notebook instance:

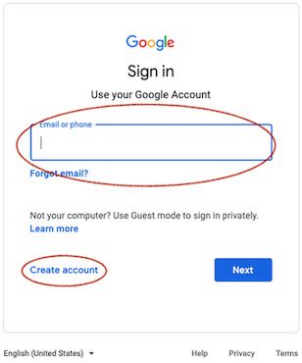
- Pre-Installed Libraries
- Saved on the Cloud
- Collaboration
- Free GPU and TPU Use

Google Colab Setup

1. Visit the Google Colab (http://colab.research.google.com/) page, which will direct you to the Google Colaboratory Welcome Page.
2. Click the **Sign in** button on the right top.



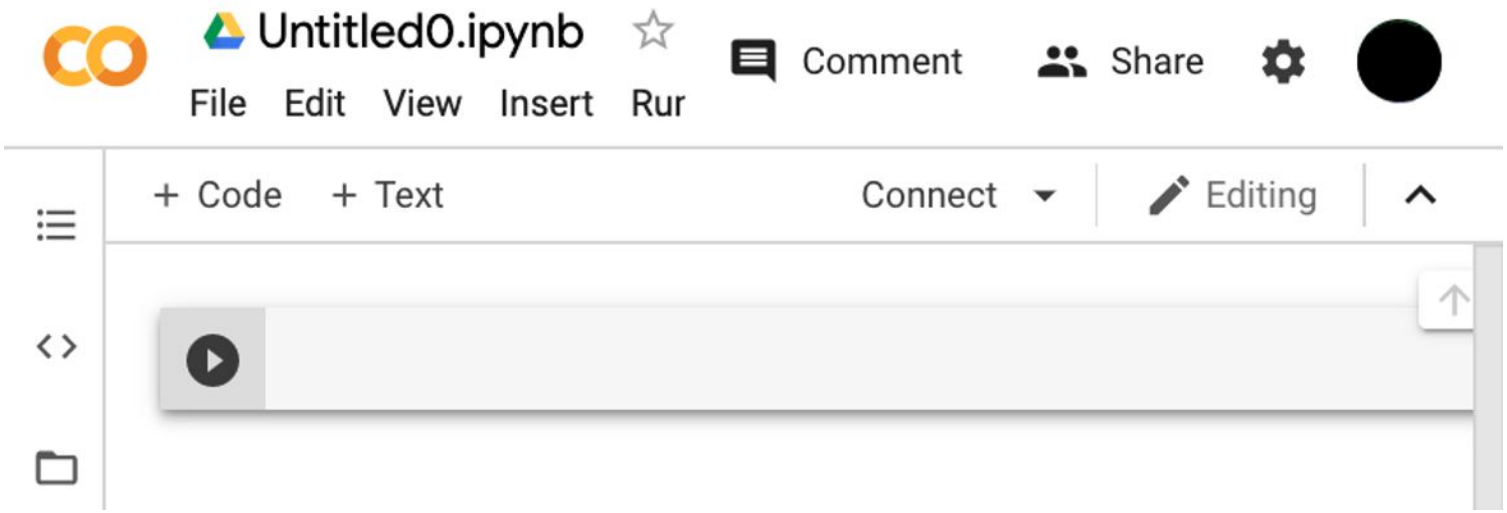
3. Sign in with your Gmail account.



Google Colab Setup

4. As soon as you complete the sign-in process, you are ready to use Google Colab.

5. You may easily create a new Colab notebook on this page by clicking *File> New notebook*.



Summary on running Python

Method 1 — Python Interpreter

- Install Anaconda Distribution in https://www.anaconda.com/download
- Open a terminal or command prompt.
- Type `python` and press Enter.
- Enter Python commands one by one and see the output immediately.

Method 2 — Colab

- Go to colab.research.google.com.
- Sign in with your Google account.
- Create a new notebook.
- Write and run Python code in cells using Shift+Enter.

Method 3 — Jupyter notebook

- Run `jupyter notebook` in terminal.
- Open the browser interface.
- Create a new notebook and run cells with Shift+Enter.

Method 4 — Script Mode

- Write code in a text editor and save it as a `.py` file.
- Run it from the terminal using `python filename.py`.
- Example: `python main.py`.

Python Interpreter

- Fast to start, with low resource use
- Small Experiments Quick feedback
- Better to test python library
- Use PC CPU/GPU Resources.

Colab

- Runs in the cloud -> no local setup
- Easy to share and collaborate
- Free GPU/TPU Resources
- Data store in cloud with lower security
- Works well with Google Drive

Jupyter notebook

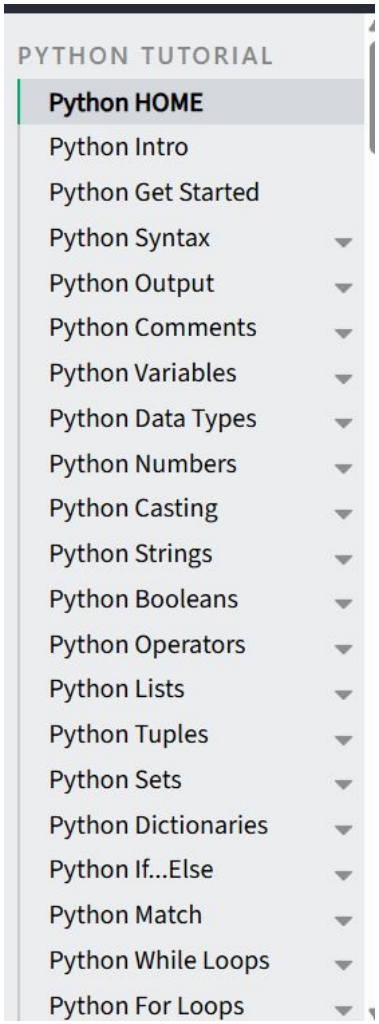
- Better interactive exploration.
- Open the browser interface.
- Better Documentation along code
- Data store locally, great security.

Method 4 — Script Mode

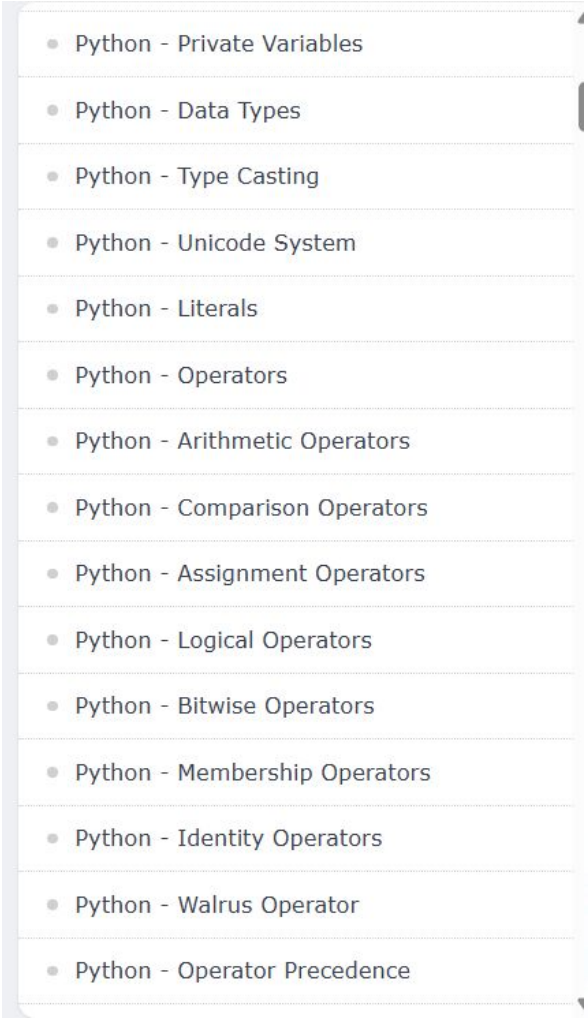
- Best for larger programs because code is saved in a structured file
- Easier to reuse, test, and deploy as a real application or module

Additional basic python learning

https://www.w3schools.com/python/



https://www.tutorialspoint.com/python/index.htm

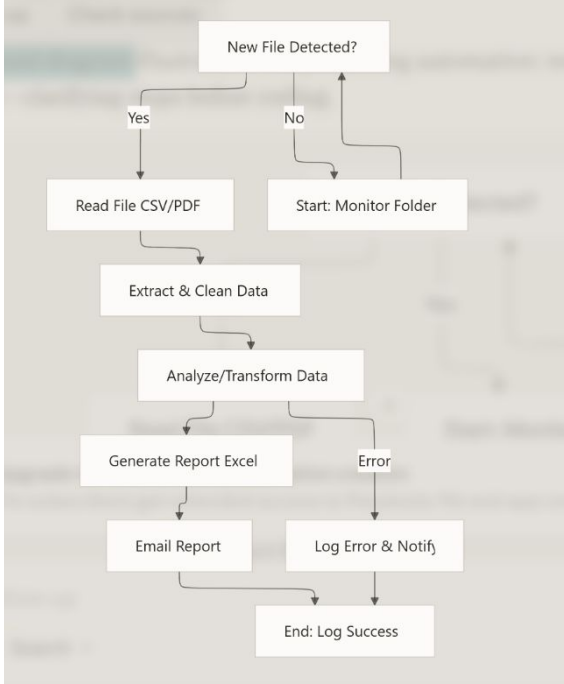


Python Process Automation Overview

Python streamlines working processes by automating tasks like data extraction and reporting, minimizing manual effort.

The objective of drawing the flowchart is to visually map the automation sequence, aiding in design, debugging, and stakeholder communication for reliable implementation.

Methodology of Workflow Flowchart: Mermaid Diagram



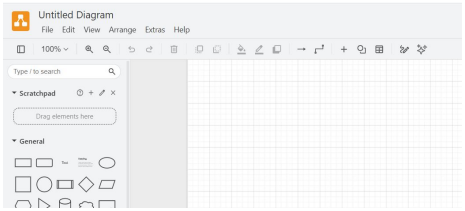
Software to draw Flowchart : draw.io

Key Benefits

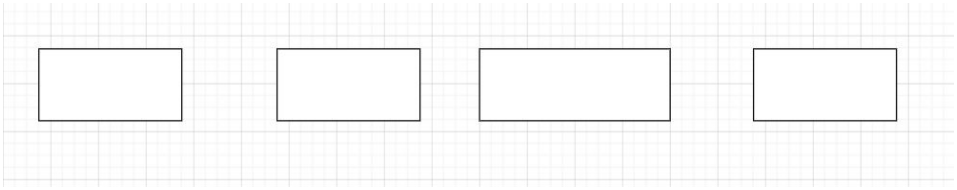
- draw.io (now diagrams.net) is a free, web-based diagramming tool ideal for creating flowcharts, offering drag-and-drop shapes, templates.

Quick Steps to Create a Flowchart

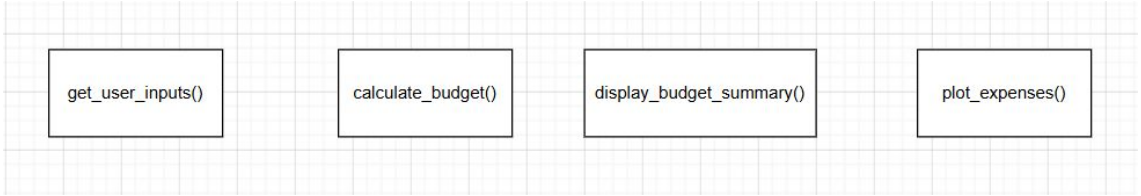
1. Go to app.diagrams.net.



2. Conduct an python flow chart with title “How to Manage Your Budget”
3. Drag 4 shapes “Rectangle” from left panel into main canvas

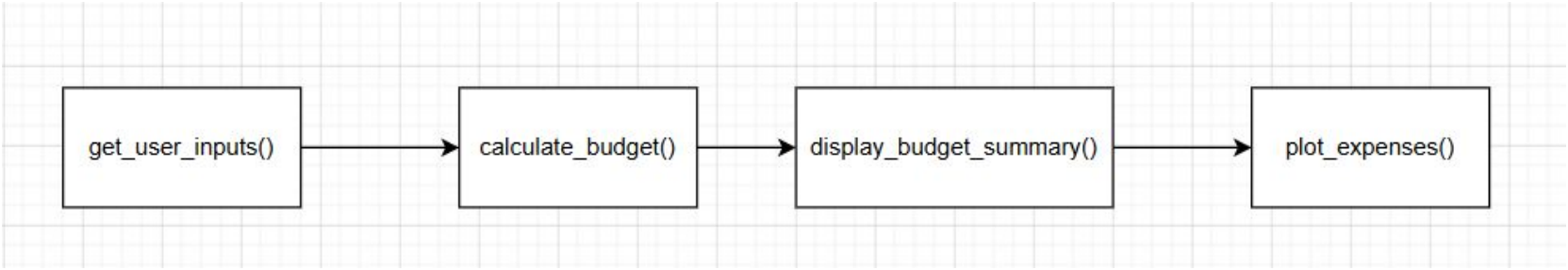


4. Add labels by double-clicking each shapes.

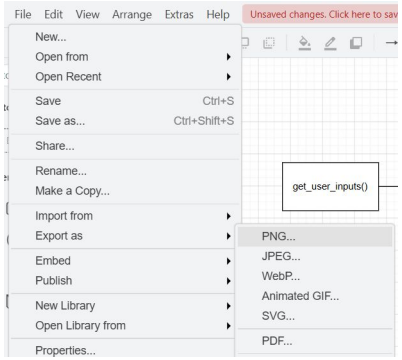


Software to draw Flow Charts: draw.io

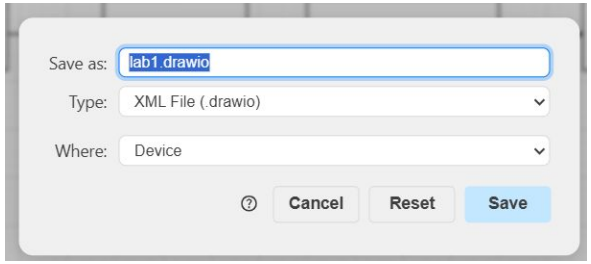
5. Connect with arrows



6. Click ”File” -> “Export As” and select either “PNG/PDF”



7. or Click “Save As” and assign an file name “lab1.drawio” and save to Device/Google Drive/OneDrive..



How to Manage Your Budget with a Simple Python Script

Open Jupyter notebook and enter main function

Main Function

```
income, expenses = get_user_input()
total_expenses, balance = calculate_budget(income, expenses)
display_budget_summary(income, total_expenses, balance)
plot_expenses(expenses)
```

Core Concepts

Each line represents a sequential execution: gather input, compute results, display output, and visualize data. Variables like `income` and `expenses` hold **values**, passed as **arguments** to functions for processing.

Values Examples

- `income = 5000` (an integer value).
- `expenses = [1000, 2000, 500]` (a list value).
- `balance = 1500` (computed result value).

Arguments Examples

- `income` is the first argument.
- `expenses` is the second argument.
- balance is the third argument.

How to Manage Your Budget with a Simple Python Script From Flow Chart

1. Getting User Input

```
import pandas as pd
import matplotlib.pyplot as plt

def get_user_input():
    """Get user input for income and expenses."""
    income = float(input("Enter your income: "))
    # Expecting expenses to be a dictionary input
    expenses = {}
    while True:
        category = input("Enter expense category (or 'done' to finish): ")
        if category.lower() == 'done':
            break
        amount = float(input(f"Enter amount for {category}: "))
        expenses[category] = amount
    return income, expenses
```

How to Manage Your Budget with a Simple Python Script

2. Calculating the Budget

```
def calculate_budget(income, expenses):
    """Calculate total expenses and balance."""
    total_expenses = sum(expenses.values())
    balance = income - total_expenses
    return total_expenses, balance
```

3. Displaying the Budget Summary

```
def display_budget_summary(income, total_expenses, balance):
    """Display the budget summary."""
    print(f"Income: ${income:.2f}")
    print(f"Total Expenses: ${total_expenses:.2f}")
    print(f"Balance: ${balance:.2f}")
```

How to Manage Your Budget with a Simple Python Script

4. Plotting the Expenses

```
def plot_expenses(expenses):
    """Plot the expenses as a bar chart."""
    df = pd.DataFrame(list(expenses.items()), columns=['Category', 'Amount'])
    df.plot(kind='bar', x='Category', y='Amount', legend=False)
    plt.ylabel('Amount ($)')
    plt.title('Expense Distribution')
    plt.show()
```

How to Manage Your Budget with a Simple Python Script

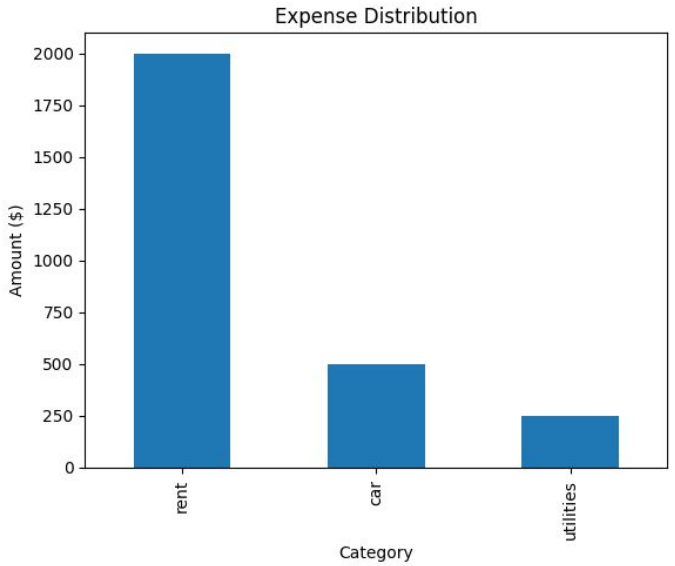
Run above Code and following is displayed

Main Input

Enter your income: 10000
Enter expense category (or 'done' to finish): rent
Enter amount for rent: 2000
Enter expense category (or 'done' to finish): car
Enter amount for car: 500
Enter expense category (or 'done' to finish): utilities
Enter amount for utilities: 250
Enter expense category (or 'done' to finish): done

Result

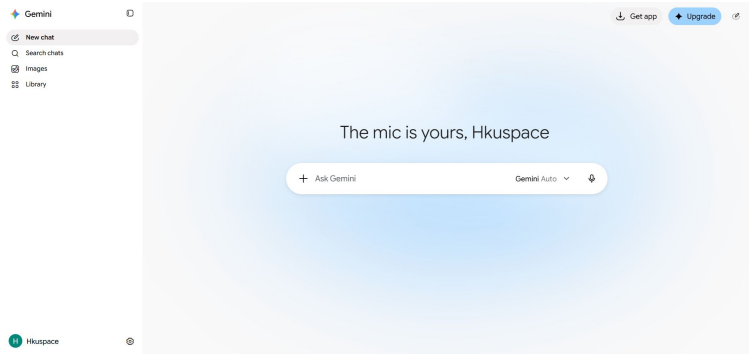
Income: $10000.00
Total Expenses: $2750.00
Balance: $7250.00



AI Tools to Generate Code

Quick flow to copy:

1. Open gemini.google.com
2. Click Gemini Auto and Select “Medium” or “High”
3. Enter Prompt “write python to Manage Your Budget with 4 function”



Here is a clean, easy-to-use Python script for managing a personal budget using **four distinct functions**:

1. `add_income()` – Records incoming money.
2. `add_expense()` – Records money spent.
3. `calculate_balance()` – Computes totals and net balance.
4. `display_summary()` – Prints a formatted overview.

Budget Management Python Script

Python

```python
def add_income(budget, amount, description):
    """Adds income to the budget record."""
    budget['income'].append({'amount': amount, 'description': description})
    print(f"Added income: ${amount:.2f} ({description})")

def add_expense(budget, amount, description):
    """Adds an expense to the budget record."""
    budget['expenses'].append({'amount': amount, 'description': description})
```

Extra Exercise - Generate Code for changing file name in special folder

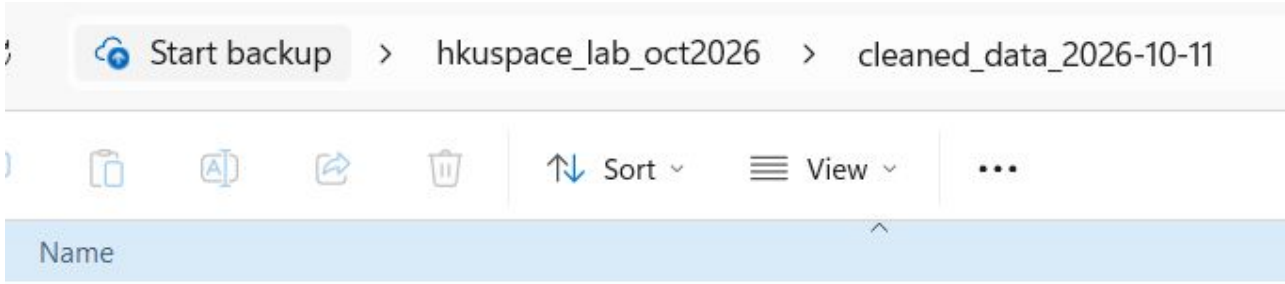
Quick flow to copy:

1. Open gemini.google.com
2. Click Gemini Auto and Select “Medium” or “High”
3. Enter Prompt “write python that input local directory url like "C:\Users\Eddie\hkuspace_lab_oct2026" and get all data inside to generate master csv file with heading "id, local_directory_url, local_file_path,filename, format, published_date, published_time" in standard format and local_file_path is like "C:\Users\Eddie\hkuspace_lab_oct2026\b 3.pdf", then add 2 more columns "cleaned_file_path, cleaned_filename" like "like "C:\Users\Eddie\hkuspace_lab_oct2026\b_3.pdf" that remove space in filename.
4. Copy the code into jupyter notebook to run it.
5. Enter another prompt “Modify code that rename original filename with space to new filename based on csv list column "cleaned_filename" that save the new file in new folder named "cleaned_data_{today}".

Extra Exercise - Modify All filename in 1 folder

Procedure

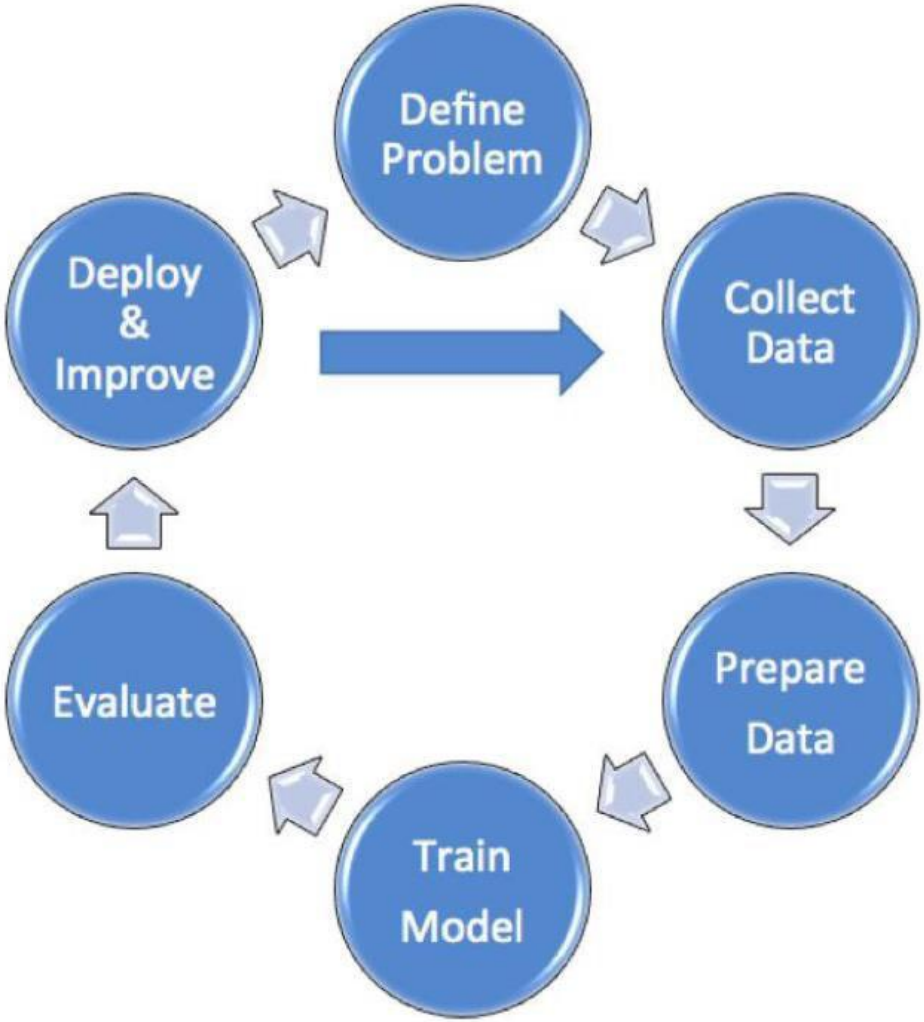
1. Download the following 3 files
   https://github.com/innoviai/IPA_Courses-October2026/blob/main/cleanedfilename.py
   https://github.com/innoviai/IPA_Courses-October2026/blob/main/filename_change.py
   https://github.com/innoviai/IPA_Courses-October2026/blob/main/master_files.csv
2. Create a folder named “hkuspace_lab_oct2026” at Desktops
3. Running cleanedfilenname.py to create Master List in CSV named “master_files.csv” that include directory, filename, format, cleaned file path and cleaned filename with heading “id,local_directory_url,local_file_path,filename,format,published_date,published_time, cleaned_file_path,cleaned_filename”
4. Conduct filename validation that cleaned file name have no space in master_files.csv
5. Run the filename_change.py to create a folder and change the original filename to new one.



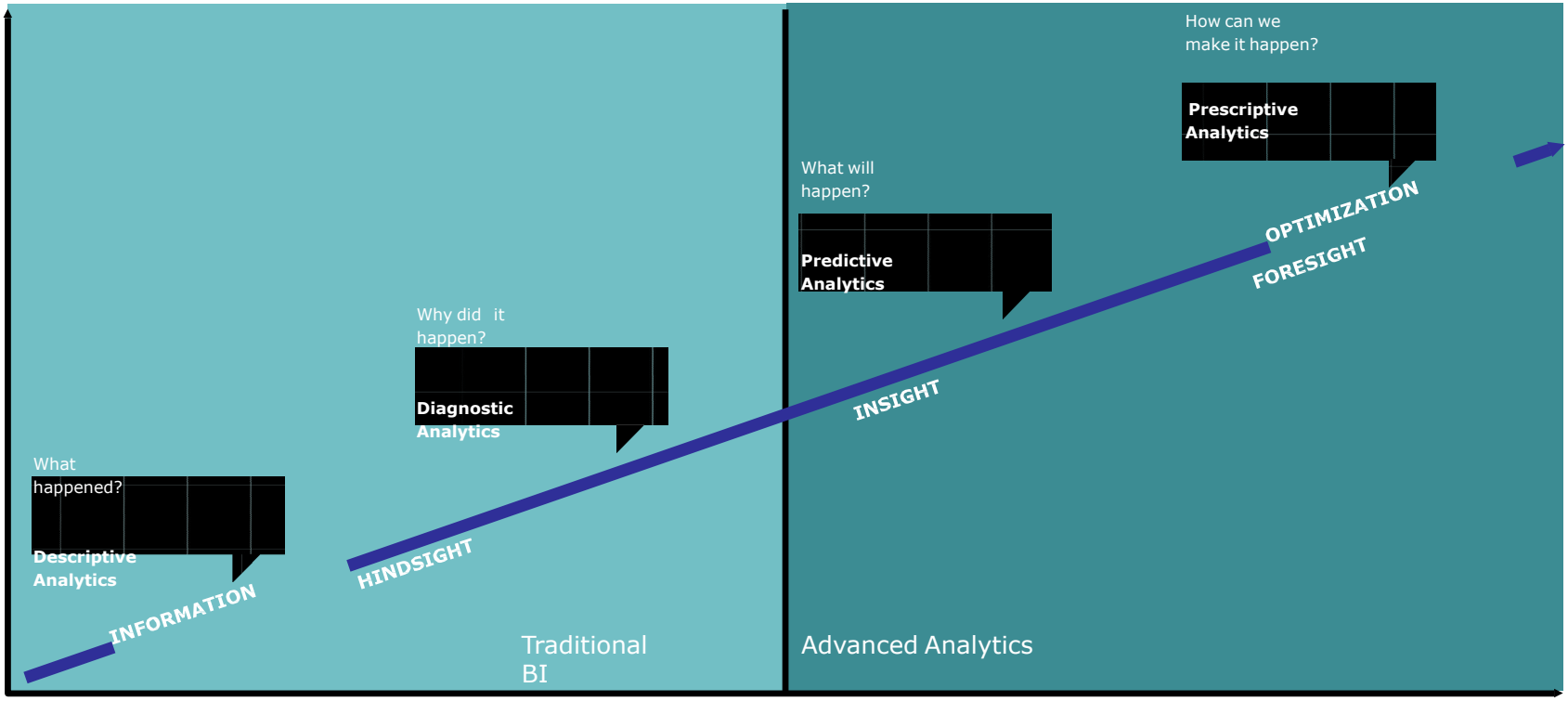
HKUSPACE
香港大學專業進修學院
HKU School of Professional and Continuing Education

- What is Process Automation, RPA, IPA?
- Overview of the technological building blocks related to process automation
- Contemporary tools for process automation
- Challenges and opportunities of business process automation
- Business implications of process automation

What is Business Analytics & Machine Learning?



From Digitalization To Data-Driven Decision Making

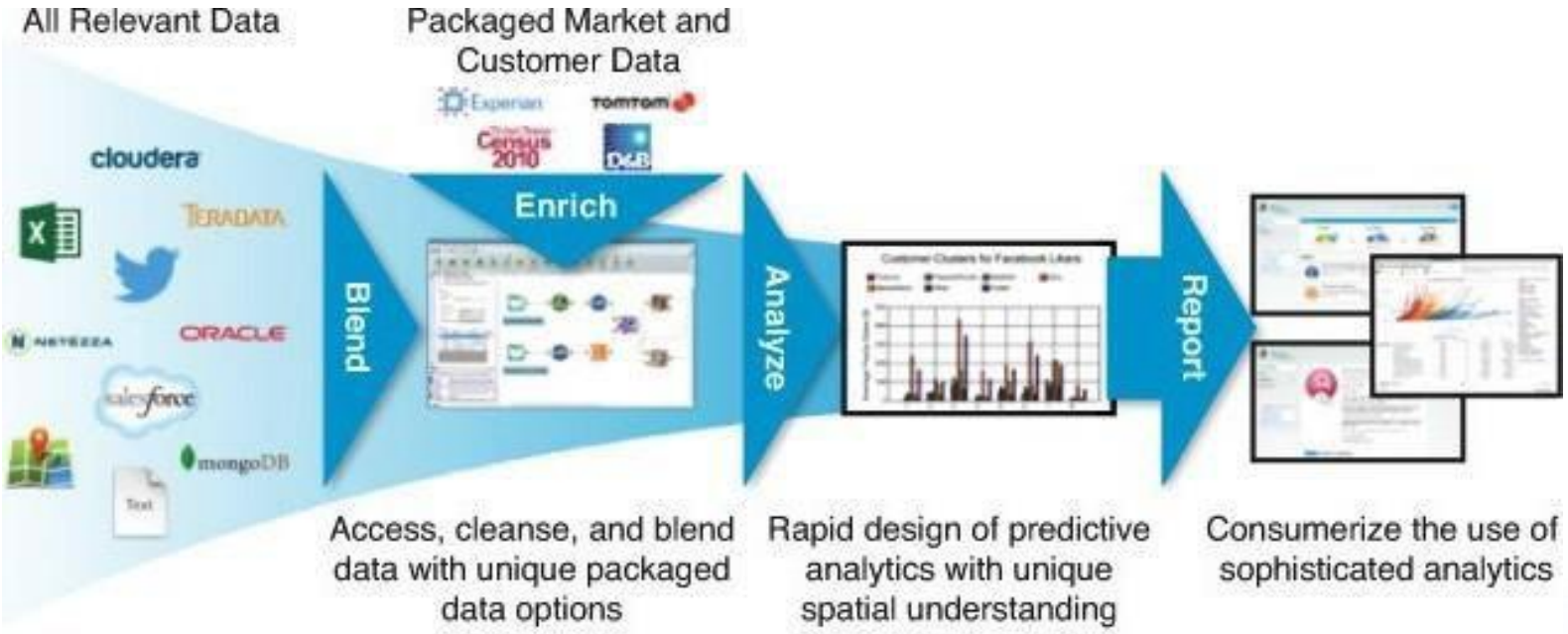


Modern BI Process - Blend, Enrich, Analyze, and Report

Predictive Analytics: By applying machine learning algorithms to historical project data, organizations can forecast outcomes more accurately and mitigate risks in project delivery

.

Business Intelligence: Automating the preparation of data for analytics tools allows organizations to gain actionable insights more rapidly, enhancing their decision-making capabilities.

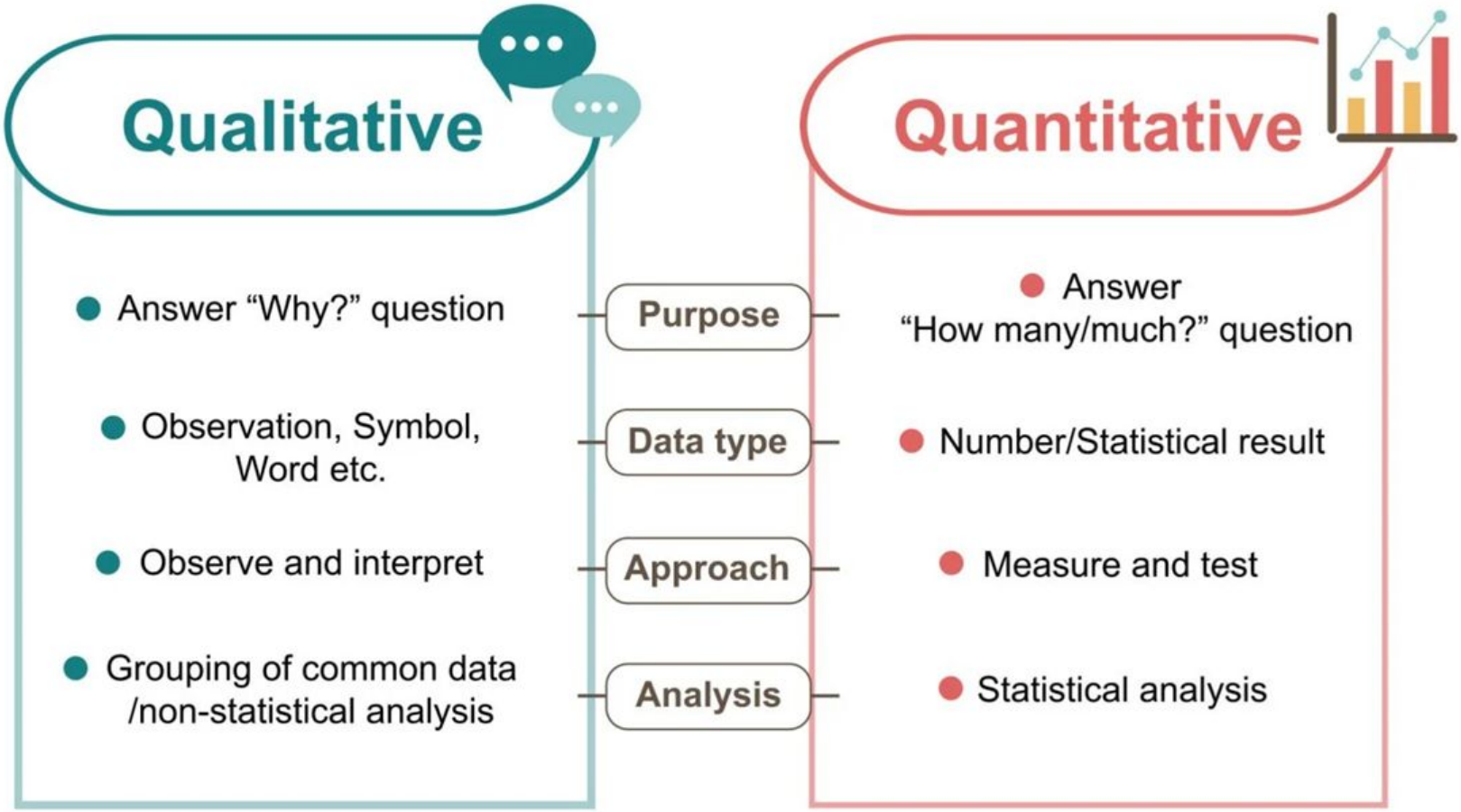


Data-Driven Business Strategy

A **data-driven business strategy** is a systematic plan that guides business decisions, operational improvements, and growth using verified data analysis rather than habit or intuition.



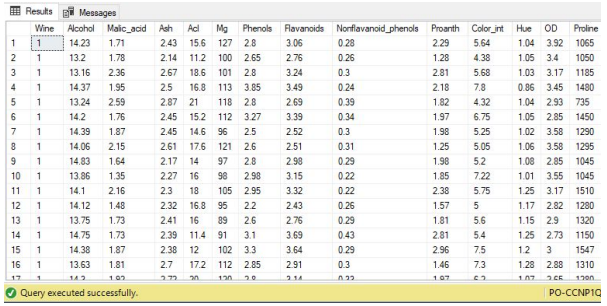
Type of Research Design



Concept of Structured, Unstructured Data

Structured Data

- Fixed schema (rows/columns, E.g. Tables, CSV, Excel)
- Easy to query/search (names, prices)
- Use VBA, SQL or ML techniques



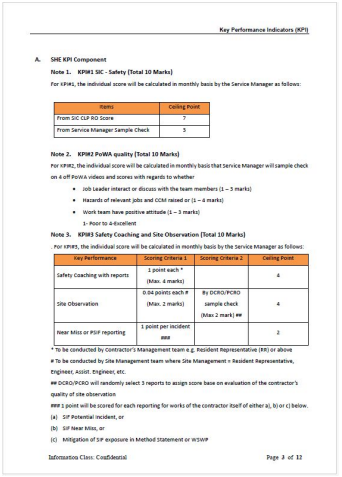
Results | Messages

	Wine	Alcohol	Malic_acid	Ash	Acl	Mg	Phenols	Flavanoids	Nonflavanoid_phenols	Proanth	Color_int	Hue	OD	Proline
1	1	14.23	1.71	2.43	15.6	127	2.8	3.06	0.28	2.29	5.64	1.04	3.92	1065
2	1	13.2	1.78	2.14	11.2	100	2.65	2.76	0.26	1.28	4.38	1.05	3.4	1050
3	1	13.16	2.36	2.67	18.6	101	2.8	3.24	0.3	2.81	5.68	1.03	3.17	1185
4	1	14.37	1.95	2.5	16.8	113	3.85	3.49	0.24	2.18	7.8	0.86	3.45	1480
5	1	13.24	2.59	2.87	21	118	2.8	2.69	0.39	1.82	4.32	1.04	2.93	735
6	1	14.2	1.76	2.45	15.2	112	3.27	3.39	0.34	1.97	6.75	1.05	2.85	1450
7	1	14.39	1.87	2.45	14.6	96	2.5	2.52	0.3	1.98	5.25	1.02	3.58	1290
8	1	14.06	2.15	2.61	17.6	121	2.6	2.51	0.31	1.25	5.05	1.06	3.58	1295
9	1	14.83	1.64	2.17	14	97	2.8	2.98	0.29	1.98	5.2	1.08	2.85	1045
10	1	13.86	1.35	2.27	16	98	2.98	3.15	0.22	1.85	7.22	1.01	3.55	1045
11	1	14.1	2.16	2.3	18	105	2.95	3.32	0.22	2.38	5.75	1.25	3.17	1510
12	1	14.12	1.48	2.32	16.8	95	2.2	2.43	0.26	1.57	5	1.17	2.82	1280
13	1	13.75	1.73	2.41	16	89	2.6	2.76	0.29	1.81	5.6	1.15	2.9	1320
14	1	14.75	1.73	2.39	11.4	91	3.1	3.69	0.43	2.81	5.4	1.25	2.73	1150
15	1	14.38	1.87	2.38	12	102	3.3	3.64	0.29	2.96	7.5	1.2	3	1547
16	1	13.63	1.81	2.7	17.2	112	2.85	2.91	0.3	1.46	7.3	1.28	2.88	1310

Query executed successfully. PO-CCNP1C

Unstructured Data

- No fixed format (text, images, videos, documents)
- Needs deep learning vs NLP/ML techniques
- Or use LLM / Foundation Model



Key Performance Indicators (KPI)

A. SHE KPI Component

Note 1: KPIM1 SIC - Safety (Total 10 Marks)

For KPIM1, the individual score will be calculated in monthly basis by the Service Manager as follows:

Item	Ceiling Point
From SIC CLP RO Score	7
From Service Manager Sample Check	3

Note 2: KPIM2 PoWA quality (Total 10 Marks)

For KPIM2, the individual score will be calculated in monthly basis that Service Manager will sample check on 4 off PoWA videos and scores with regards to whether

- Job Leader interact or discuss with the team members (1 – 5 marks)
- Hazards of relevant jobs and CCM raised or (1 – 4 marks)
- Work team have positive attitude (1 – 3 marks)

1- Poor to 4-Excellent

Note 3: KPIM3 Safety Coaching and Site Observation (Total 10 Marks)

For KPIM3, the individual score will be calculated in monthly basis by the Service Manager as follows:

Key Performance	Scoring Criteria 1	Scoring Criteria 2	Ceiling Point
Safety Coaching with reports	1 point each * (Max. 4 marks)		4
Site Observation	0.04 points each # (Max. 2 marks)	By DCRO/PCRO sample check (Max 2 mark) ##	4
Near Miss or PSIF reporting	1 point per incident ###		2

* To be conducted by Contractor's Management team e.g. Resident Representative (RR) or above

To be conducted by Site Management team where Site Management = Resident Representative, Engineer, Asstd. Engineer, etc.

DCRO/PCRO will randomly select 3 reports to assign score base on evaluation of the contractor's quality of site observation

1 point will be scored for each reporting for works of the contractor itself of either a), b) or c) below.

(a) SIF Potential Incident, or

(b) SIF Near Miss, or

(c) Mitigation of SIF exposure in Method Statement or WSWP

Information Class: Confidential Page 3 of 12



Concept of Structured, Unstructured, Discrete, Continuous Data

Structured Data

- Fixed schema (rows/columns, e.g., SQL tables)
- Easy to query/search (names, prices)
- needs traditional ML models such as XGBoost, Random Forests techniques

Unstructured Data

- No fixed format (text, images, videos)
- needs deep learning vs NLP/ML techniques
- Or use LLM / Foundation Model

Discrete Data

- Countable, distinct values (e.g., 0,1,2 items)
- needs traditional ML classification models like Logistic Regression or Decision Trees

Continuous Data

- Any value in range (e.g., 1.75m height)
- No gaps; line/histograms
- needs traditional ML regression models such as Linear Regression or Gradient Boosting

Data Processing & Analysis
Practise 2

Which Data Type is Structured/Unstructured, Discret/Continuous?

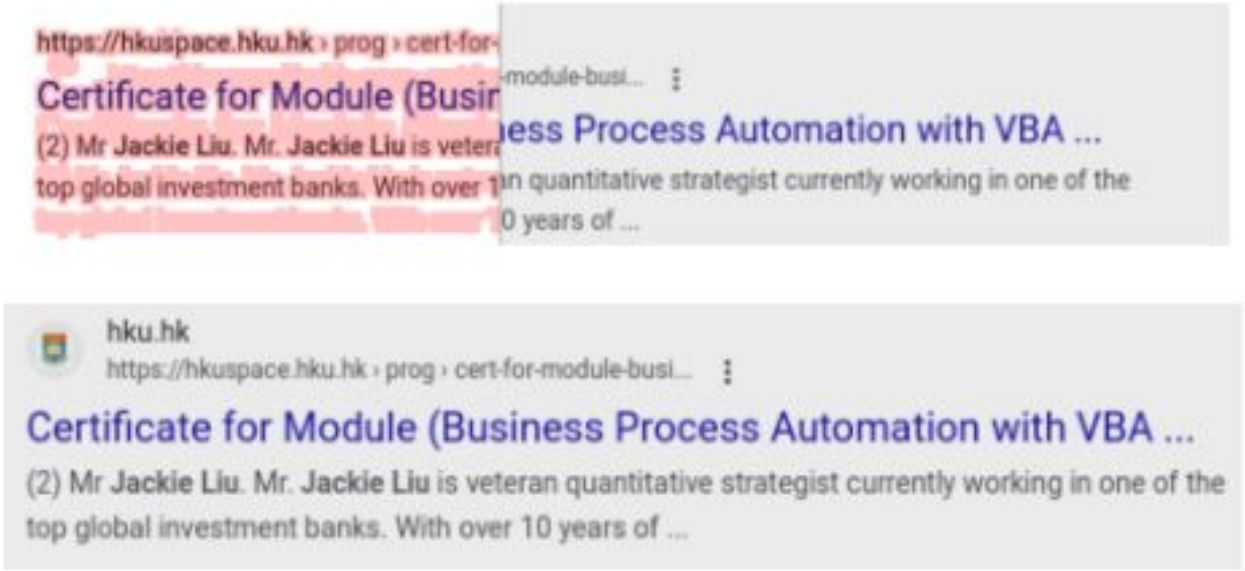
E-mails	
Digital Images	
Stock Market Logs	
Historical Gold Prices	
Credit Approval Records	
Social Media Friend Relationships	
Tweets & Trending Topics	
Sales Records	

Source: Practical Data Analysis - Second Edition Hector Cuesta, Dr. Sampath Kumar OACKT Publishing BIRMINGHAM - MUMBAI

Tools for Automation – Getting a taste

- **Visualping (https://visualping.io/)**
  - Websites monitoring

visualping



Tools for Automation – Considerations

- **Cost**
  - **One-time cost**
    - Initial effort of adaptation
    - Implementation and rollout
  - **Running cost**
    - License, e.g., Power Automate: $Millions/year[1]
    - Infrastructure, e.g., Cloud storage
    - Training, e.g., Automation Anywhere, 10+ hours[2]
    - Support, e.g., bot re-configuration



1. *RPA Pricing: Comparison of Leading RPA Vendors' Fees in 2023.* https://research.aimultiple.com/rpa-pricing/
2. *RPA Course Learning Trails | Automation Anywhere University.* https://university.automationanywhere.com/training/rpa-learning-trails/

Tools for Automation – Considerations

- **Flexibility**
  - Specialized vs Generalized, e.g., XCEPTOR ➔ Tax
- **Scalability**
  - Desktop vs Enterprise, e.g., Excel VBA
  - Version Control, e.g., Use of repository
- **Time to delivery**
  - "Perfect" solution does not exist

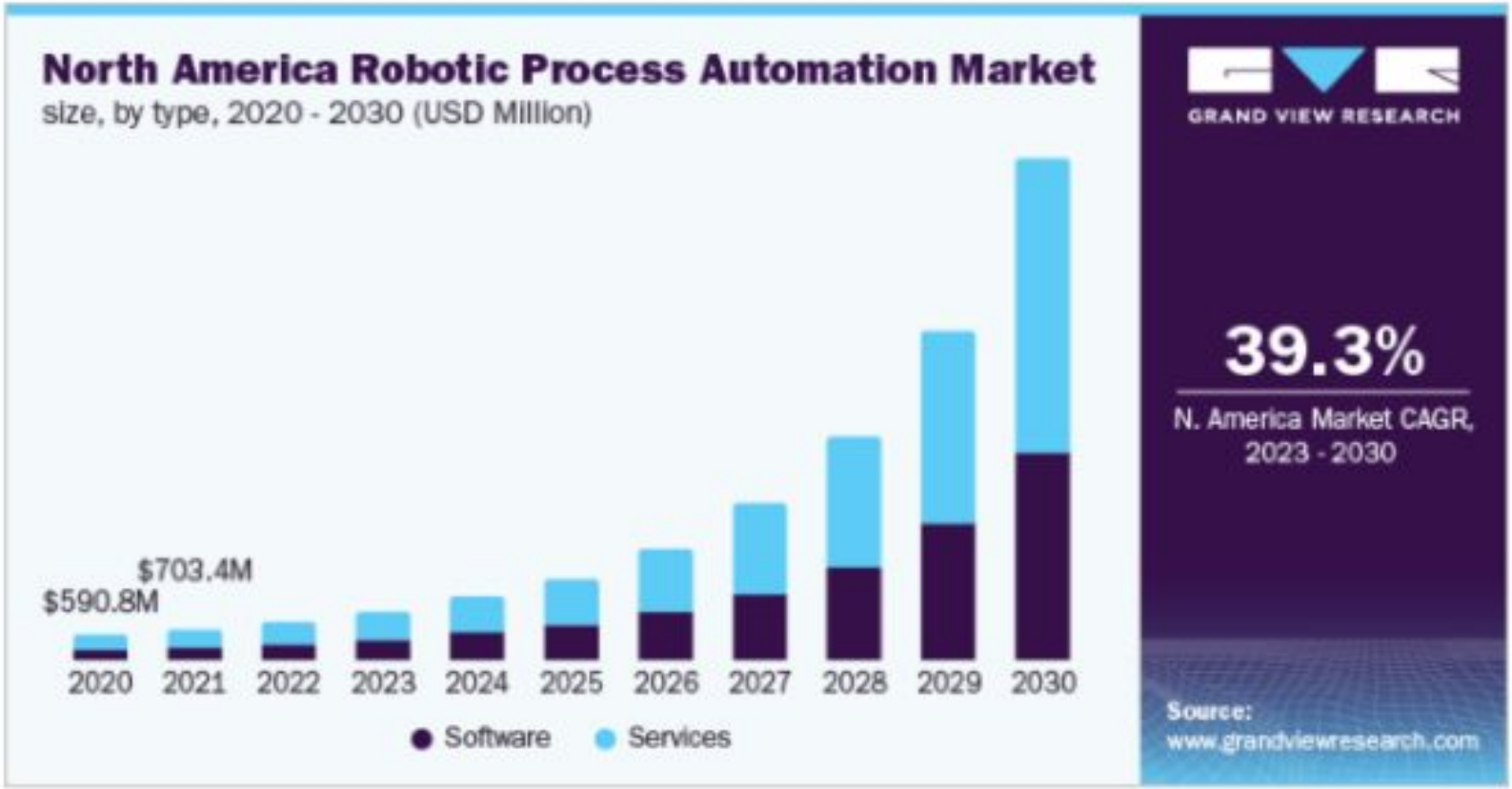






1. Smeets, M., Erhard, R., & Kaußler, T. (2021, July 30). *Robotic Process Automation (RPA) in the Financial Sector: Technology - Implementation - Success for Decision Makers and Users*. Springer. https://doi.org/10.1007/978-3-658-32974-7

Opportunities of BPA – The Money



1. *Robotic Process Automation Market Size & Share Report 2030*. Robotic Process Automation Market Size & Share Report 2030. https://www.grandviewresearch.com/industry-analysis/robotic-process-automation-rpa-market

Opportunities of BPA – Sectors

Industries	% of RPA Solutions
Banking, Financial Services & Insurance	51 %
Business Process Outsourcing (BPO)	14 %
Manufacturing – Consumer Packaged Goods (CPG)	7 %
Professional, Legal & Accountancy Services	7 %
Retail Trade	7 %
Technology (IT, Internet, SAAS)	7 %
Utilities	7 %

1. *Top 15 RPA Use Cases & Examples in Banking in 2023.* https://research.aimultiple.com/banking-rpa/

Opportunities of BPA – Sectors

- **Healthcare**
  - Monitoring, drug dispensing
- **Retail**
  - Auto-checkout, smart shelf, inventory management
- **Marketing**
  - Lead generation
  - social media management
- **Finance**
  - Fraud detection, Credit management, Regtech



Opportunities of BPA – Regtech

- **Follow the money**

CNN

Wells Fargo ordered to pay $3.7 billion for 'illegal activity' including unjust foreclosures and vehicle repossessions

The CFPB said the more than $2 billion in customer refunds Wells Fargo has been ordered to pay includes more than $1.3 billion to consumers hurt...

20 Dec 2022



Spiceworks

SEC Penalizes Major Wall Street Firms $1.97B For Using Unauthorized Messaging Apps

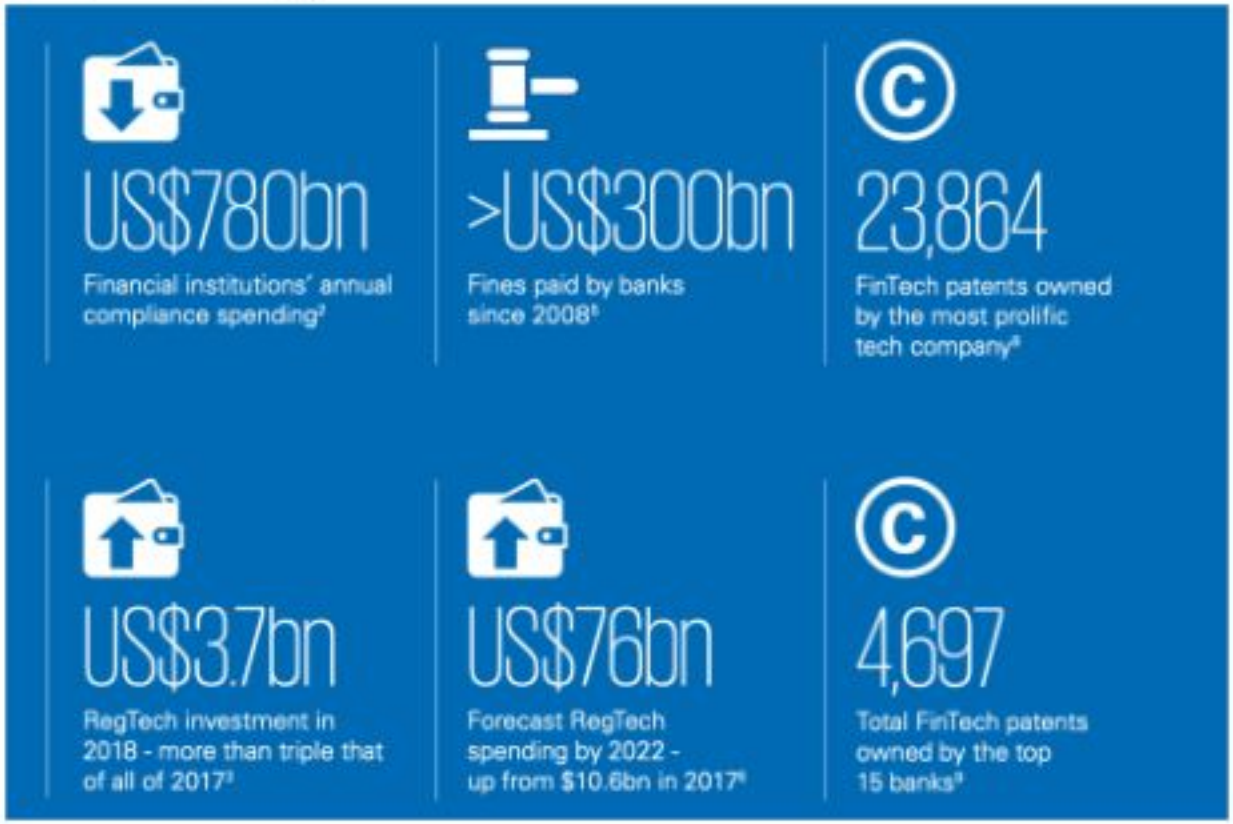
If there are allegations of wrongdoing or misconduct, we must be ... Morgan Stanley recently agreed to a $35 million fine by the SEC for...

28 Sept 2022



Opportunities of BPA – Regtech

- **Follow the money**



1. KPMG. (2019, June). *There's a revolution coming.* https://assets.kpmg.com/content/dam/kpmg/cn/pdf/en/2019/06/embracing-the-challenge-of-the-new-regtech-era.pdf

Opportunities of BPA – Regtech

- **Common themes**

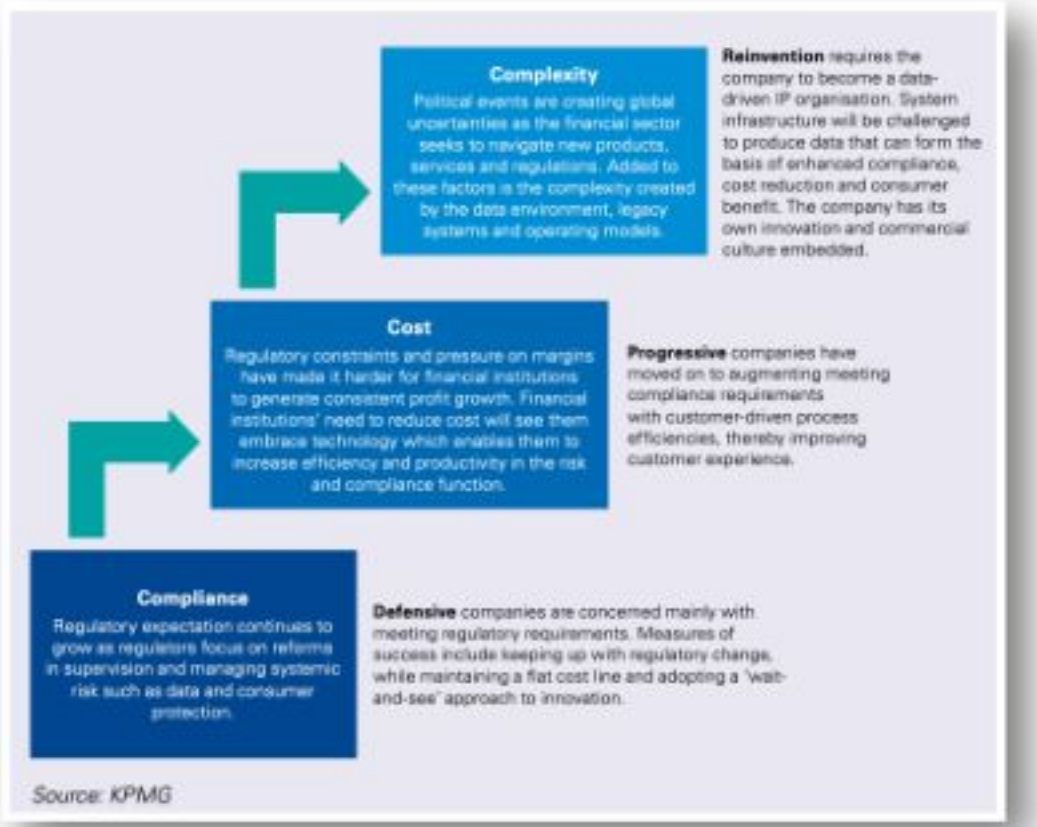


1. KPMG. (2022, November). https://assets.kpmg.com/content/dam/kpmg/uk/pdf/2022/11/innovate-finance-regtech-industry-and-adoption.pdf. https://assets.kpmg.com/content/dam/kpmg/uk/pdf/2022/11/innovate-finance-regtech-industry-and-adoption.pdf

Opportunities of BPA – Regtech

Adoption phases:

1. Defensive
2. Progressive
3. Reinvention



1. KPMG. (2019, June). *There's a revolution coming.* https://assets.kpmg.com/content/dam/kpmg/cn/pdf/en/2019/06/embracing-the-challenge-of-the-new-regtech-era.pdf

Opportunities of BPA – Regtech

- **HKMA 3-Year Roadmap**
  - **RPA**: automating rule-based processes
  - **Network Analysis**: Company holdings and exposures



Illustrative Network Analysis Solution

1. HKMA. (2021, June). *Regtech Watch Issue No. 7.* https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2021/20210617e1a1.pdf

Opportunities of BPA – Regtech

- **HKMA 3-Year Roadmap**
  - **Speech-to-text (STT):** Audio to time-stamped text
  - **Sentiment Analysis**: Realtime news monitoring

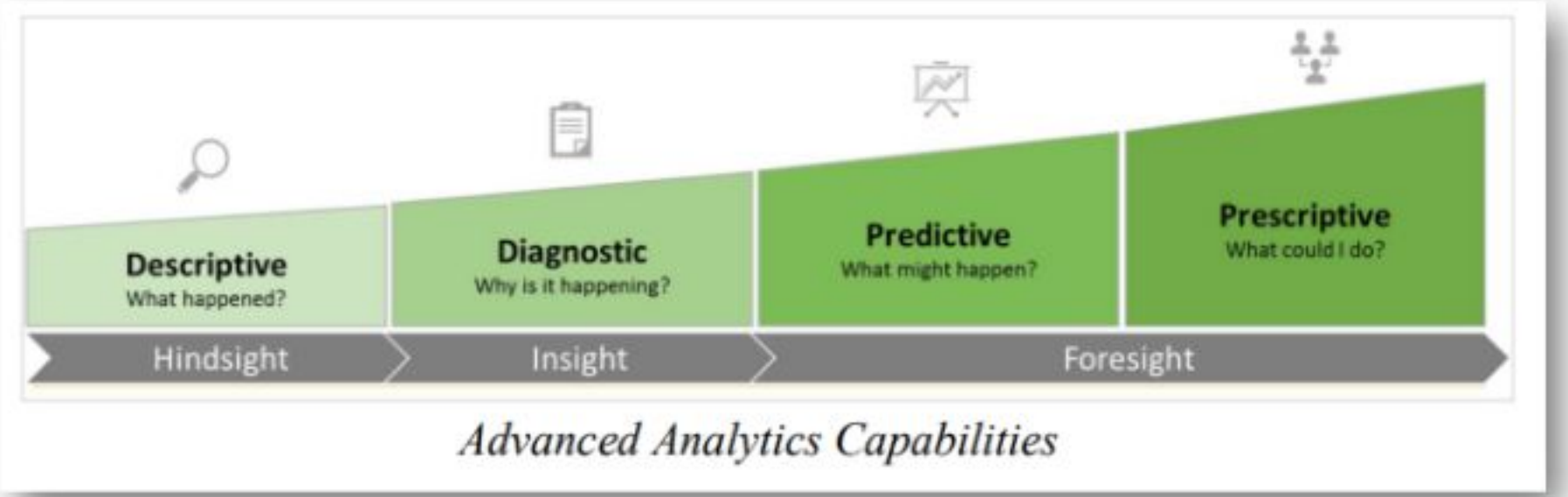


Illustrative Sentiment Analysis Heat Map

1. HKMA. (2021, June). *Regtech Watch Issue No. 7.* https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2021/20210617e1a1.pdf

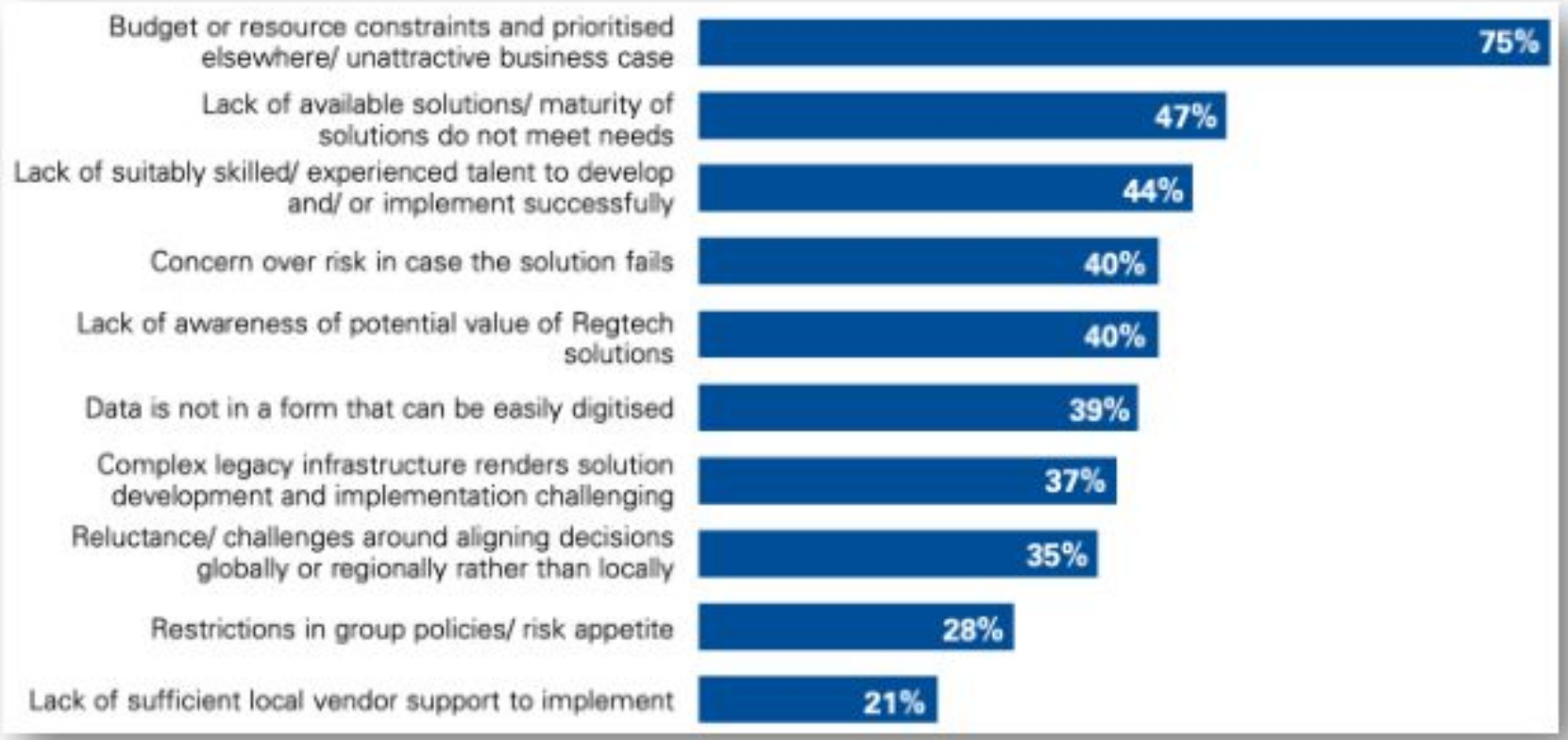
Opportunities of BPA – Regtech

- **Analytics enabled by BPA**



1. HKMA. (2021, June). *Regtech Watch Issue No. 7.* https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2021/20210617e1a1.pdf

Challenges of BPA – Insight from the banks



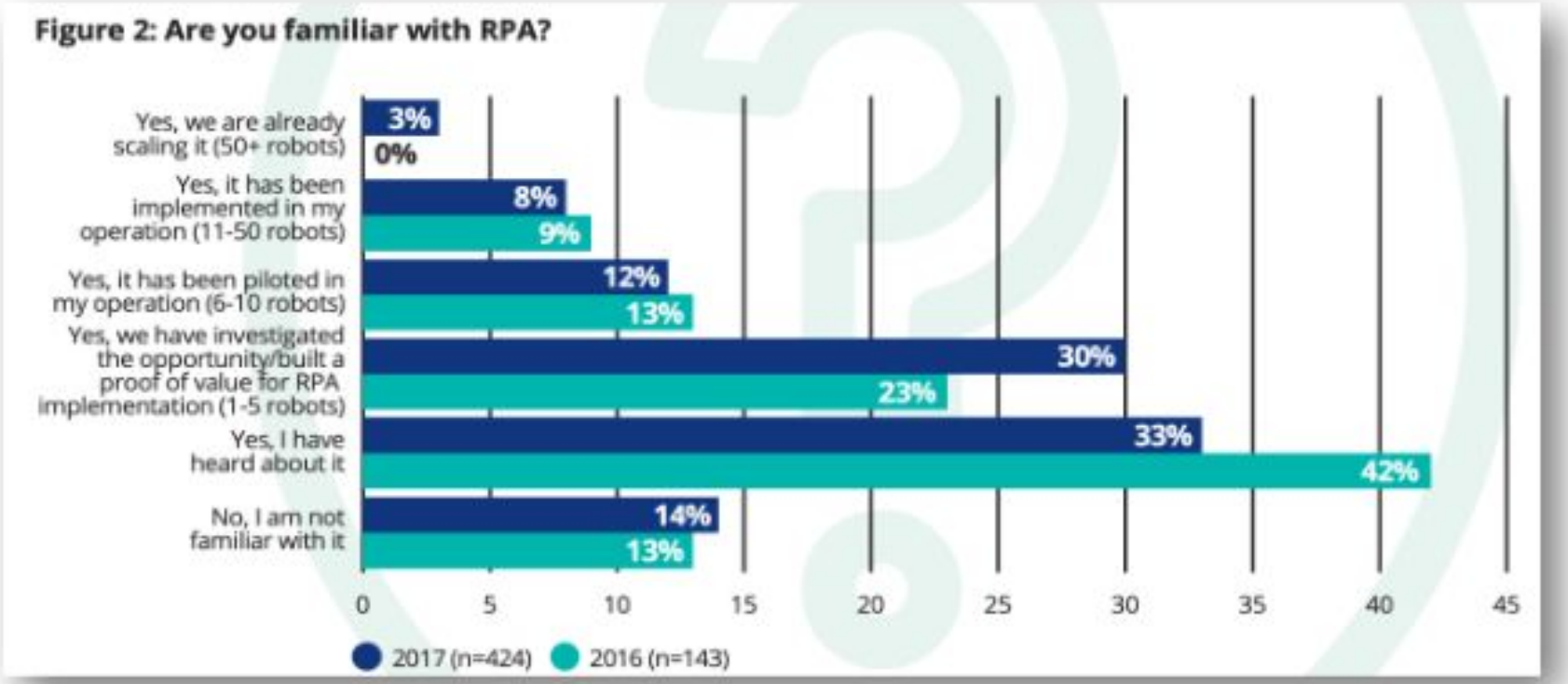
1. KPMG & HKMA. (2020). *Transforming Risk Management and Compliance: Harnessing the Power of Regtech*. https://www.hkma.gov.hk/media/chi/doc/key-information/press-release/2020/20201102c3a1.pdf

Challenges of BPA – Worries from the workers?



1. Deloitte. (2018). *The robots are ready. Are you?* https://www2.deloitte.com/content/dam/Deloitte/tr/Documents/technology/deloitte-robots-are-ready.pdf

Challenges of BPA – Hesitant to scale up



1. Deloitte. (2018). *The robots are ready. Are you?* https://www2.deloitte.com/content/dam/Deloitte/tr/Documents/technology/deloitte-robots-are-ready.pdf

Challenges of BPA

- **Shortage**
  - Budget / Resource/ Talent
- **Concerns**
  - Risk of rushing / failing
  - Moral / Structural unemployment

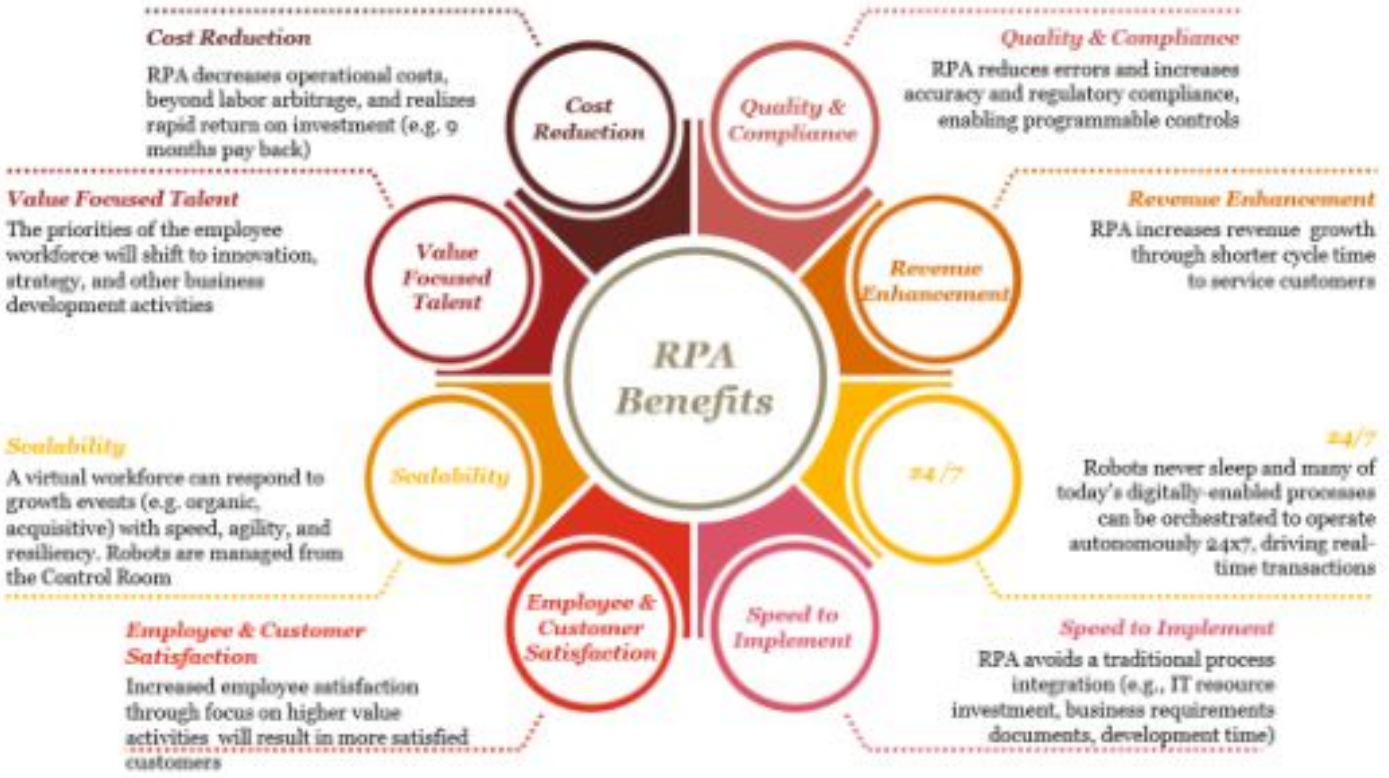




* Success depends on planning
(More on project management later)

Implications of BPA – The Good

- Cost, Efficiency, Customer Experience



1. *PwC Robotics Process Automation Solutions*. PwC. https://www.pwchk.com/en/services/entrepreneurial-and-private-business/new-technology-digitalisation-and-transformation/process-automation-solutions.html

Implications of BPA – The Bad

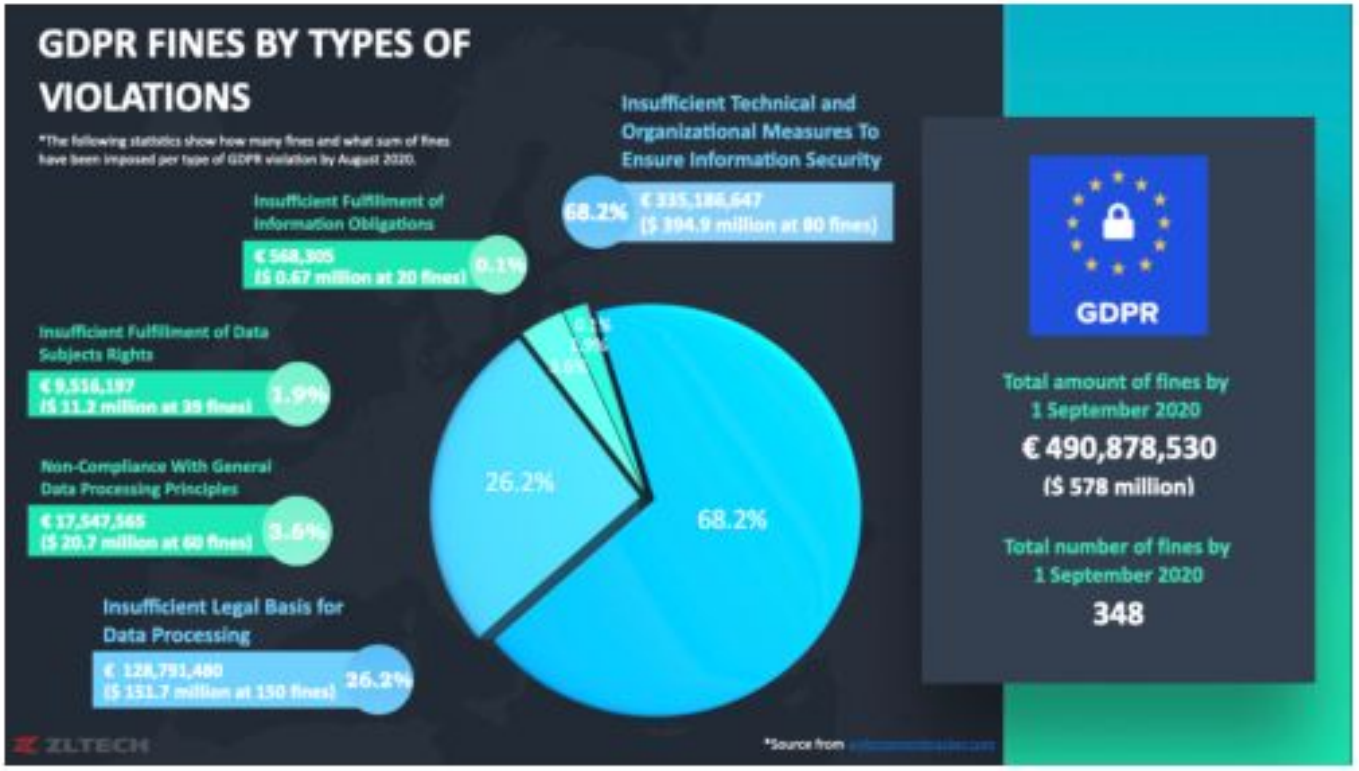
- Automation complacency / dependency
- Risk of Technical Issues / Downtime



Image: Civil Aviation Safety Authority

Implications of BPA – The Ugly

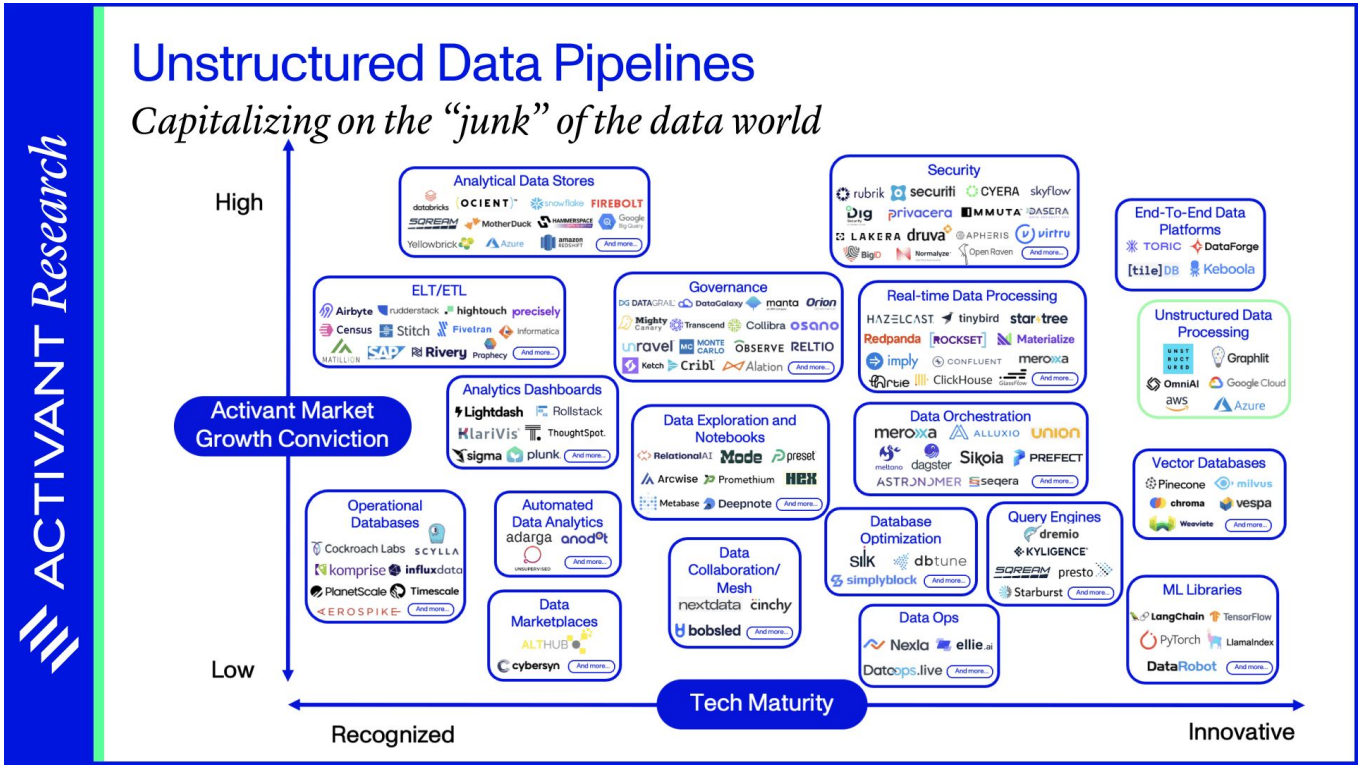
- Data Collection and Analysis [1]
- Vulnerability to Cyberattacks



1. Chen, B. (2020, September 9). *GDPR Recap: 28 Nations, 348 Fines, Half a Billion Euros*. ZL Tech. https://www.zlti.com/blog/gdpr-recap-28-nations-343-fines-half-a-billion-euros/

Intelligent Process Automation (IPA) – Data Science

Intelligent Process Automation (IPA)'s integration with data science is particularly impactful in several areas: Unstructured Data Processing: IPA can convert unstructured data into structured formats suitable for analysis, enabling businesses to leverage insights from previously untapped data source.

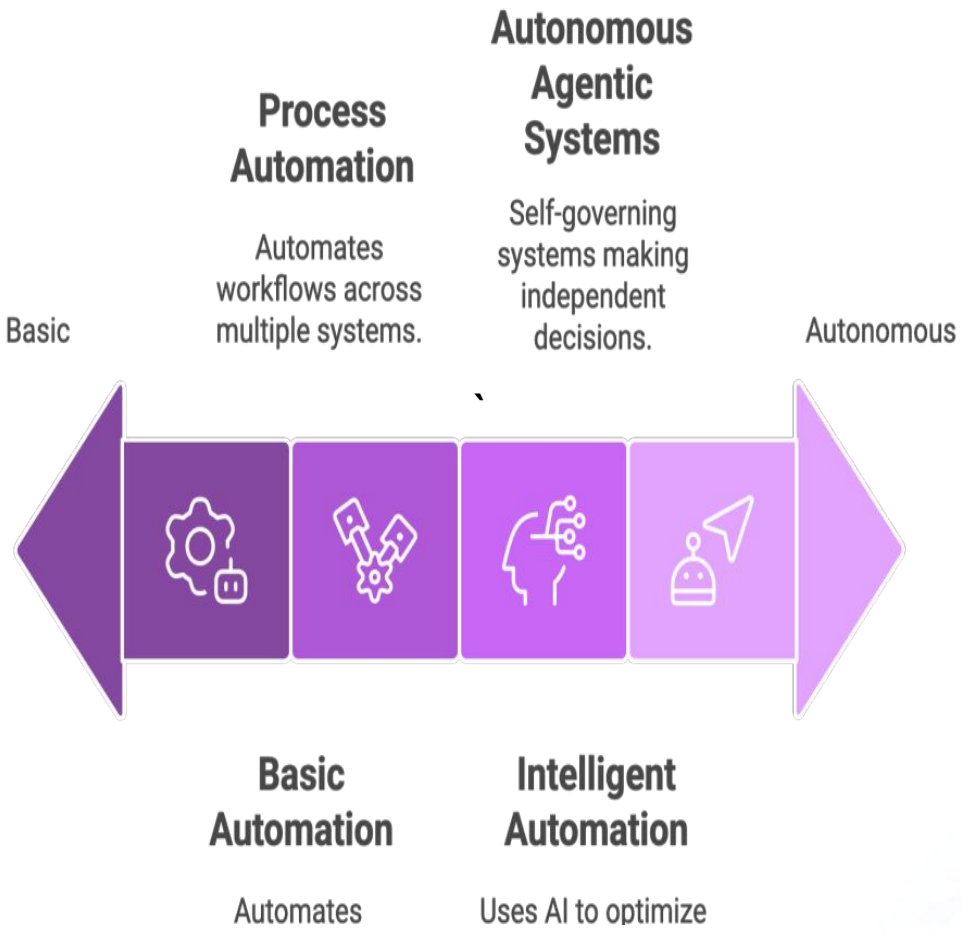


HKUSPACE
香港大學專業進修學院
HKU School of Professional and Continuing Education

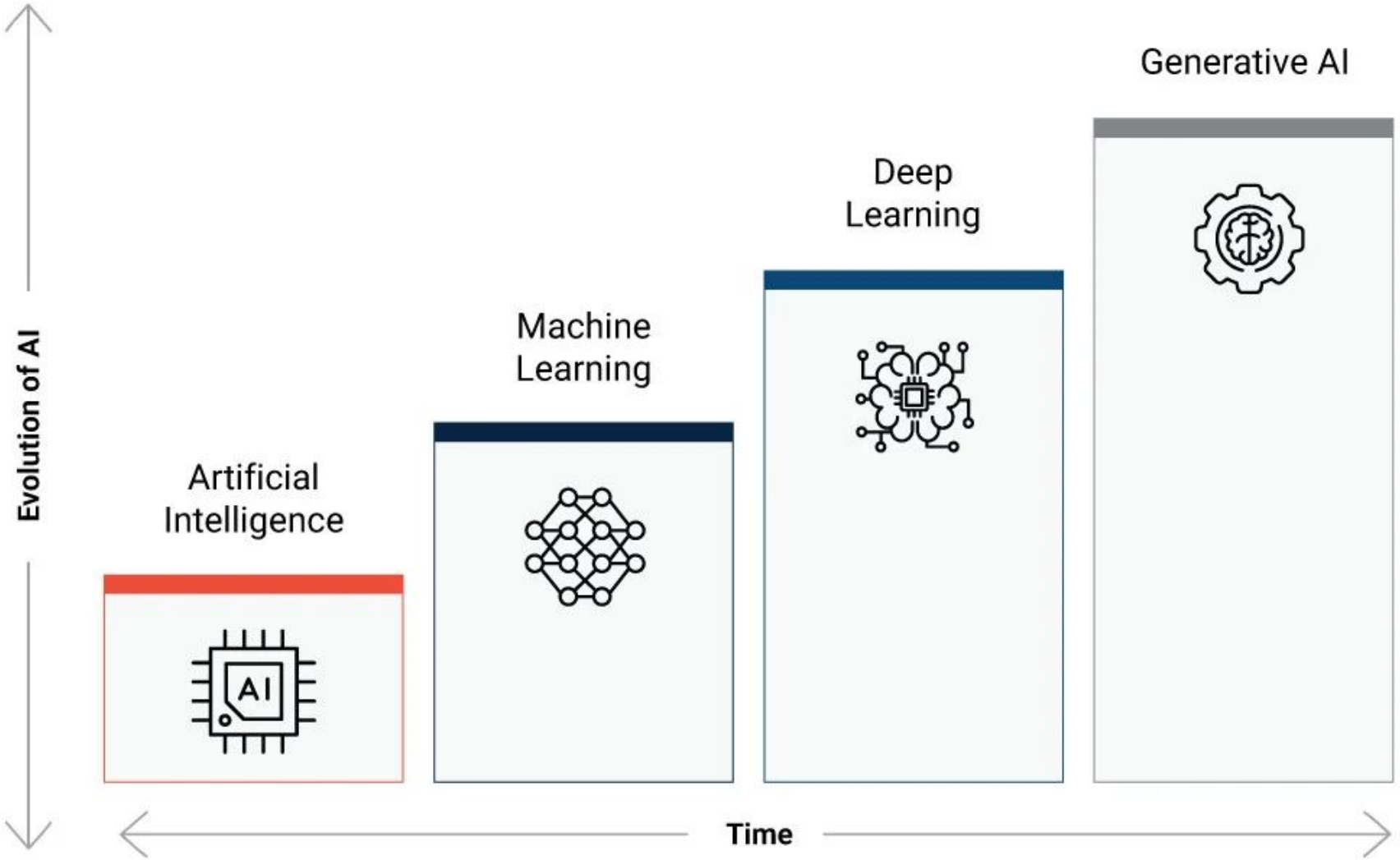
- What is Process Automation, RPA, IPA?
- Overview of the technological building blocks related to process automation
- Contemporary tools for process automation
- Challenges and opportunities of business process automation
- Business implications of process automation

Four Level of Business Automation

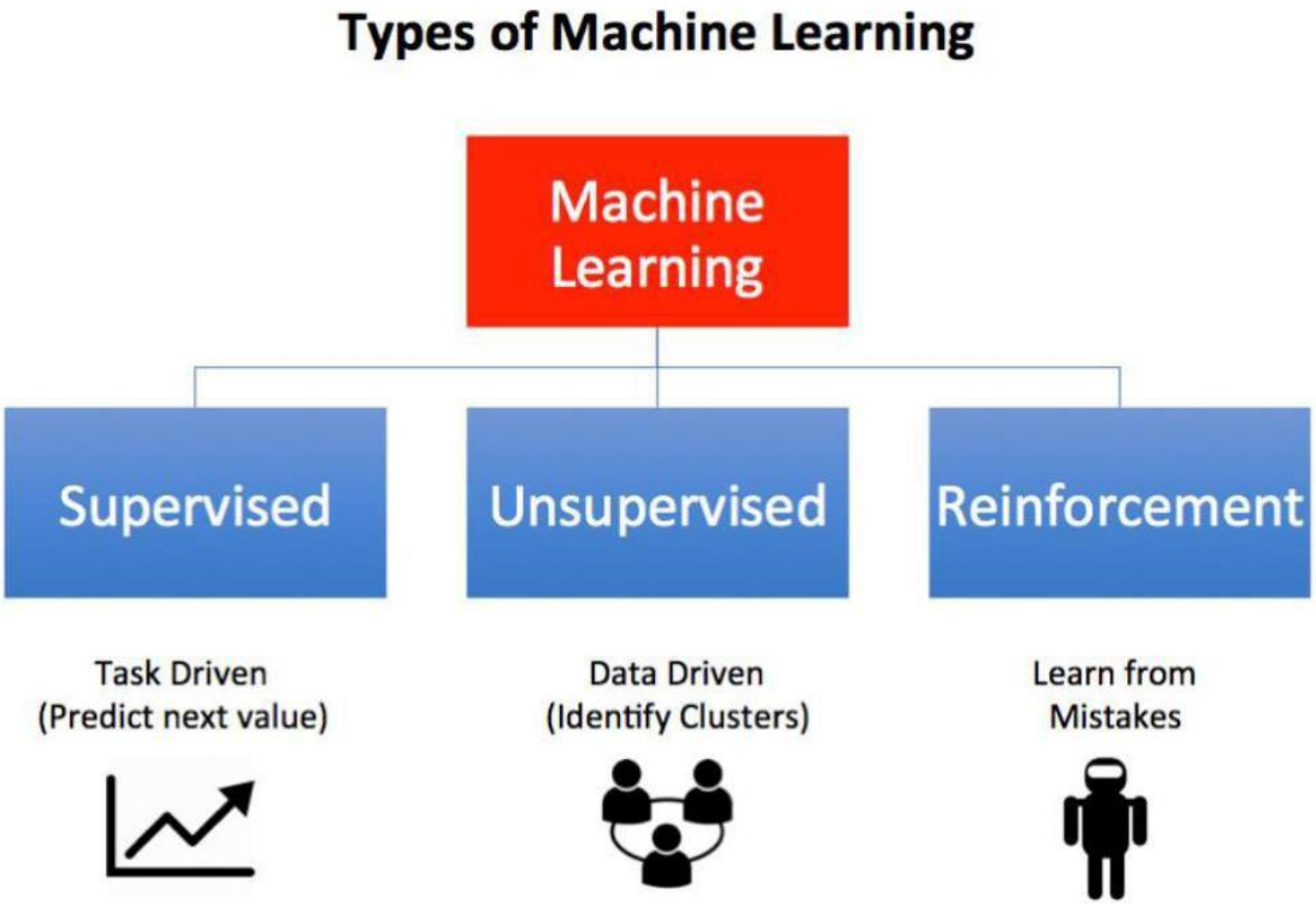
1. Basic Automation
2. Process Automation
3. Intelligent Process Automation
4. Antonomous Agentic Systems



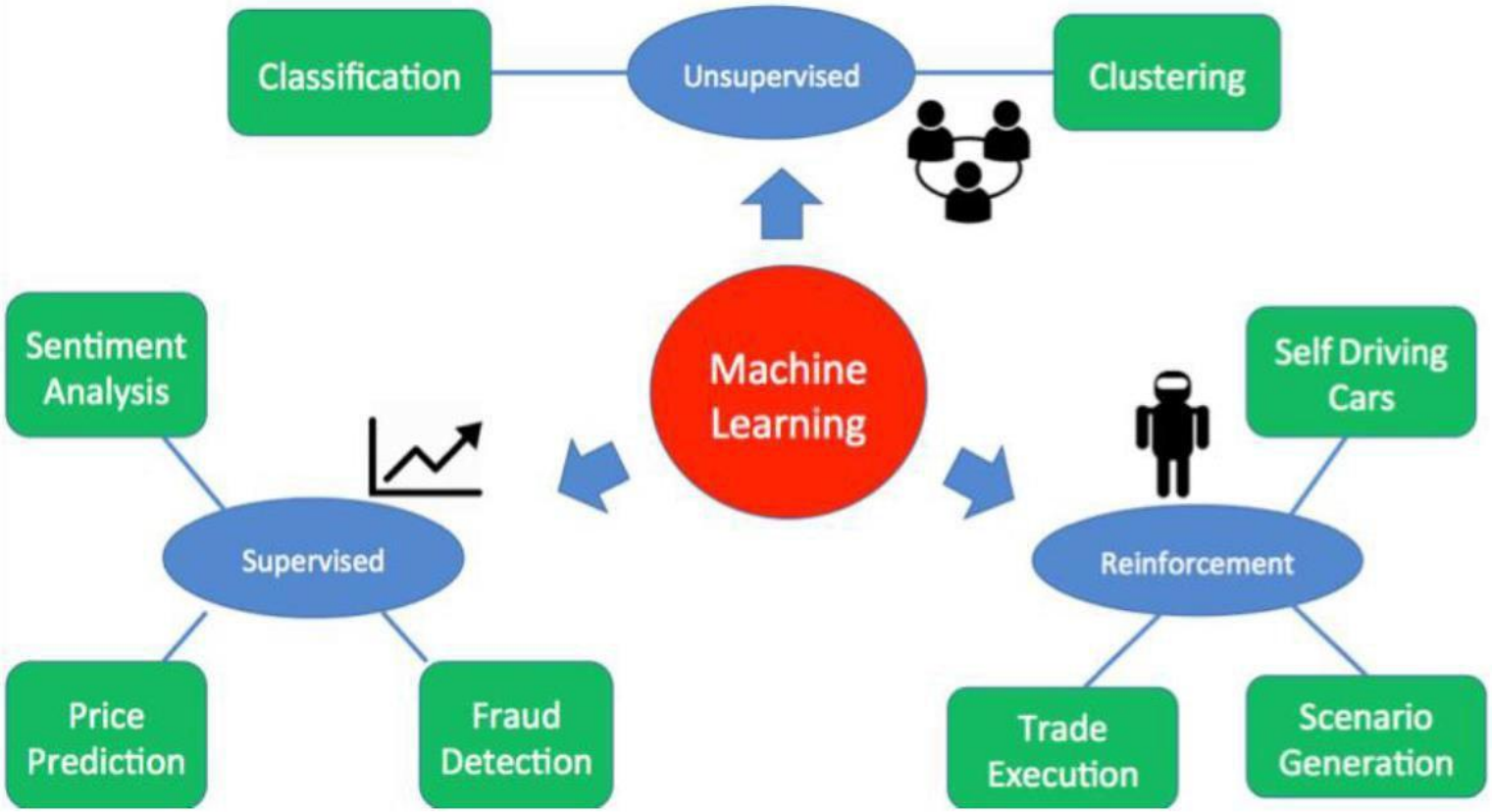
Overall Artificial Intelligence Journey



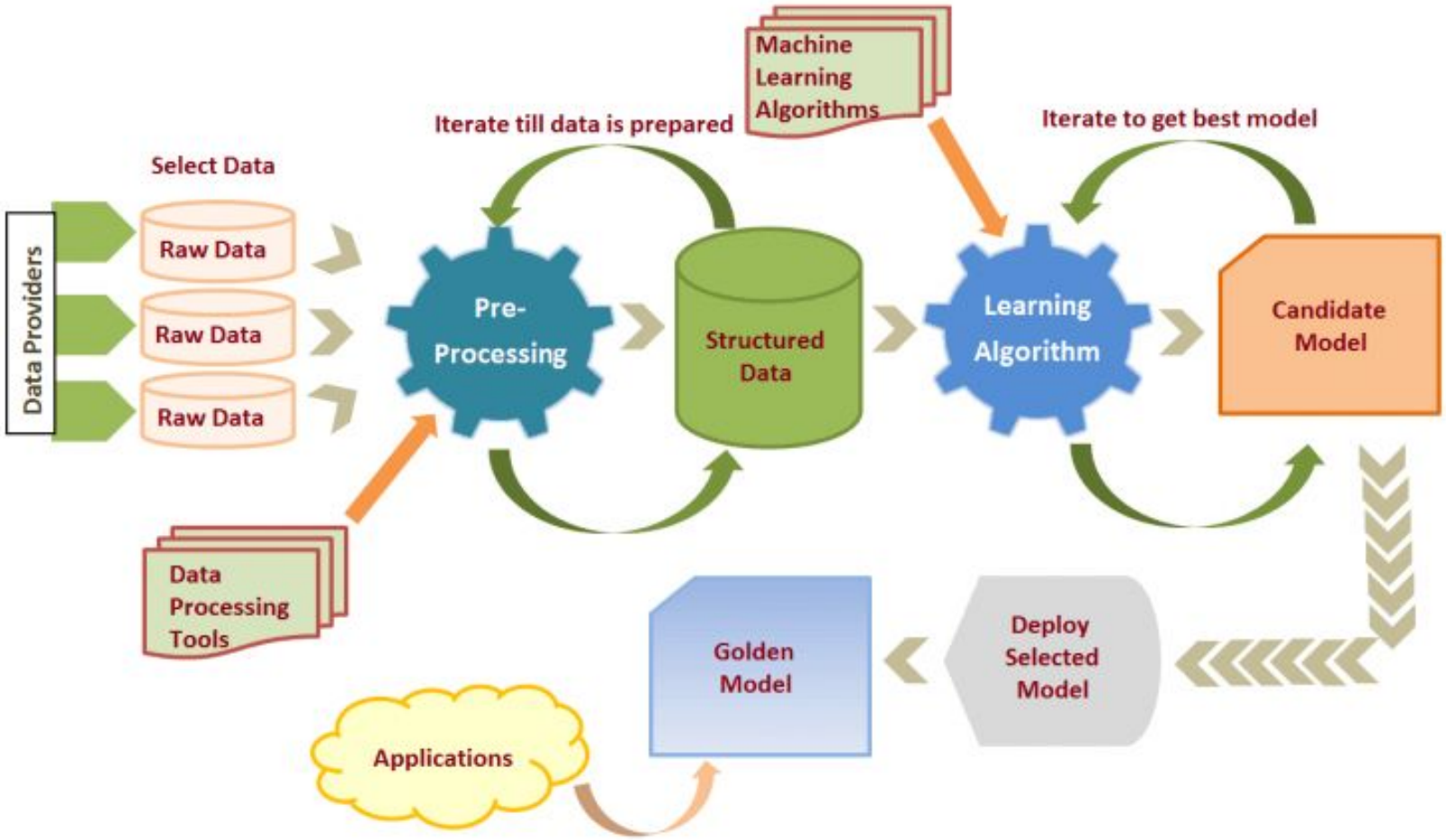
What is Machine Learning?



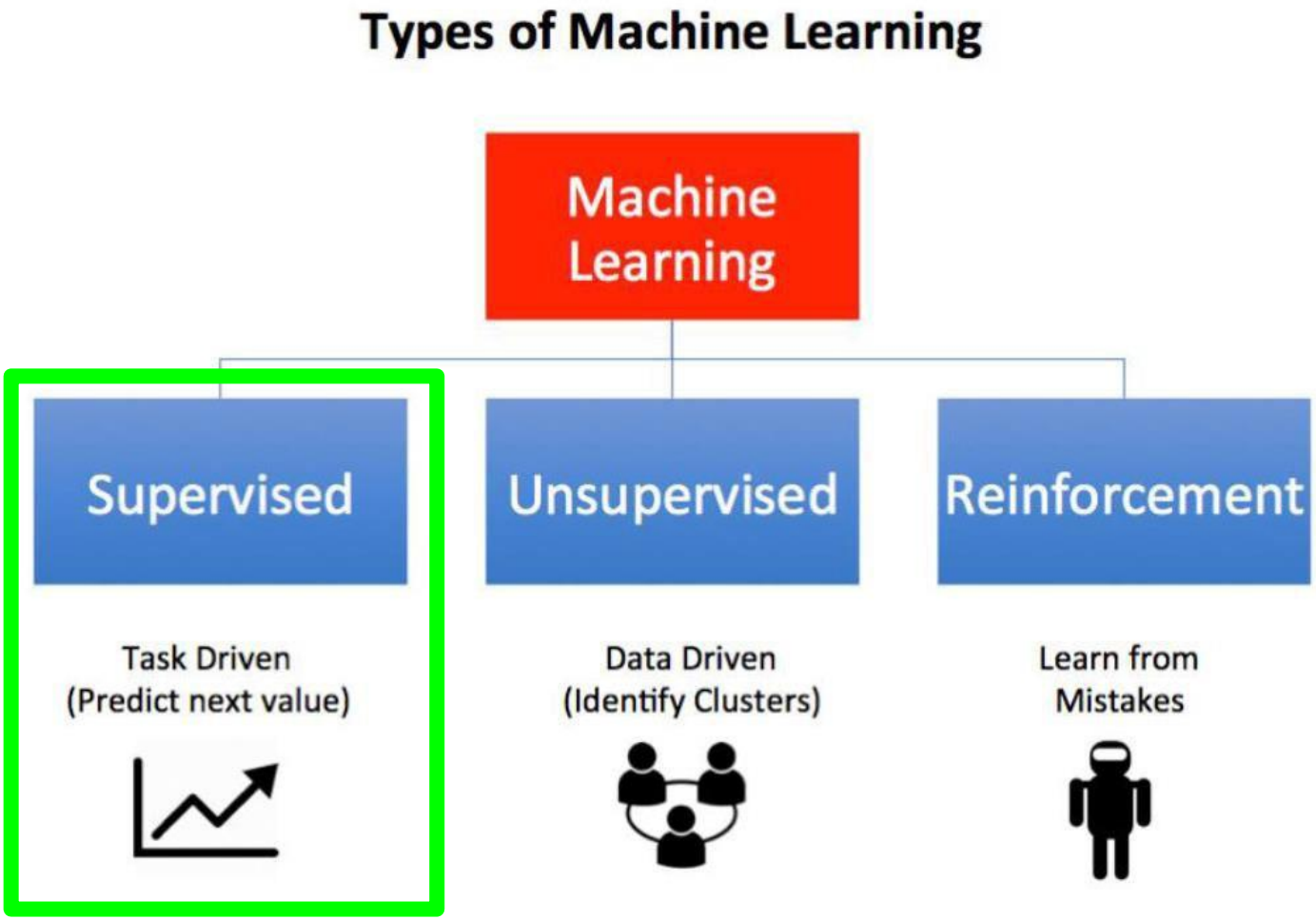
What is Business Analytics & Machine Learning?



Applied Machine Learning Pipeline Overview



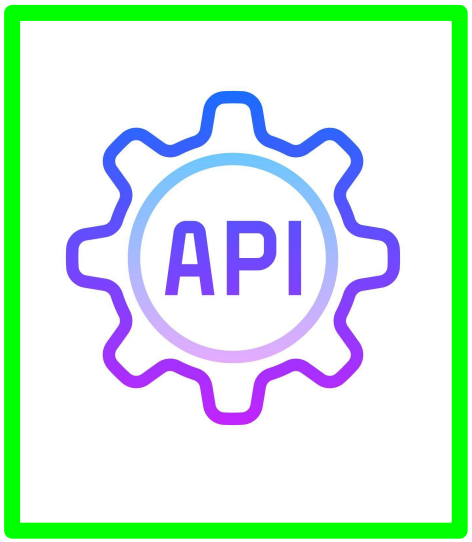
What is Business Analytics & Machine Learning?



Structured Data Collection Method

Main Data collection are get from Google Sheets, API and Web Scraping







kaggle

https://www.kaggle.com/

開放數據平台
DATA.GOV.HK

https://data.gov.hk/en/

What is Data Dictionary from API

Data Dictionary from API = single source of truth that tells developers what each field means, its type, format, rules, and example - so API JSON can be validated and stored correctly.

Definition

structured metadata describing API data fields, types, formats.

- Describes each data field available in the API response
- Defines data type, format, validation, and requirement rules
- Ensures consistent usage across developers and systems

Example — /users Endpoint

Field Name	Data Type	Description	Required	Sample Value
id	integer	Unique identifier for the user	Yes	12345
name	string	Full name of the user	Yes	Jane Doe
email	string (email)	User's email address for contact	Yes	jane.doe@example.com

Purpose: Provides a single source of truth for API data structure to improve clarity, validation, and onboarding

API Documentation • Data Governance

Data Dictionary - Real API JSON Response Mapping

API Endpoint: **GET /api/products**
— Retrieves a list of products from the catalog via REST API

Data Dictionary Table

Field	Type	Format	Description	Example
product_id	string	UUID	Unique identifier for the product	550e8400-e29b-41d4-a716-446655440001
name	string	–	Product name	
price	number	decimal (2dp)	Product price in USD, 2 decimal places	199.99
published_date	date	ISO YYY-MM-DD	Date the product was published in ISO 8601	2024-03-12

API → Data Dictionary Flow



Validation Rules: UUID format, 2 decimal precision for price, ISO 8601 date

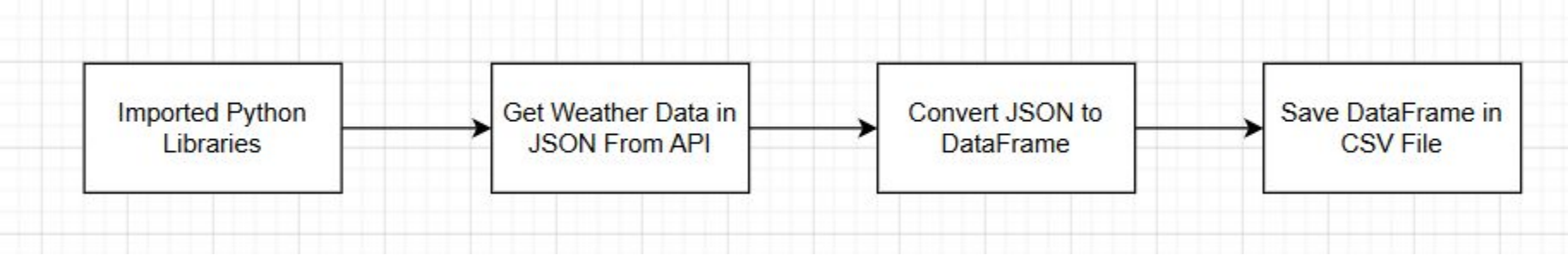
Mapping Purpose: Align API JSON fields to internal data model

Example corresponds to a single product object from the response array

** Note: Actual API response returns a JSON array of product objects. Example shown is one item for clarity.*

Data Collection - Reading HKO Weather Data from API

Below are the Flow Chart with the Process of Getting Weather Data from HKO API



Following Code contain 4 steps with the hypothesis
"Read Weather Forecasting Data from HKO API and Save in CSV File"

Step 1. Import the required libraries:

```
#import necessary library
import requests
import pandas as pd
```

Data Collection - Reading HKO Weather Data from API

Step 2: Load the weather forecasting data from 'https://data.weather.gov.hk/weatherAPI/opendata/'
Before writing code, please take a look on API documentation at Hong Kong Observatory
Study the Weather API data dict Documentation in
https://www.hko.gov.hk/en/weatherAPI/doc/files/HKO_Open_Data_API_Documentation.pdf

1. Weather Information API

Dataset

- 9-day Weather Forecast
- Current Weather Report
- Local Weather Forecast
- Weather Warning Information
- Weather Warning Summary
- Special Weather Tips

API URL

https://data.weather.gov.hk/weatherAPI/opendata/weather.php

Please include valid parameters in API request. For valid parameters, please refer to Request table in this section.

HTTP Request Method

GET

Return Type

JSON

Request Example

https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType=flw&lang=en

Parameter

1. **dataType**
2. **lang**

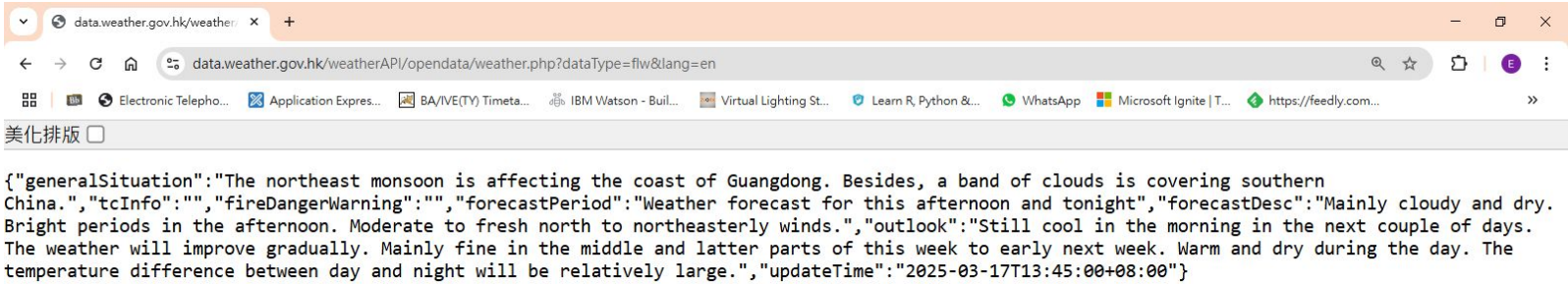
Request

Parameter	Accepted values	Description
dataType	flw	flw: Local Weather Forecast
	fnd	fnd: 9-day Weather Forecast
	rhrread	rhrread: Current Weather Report
	warnsum	warnsum: Weather Warning Summary
	warningInfo	warningInfo: Weather Warning Information
	swt	swt: Special Weather Tips
lang	en	en: English
	tc	tc: Traditional Chinese
	sc	sc: Simplified Chinese
		Default language: en

4

Data Collection - Reading HKO Weather Data from API

When browser load https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType=flw&lang=en



generalSituation	tcInfo	fireDangerWarning	forecastPeriod	forecastDesc	outlook	updateTime
":"The northeast monsoon is affecting the coast of Guangdong. Besides, a band of clouds is covering southern China			Weather forecast for this afternoon and tonight	Mainly cloudy and dry. Bright periods in the afternoon. Moderate to fresh north to northeasterly winds	Still cool in the morning in the next couple of days. The weather will improve gradually. Mainly fine in the middle and latter parts of this week to early next week. Warm and dry during the day. The temperature difference between day and night will be relatively large.	2025-03-17T13:45:00+08:00

Data Collection - Reading HKO Weather Data from API

#Continue from previous page

Step 2: Load the weather forecasting data from 'https://data.weather.gov.hk/weatherAPI/opendata/'

```
# Initialize Parameters for the API
datatype = 'fnd'
lang = 'en'
apiurl = f'https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType={datatype}&lang={lang}'

# Fetching the JSON data
response = requests.get(apiurl)
response.raise_for_status()  # Check for request errors

# Converting JSON to a DataFrame
json_data = response.json()
```

Step 3: Converting JSON Data to a DataFrame

```
weatherforcast_df = pd.DataFrame(json_data['weatherForecast'])

# Display the DataFrame
print(weatherforcast_df.head())
```

```
  forecastDate       week                                        forecastWind  \
0     20250115  Wednesday                                    North force 4 to 5.
1     20250116   Thursday      North to northeast force 4, force 5 at first.
2     20250117     Friday            Northeast force 4, east force 5 later.
3     20250118   Saturday  East to northeast force 4 to 5, occasionally f...
4     20250119     Sunday                  East to northeast force 2 to 3.

                                  forecastWeather  \
0  Mainly cloudy with a few light rain patches at...
1  Fine. Rather cool in the morning. Very dry dur...
2  Fine. Rather cool in the morning. Very dry dur...
3                        Dry with sunny periods.
4                           Mainly fine and dry.

            forecastMaxtemp            forecastMintemp  \
0  {'value': 20, 'unit': 'C'}  {'value': 15, 'unit': 'C'}
1  {'value': 19, 'unit': 'C'}  {'value': 13, 'unit': 'C'}
2  {'value': 19, 'unit': 'C'}  {'value': 13, 'unit': 'C'}
3  {'value': 20, 'unit': 'C'}  {'value': 15, 'unit': 'C'}
4  {'value': 21, 'unit': 'C'}  {'value': 16, 'unit': 'C'}

                     forecastMaxrh                     forecastMinrh  \
0  {'value': 80, 'unit': 'percent'}  {'value': 40, 'unit': 'percent'}
1  {'value': 65, 'unit': 'percent'}  {'value': 35, 'unit': 'percent'}
2  {'value': 65, 'unit': 'percent'}  {'value': 30, 'unit': 'percent'}
3  {'value': 70, 'unit': 'percent'}  {'value': 40, 'unit': 'percent'}
4  {'value': 70, 'unit': 'percent'}  {'value': 40, 'unit': 'percent'}

   ForecastIcon  PSR
0            51  Low
1            92  Low
2            50  Low
3            51  Low
4            50  Low
```

Data Collection - Reading HKO Weather Data from API

#Continue from previous page

Step 4. Saving Weather Forecasting DataFrame to a CSV

```
csv_file = 'weatherForecast.csv'  # Desired output CSV file
weatherforcast_df.to_csv(csv_file, index=False)

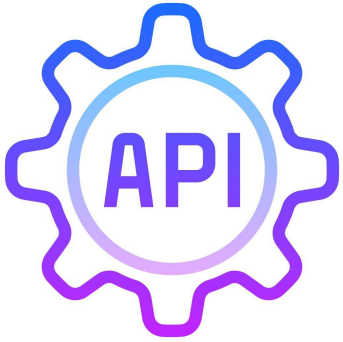
# Output successful message for saving weather forecasting data.
print(f"Data has been successfully saved to {csv_file}")
```

A	B	C	D	E	F	G	H	I	J
forecastDate	week	forecastWind	forecastWeather	forecastMaxtemp	forecastMintemp	forecastMaxrh	forecastMinrh	ForecastIcon	PSR
20250115	Wednesday	North force 4 to 5.	Mainly cloudy with a fe	{'value': 20, 'unit':	{'value': 15, 'unit':	{'value': 80, 'un:	{'value': 40, 'un	51	Low
20250116	Thursday	North to northeast force	Fine. Rather cool in the r	{'value': 19, 'unit':	{'value': 13, 'unit':	{'value': 65, 'un:	{'value': 35, 'un	92	Low
20250117	Friday	Northeast force 4, east	Fine. Rather cool in the r	{'value': 19, 'unit':	{'value': 13, 'unit':	{'value': 65, 'un:	{'value': 30, 'un	50	Low
20250118	Saturday	East to northeast force	Dry with sunny periods.	{'value': 20, 'unit':	{'value': 15, 'unit':	{'value': 70, 'un:	{'value': 40, 'un	51	Low
20250119	Sunday	East to northeast force	Mainly fine and dry.	{'value': 21, 'unit':	{'value': 16, 'unit':	{'value': 70, 'un:	{'value': 40, 'un	50	Low
20250120	Monday	East to northeast force	Fine and dry.	{'value': 22, 'unit':	{'value': 16, 'unit':	{'value': 70, 'un:	{'value': 40, 'un	50	Low
20250121	Tuesday	East to northeast force	Fine and dry.	{'value': 22, 'unit':	{'value': 17, 'unit':	{'value': 70, 'un:	{'value': 40, 'un	50	Low
20250122	Wednesday	East force 4, occasiona	Sunny periods.	{'value': 21, 'unit':	{'value': 17, 'unit':	{'value': 75, 'un:	{'value': 50, 'un	51	Low
20250123	Thursday	East force 4, occasiona	Sunny intervals.	{'value': 21, 'unit':	{'value': 17, 'unit':	{'value': 75, 'un:	{'value': 50, 'un	52	Low

Structured Data Collection Method

Main Data collection are get from Google Sheets, API and Web Scraping







kaggle

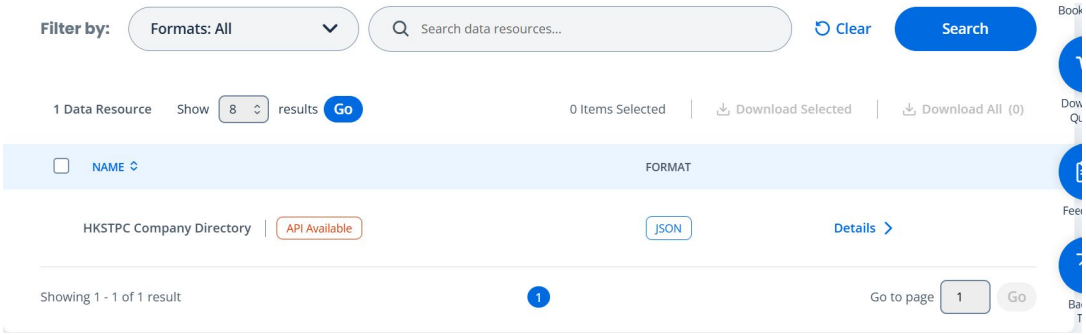
https://www.kaggle.com/

開放數據平台
DATA.GOV.HK

https://data.gov.hk/en/

Data.gov.hk Data Collection

1. Open https://data.gov.hk/en/
2. Click Datasets -> Select “Commerce and Industry”
3. Select “HKSTPC Company Directory”



4. Click “Details” -> Click URL “https://opendata.hkstp.org/corporate/companydirectory/v1/”

{"value":[{"type":null,"type_Name_EN":null,"type_Name_TC":null,"type_Name_SC":null,"cluster":"Information Communications Technology","cluster_TC":"資訊及通訊科技","cluster_SC":"资讯及通讯科技","telephone":null,"fax":null,"email":null,"website":"https://www.omnichat.ai/","name_EN":"Omnichat Limited","address_EN":null,"contact_EN":null,"introduction_EN":"Founded in 2017 and headquartered in Hong Kong, Omnichat is a leading provider of AI-powered conversational commerce solutions for the Asia-Pacific region. Uniquely positioned as the sole Meta Business Partner and LINE Biz-Solutions Tech Partner in APAC with integrated WhatsApp, Facebook, Instagram, LINE, WeChat, and KakaoTalk capabilities, Omnichat empowers brands to centralise communications, deliver personalised experiences, and drive digital transformation across their operations. Over 5,000 brands, including international retailers and enterprises, leverage Omnichat to optimise their customer journeys.","product_EN":" ","name_TC":"Omnichat Limited","address_TC":null,"contact_TC":null,"introduction_TC":"Omnichat 成立於 2017 年，總部設於香港，為亞太區領先的 AI 對話式商務方案供應商。作為亞太區唯一同時獲 Meta 和 LINE 認證的合作夥伴，Omnichat 無縫整合 WhatsApp、Facebook、Instagram、WeChat 及 KakaoTalk 等多個通訊平台，協助品牌集中管理訊息、打造個人化體驗，並推動數碼轉型。現時已有超過 5,000 個品牌，包括國際知名的零售商及企業，採用 Omnichat 優化顧客互動旅程。","product_TC":" ","name_SC":"Omnichat Limited","address_SC":null,"contact_SC":null,"introduction_SC":"Omnichat 成立于 2017 年，总部位于香港，是亚太区领先的 AI 对话式商务解决方案供应商。作为亚太区唯一同时获得 Meta 和 LINE 官方认证的合作伙伴，Omnichat 无缝整合 WhatsApp、Facebook、Instagram、WeChat 及 KakaoTalk 等多个通讯平台，协助品牌集中管理信息、打造个性化体验并推动数字化转型。目前，包括国际知名零售商及企业在内的超过 5,000 个品牌，正采用 Omnichat 优化其客户互动旅程。","product_SC":" "},{"type":null,"type_Name_EN":null,"type_Name_TC":null,"type_Name_SC":null,"cluster":"Electronics","cluster_TC":"電子","cluster_SC":"电子","telephone":null,"fax":null,"email":null,"website":"http://www.xtxtech.com","name_EN":"XTX Technology Limited","address_EN":null,"contact_EN":null,"introduction_EN":" ","product_EN":" ","name_TC":"XTX Technology Limited","address_TC":null,"contact_TC":null,"introduction_TC":" ","product_TC":" ","name_SC":"XTX Technology Limited","address_SC":null,"contact_SC":null,"introduction_SC":" ","product_SC":" "},{"type":null,"type_Name_EN":null,"type_Name_TC":null,"type_Name_SC":null,"cluster":"Biomedical Technology","cluster_TC":"生物醫療","cluster_SC":"生物医疗","telephone":"59331197","fax":null,"email":"terry@raphaihealth.com","website":" ","name_EN":"Raphai Health Technologies Limited","address_EN":"Unit 518, 5/F, Building 19W,\nNo. 19 Science Park West Avenue,\nHong Kong Science Park, Pak Shek Kok, N.T.\n","contact_EN":"Mr. Terry Li","introduction_EN":"Founded in August 2023 by Mr. Oscar Wong and Mr. Terry Li, Oscar, an engineering expert, has been working in AI-assisted skeletal sensing technology since 2016 and managed an R&D team based in the Greater Bay Area (GBA). Terry, with over 30 years of banking and insurance executive experience, has worked in various Asian markets, including Vietnam, Hong Kong, Singapore, and Indonesia. Both Oscar and Terry are EMBA graduates from Ivey Business School, Western University, Canada.","product_EN":"Raphai has developed skeletal-sensing technology that enables real-time monitoring of Musculoskeletal (MSK) Medicine Exercise-Prescription (ExRx) and patient interaction. This cloud-based solution can help MSK caregivers evolve into a highly interactive 'phygital' caregiver and patient journey, thereby improving recovery time. The overall healthcare system can enhance MSK care accessibility at a lower cost.","name_TC":"金匱醫學科技有限公司","address_TC":"Unit 518, 5/F, Building 19W,\nNo. 19 Science Park West Avenue,\nHong Kong Science Park, Pak Shek Kok, N.T.\n","contact_TC":"Mr. Terry Li","introduction_TC":"金匱醫學科技成立於2023年8月，由黃俊先生（Oscar Wong）和李鴻榮先生（Terry Li）共同創辦。黃先生是工程專家，自2016年以來一直從事人工智慧輔助骨骼感應技術的研發工作，並管理著一支位於大灣區（GBA）的研發團隊。李先生在銀行和保險行業擁有超過30年的高管經驗，曾於越南、香港、新加坡和印尼等亞洲市場工作。黃先生和李先生都是加拿大Western University, Ivey Business School的高級行政工商管理碩士（EMBA）畢業生。","product_TC":"金匱醫學科技自主研發的骨骼感應技術，可以實現即時監控肌肉骨骼（MSK）醫學運動處方（ExRx）和患者互動。這個基於雲的解決方案可以幫助MSK護理提供者轉變為高度互動的“實數結合”護理提供者和患者旅程，從而提高康復速度。整體醫療系統可以在更低的成本下擴大MSK護理的可及性。","name_SC":"金匮医学科技有限公司","address_SC":"Unit 518, 5/F, Building 19W,\nNo. 19 Science Park West Avenue,\nHong Kong Science Park, Pak Shek Kok, N.T.\n","contact_SC":"Mr. Terry Li","introduction_SC":"金匮医学科技成立于2023年8月，由黄俊先生（Oscar Wong）和李鸿荣先生

5. Right Click and Click “Save As”

Download the data

1. Download the hko weather forecasting dataset from: https://github.com/innoviai/IPA_Courses-October2026/blob/main/weatherForecast.csv
2. Save the dataset file in a location you'll remember.

1	forecastDate	week	forecastWind	forecastWeather
2	20250115	Wednesday	North force 4 to 5.	Mainly cloudy with a few light rain patches at first. Becoming fine and dry during the day. Becoming cool at ni
3	20250116	Thursday	North to northeast force 4, force 5 at first.	Fine. Rather cool in the morning. Very dry during the day.
4	20250117	Friday	Northeast force 4, east force 5 later.	Fine. Rather cool in the morning. Very dry during the day.
5	20250118	Saturday	East to northeast force 4 to 5, occasionally force 6 offshore at first.	Dry with sunny periods.
6	20250119	Sunday	East to northeast force 2 to 3.	Mainly fine and dry.
7	20250120	Monday	East to northeast force 2 to 3.	Fine and dry.
8	20250121	Tuesday	East to northeast force 3 to 4.	Fine and dry.
9	20250122	Wednesday	East force 4, occasionally force 5 offshore.	Sunny periods.
10	20250123	Thursday	East force 4, occasionally force 5 offshore.	Sunny intervals.

Data Collection – Fetching Data using Pandas

What / Why?

- Pandas is an *open-source* Python library (anyone can contribute)
- Allows for high-performance, easy-to-use data structures and data analysis
- Unlike NumPy library which provides multi-dimensional arrays, Pandas provides 2D table object called **DataFrame**
  (akin to a spreadsheet with column names and row labels).
- Used by *a lot* of people

Data Collection – Fetching Data using Pandas

How

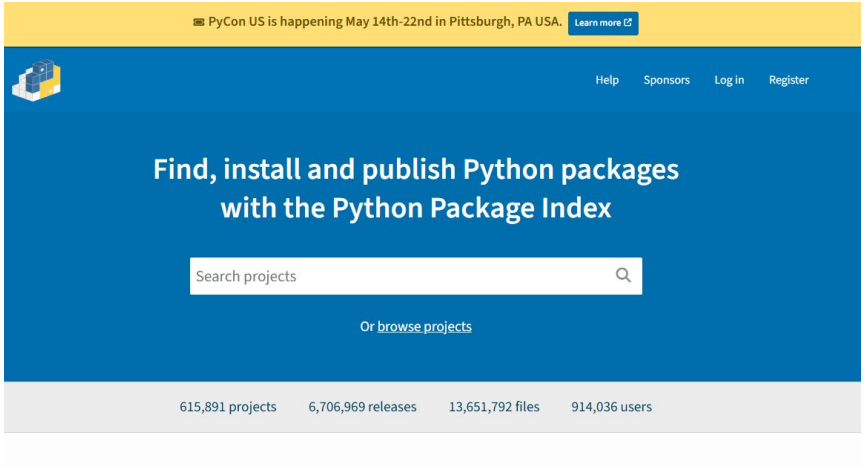
- import **pandas** library (convenient to rename it)
- Use **read_csv()** function with encoding utf-8

```
>>> import pandas as pd
>>> weather_df = pd.read_csv(rf"C:\Users\Eddie Chow\Desktop\
atherForecast.csv", encoding="utf-8")
>>> weather_df
   forecastDate       week  ... ForecastIcon  PSR
0      20250115  Wednesday  ...           51  Low
1      20250116   Thursday  ...           92  Low
2      20250117     Friday  ...           50  Low
3      20250118   Saturday  ...           51  Low
4      20250119     Sunday  ...           50  Low
5      20250120     Monday  ...           50  Low
6      20250121    Tuesday  ...           50  Low
7      20250122  Wednesday  ...           51  Low
8      20250123   Thursday  ...           52  Low

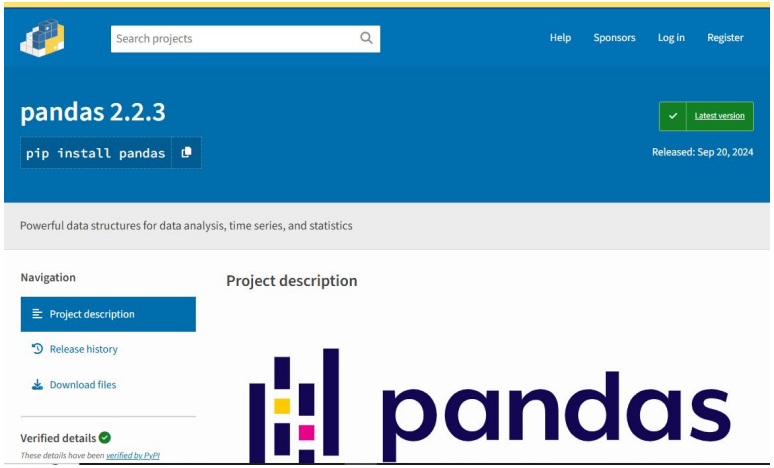
[9 rows x 10 columns]
```

Library Installation in Python

Most of the python library can be founded in https://pypi.org/

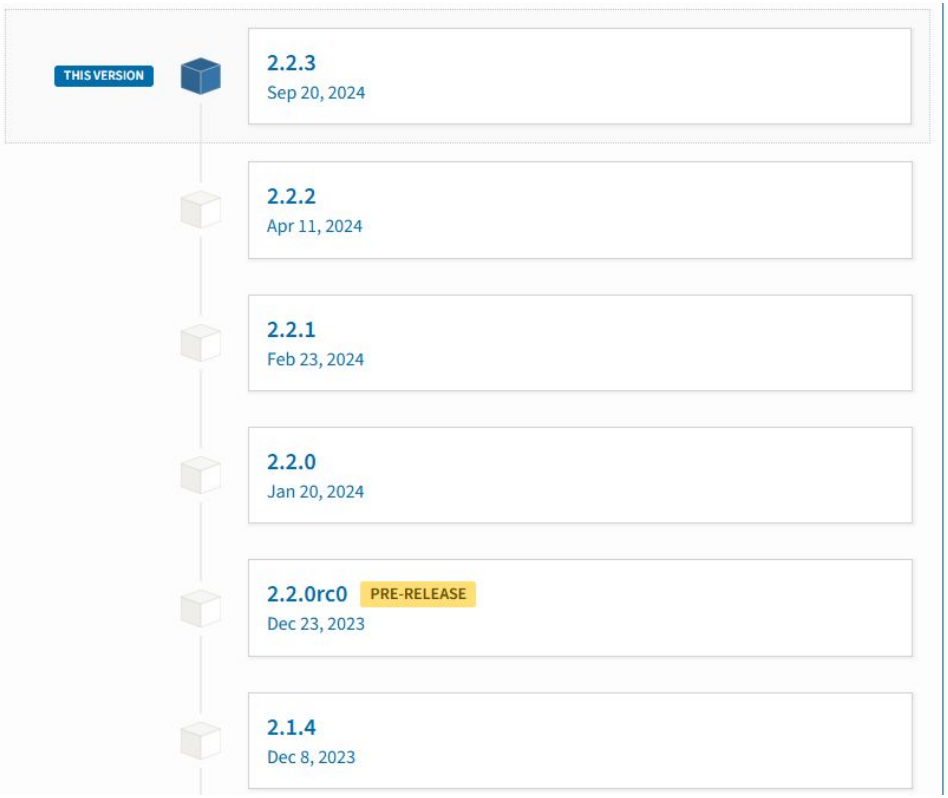


You can enter pandas to search or type https://pypi.org/project/pandas/



Library Installation in Python

Same library name have different version like 2.2.3, 2.1.0, …
The default page show the newest version.



If you use python 3.7 or 3.8, you better install the version 2.1.0



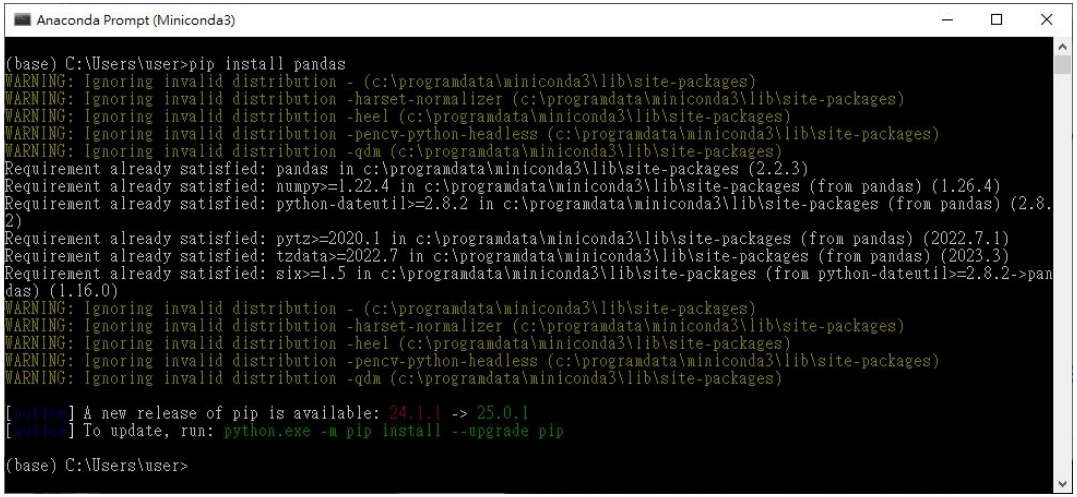
For library installation, you can go to anaconda prompt and enter
“pip install pandas” or “pip install pandas==2.1.0”

Library Installation in Python

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.



2. Install python package requests by entering “pip install pandas”



Examine the data set

```
print(df1.head())
```

#	Name	Type 1	Type 2	Total	...	Sp. Def	Speed	Stage	Legendary
					...				
1	Bulbasaur	Grass	Poison	318	...	65	45	1	False
2	Ivysaur	Grass	Poison	405	...	80	60	2	False
3	Venusaur	Grass	Poison	525	...	100	80	3	False
4	Charmander	Fire	NaN	309	...	50	65	1	False
5	Charmeleon	Fire	NaN	405	...	65	80	2	False

```
print(df1.describe())
```

	Total	HP	Attack	...	Sp. Def	Speed	Stage
count	151.00000	151.000000	151.000000	...	151.000000	151.000000	151.000000
mean	407.07947	64.211921	72.549669	...	66.019868	68.933775	1.582781
std	99.74384	28.590117	26.596162	...	24.197926	26.746880	0.676832
min	195.00000	10.000000	5.000000	...	20.000000	15.000000	1.000000
25%	320.00000	45.000000	51.000000	...	49.000000	46.500000	1.000000
50%	405.00000	60.000000	70.000000	...	65.000000	70.000000	1.000000
75%	490.00000	80.000000	90.000000	...	80.000000	90.000000	2.000000
max	680.00000	250.000000	134.000000	...	125.000000	140.000000	3.000000

Data Collection – Fetching Data using Pandas

Common Panda functions

High-level viewing:

- head() – first N observations
- tail() – last N observations
- columns() – names of the columns
- describe() – statistics of the quantitative data
- dtypes() – the data types of the columns

Data Collection – Fetching Data using Pandas

Common Panda functions

Accessing/processing:

- df[“column_name”]
- Df.column_name
- .max(), .min(), .idxmax(), .idxmin()
- <dataframe> <conditional statement>
- .loc[] – label-based accessing
- .iloc[] – index-based accessing
- .sort_values()
- .isnull(), .notnull()

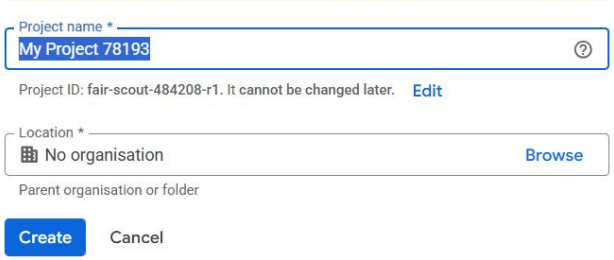
(Optional) Data Collection - Reading Data from Gmail

Enable the Google Sheets API with read permission.

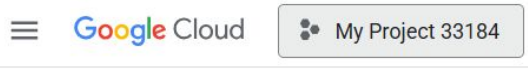
1. Open Browser and enter https://console.cloud.google.com/
2. Click “Select a project”
3. Click “New Project”



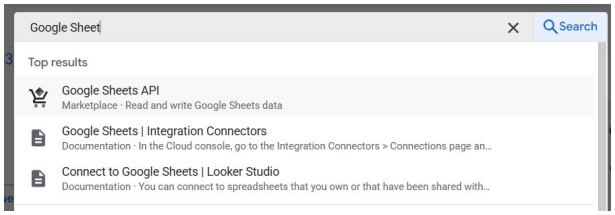
4. Enter information and click “Create”



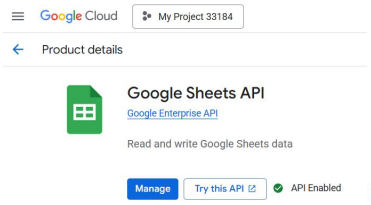
5. Select your Project that create



6. in Search Bar, enter ‘Google Sheet’



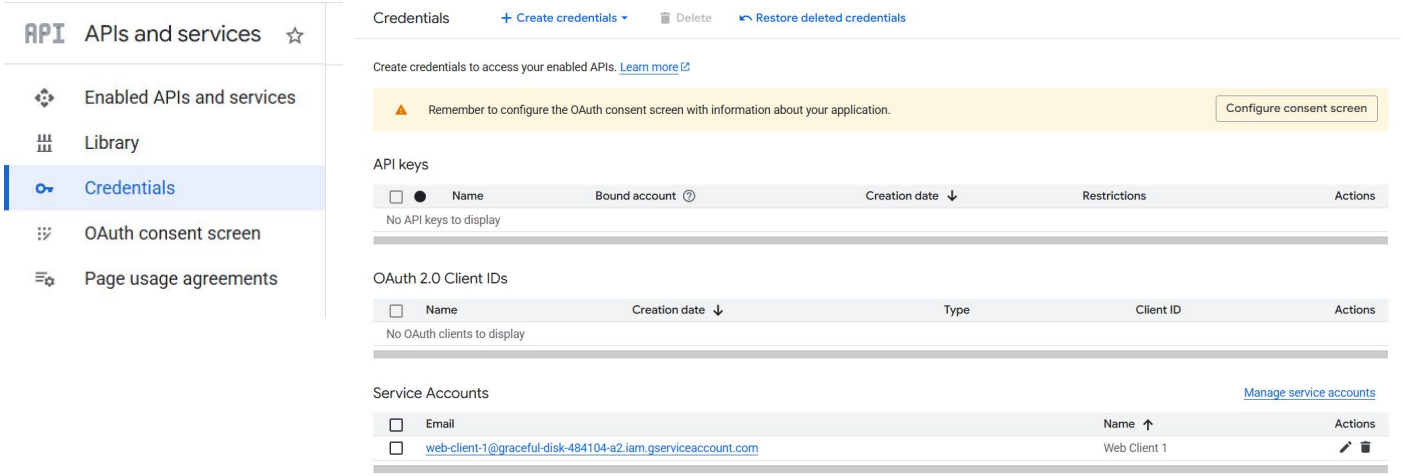
7. Select “Gmail API”
8. Click “Enable”



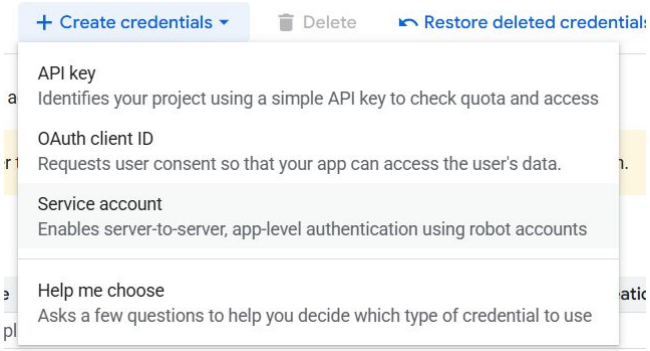
(Optional) Data Collection - Reading Data from Gmail

Credentials Setting

1. Click Credentials and whole screen is displayed



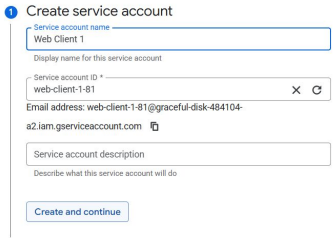
2. Click “Create credentials” and “Service Account”



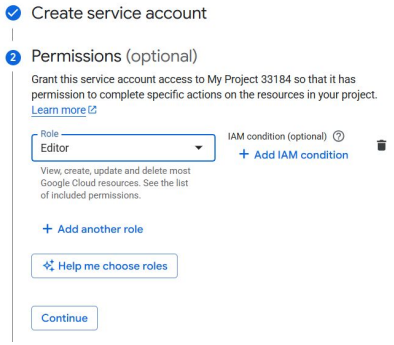
(Optional) Data Collection - Reading Data from Gmail

Credentials Setting

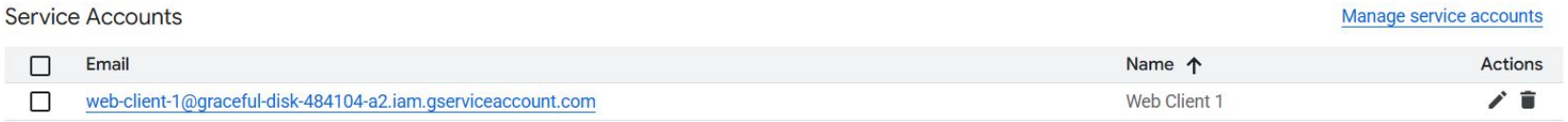
3. In ”Service account name”, enter “Web Client 1” and Click “Create and Continue”



4. In “Select a role”, enter “Editor”, click “Continue”



5. Click “Done” to set credentials.

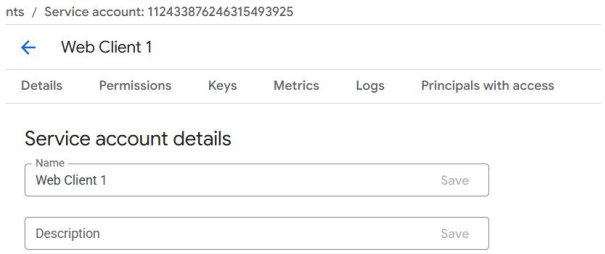


6. Copy the email like “web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com”

(Optional) Data Collection - Reading Data from Gmail

Credentials Setting

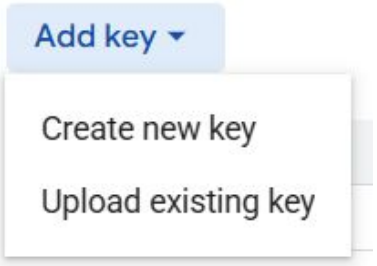
7. Click “Service Account Email”



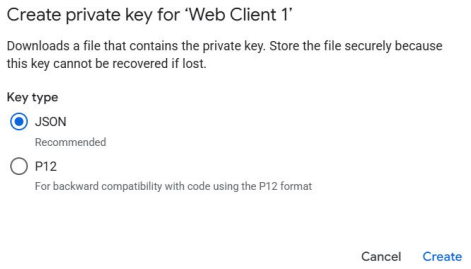
8. Click “Keys”



9. Click “Add Key” and “Create New Key”



10. Enter “JSON”,



11. A JSON Key “credentials.json” is saved in your local PC

(Optional) Data Collection - Reading Data from Gmail

Conduct Python Code Running to fetch gmail content

1. Open Anaconda Prompt, enter "`pip install beautifulsoup4 google-api-python-client google-auth-httplib2 google-auth-oauthlib`" to install google and web scraping library

```
(base) C:\Users\Eddie Chow>pip install google-api-core google-auth google-api-python-client
Defaulting to user installation because normal site-packages is not writeable
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
Requirement already satisfied: google-api-core in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.3
4.1)
Requirement already satisfied: google-auth in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.35.0)
Requirement already satisfied: google-api-python-client in c:\users\eddie chow\appdata\roaming\python\python39\site-pack
ages (2.184.0)
Requirement already satisfied: googleapis-common-protos<2.0dev,>=1.56.2 in c:\users\eddie chow\appdata\roaming\python\py
thon39\site-packages (from google-api-core) (1.70.0)
Requirement already satisfied: protobuf!=3.20.0,!=3.20.1,!=4.21.0,!=4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<4.0.0dev
,>=3.19.5 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-core) (3.20.3)
Requirement already satisfied: requests<3.0.0dev,>=2.18.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-pa
ckages (from google-api-core) (2.32.5)
Requirement already satisfied: cachetools<5.0,>=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (
4.2.2)
Requirement already satisfied: pyasn1-modules>=0.2.1 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (0
.2.8)
Requirement already satisfied: setuptools>=40.3.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (61.2
.0)
Requirement already satisfied: six>=1.9.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (1.16.0)
Requirement already satisfied: rsa<5,>=3.1.4 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (4.7.2)
Requirement already satisfied: httplib2<1.0.0,>=0.19.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-packa
ges (from google-api-python-client) (0.31.0)
Requirement already satisfied: google-auth-httplib2<1.0.0,>=0.2.0 in c:\users\eddie chow\appdata\roaming\python\python39
\site-packages (from google-api-python-client) (0.2.0)
```

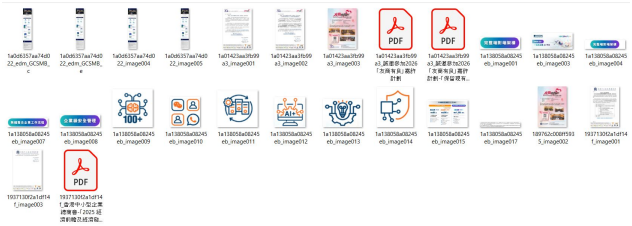
(Optional) Data Collection - Reading Data from Gmail

Conduct Python Code Running to fetch gmail content

2. Download code at https://github.com/innoviai/IPA_Courses-October2026/blob/main/single_read_gmail_csvcontent_fromsubjectname.py
3. Open single_read_gmail_csvcontent_fromsubjectname.py code
4. Edit the target_subject value in line 117.

```
117     target_subject = "香港中小型企業總商會"
118     query = f'subject:"{target_subject}"'
119
120     results = service.users().messages().list(userId='me', q=query).execute()
121     messages = results.get('messages', [])
```

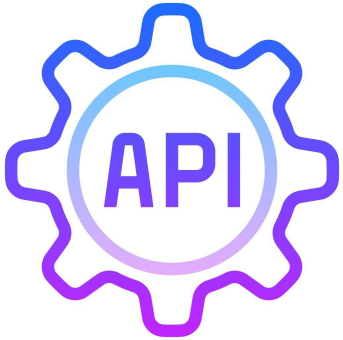
5. Open anaconda prompt to run code by entering "python single_read_gmail_csvcontent_fromsubjectname.py".
6. An email.csv, attachments folder and token.pickle file is created.



Structured Data Collection Method

Main Data collection are get from Google Sheets, API and Web Scraping







kaggle

https://www.kaggle.com/

開放數據平台
DATA.GOV.HK

https://data.gov.hk/en/

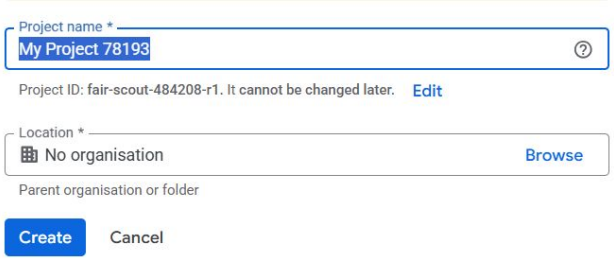
(Optional) Data Collection - Reading Weather Data from Google Sheet

Enable the Google Sheets API with readpermission.

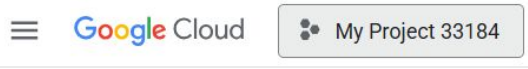
1. Open Browser and enter https://console.cloud.google.com/
2. Click “Select a project”
3. Click “New Project”



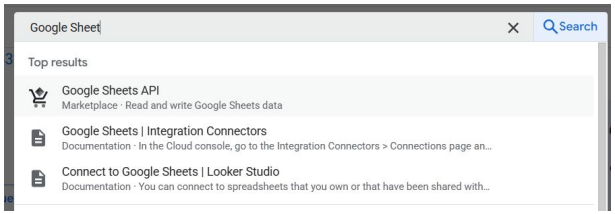
1. Enter information and click “Create”



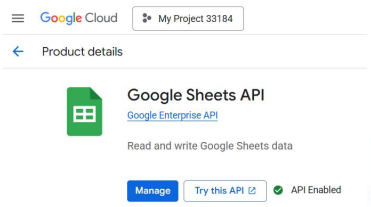
1. Select your Project that create



2. in Search Bar, enter ‘Google Sheet’



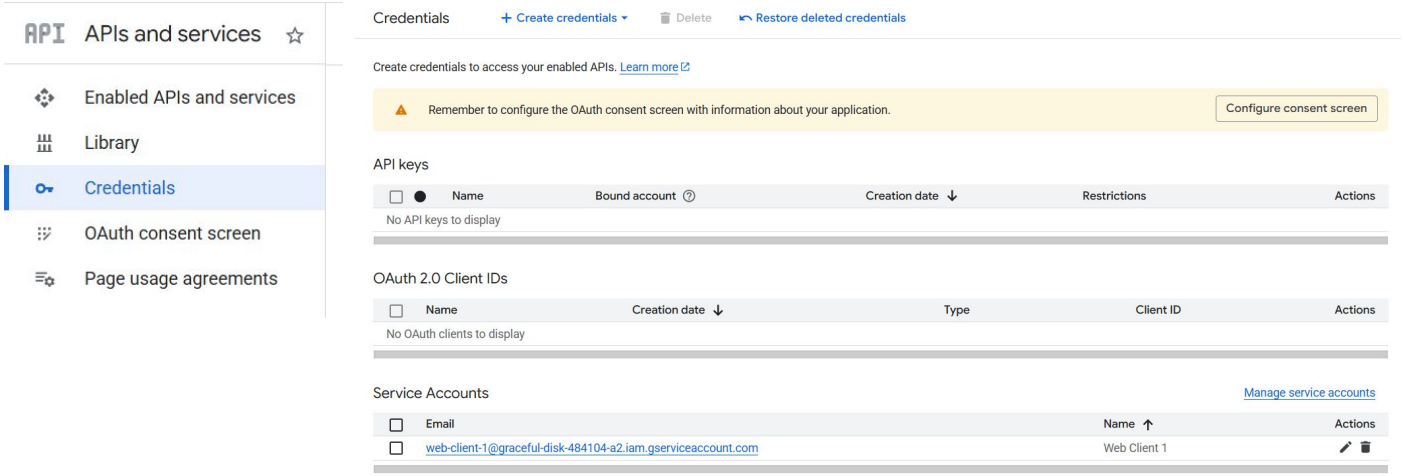
7. Select “Google Sheet API”
8. Click “Enable”



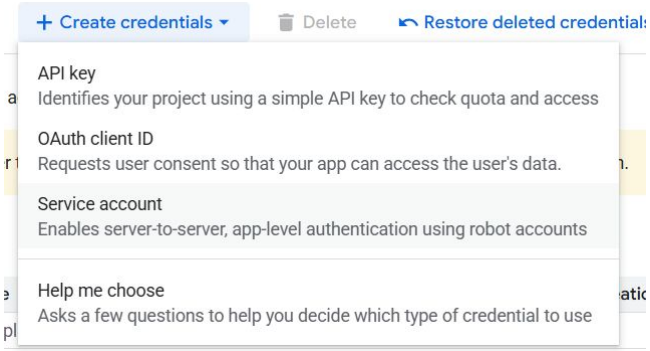
(Optional) Data Collection - Reading Weather Data from Google Sheet

Credentials Setting

1. Click Credentials and whole screen is displayed



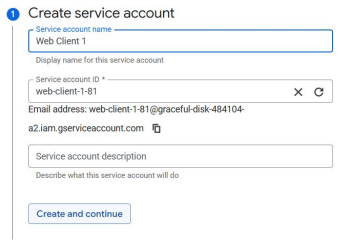
2. Click “Create credentials” and “Service Account”



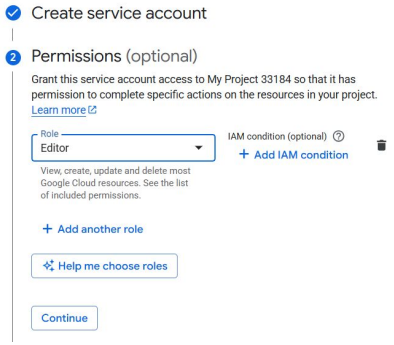
(Optional) Data Collection - Reading Weather Data from Google Sheet

Credentials Setting

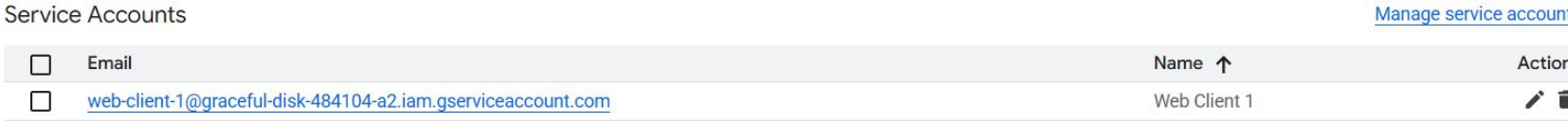
3. In ”Service account name”, enter “Web Client 1” and Click “Create and Continue”



4. In “Select a role”, enter “Editor”, click “Continue”



5. Click “Done” to set credentials.

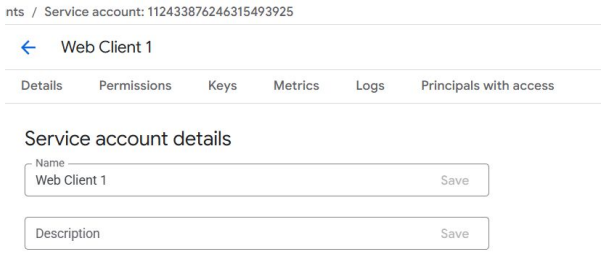


6. Copy the email like “web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com”

(Optional) Data Collection - Reading Weather Data from Google Sheet

Credentials Setting

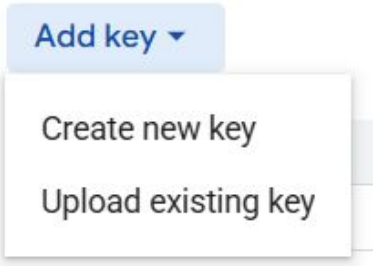
8. Click “Service Account Email”



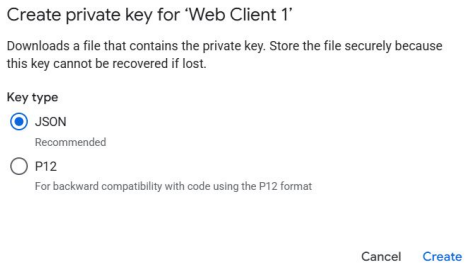
9. Click “Keys”



10. Click “Add Key” and “Create New Key”



11. Enter “JSON”,

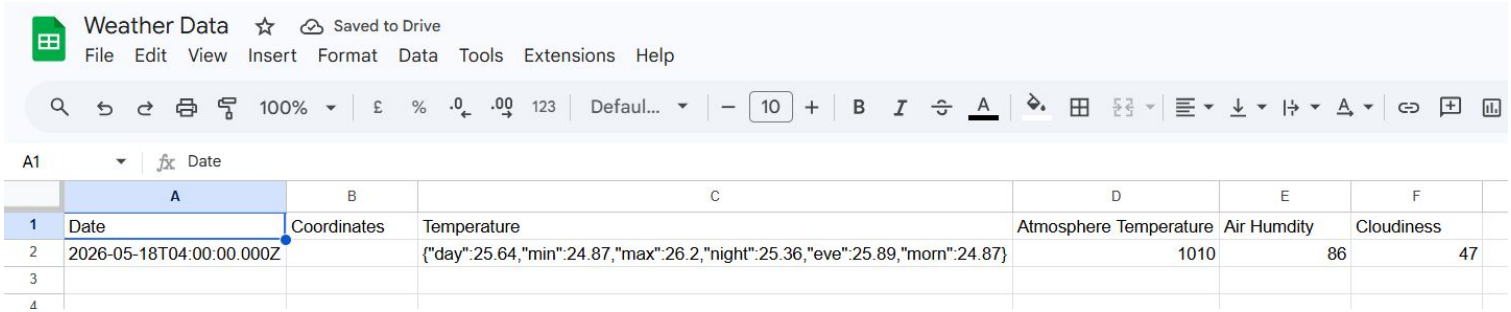


12. A JSON Key is saved in your local PC

(Optional) Data Collection - Reading Weather Data from Google Sheet

Google Sheet Settings

1. Open previous weather data google sheet



	A	B	C	D	E	F
1	Date	Coordinates	Temperature	Atmosphere Temperature	Air Humdity	Cloudiness
2	2026-05-18T04:00:00.000Z		{"day":25.64,"min":24.87,"max":26.2,"night":25.36,"eve":25.89,"morn":24.87}	1010	86	47
3						
4						

1. Click “File”, Share, “Share with Others”
2. Enter “web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com”
3. Make sure it is in editor, click “Send”
4. From the google sheet website url like “https://docs.google.com/spreadsheets/d/1BVdJkKlZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0/edit?gid=0#gid=0”
5. Copy sheet_id **“1BVdJkKlZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0”** , range_name ”**Sheet1!A1:F2** “ in notepad

(Optional) Data Collection - Reading Weather Data from Google Sheet

Python Coding

1. Open Anaconda Prompt, enter "`pip install google-api-core google-auth google-api-python-client google-auth-httplib2 google-auth-oauthlib`" to install google library

```
(base) C:\Users\Eddie Chow>pip install google-api-core google-auth google-api-python-client
Defaulting to user installation because normal site-packages is not writeable
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
Requirement already satisfied: google-api-core in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.3
4.1)
Requirement already satisfied: google-auth in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.35.0)
Requirement already satisfied: google-api-python-client in c:\users\eddie chow\appdata\roaming\python\python39\site-pack
ages (2.184.0)
Requirement already satisfied: googleapis-common-protos<2.0dev,>=1.56.2 in c:\users\eddie chow\appdata\roaming\python\py
thon39\site-packages (from google-api-core) (1.70.0)
Requirement already satisfied: protobuf!=3.20.0,!=3.20.1,!=4.21.0,!=4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<4.0.0dev
,>=3.19.5 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-core) (3.20.3)
Requirement already satisfied: requests<3.0.0dev,>=2.18.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-pa
ckages (from google-api-core) (2.32.5)
Requirement already satisfied: cachetools<5.0,>=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (
4.2.2)
Requirement already satisfied: pyasn1-modules>=0.2.1 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (0
.2.8)
Requirement already satisfied: setuptools>=40.3.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (61.2
.0)
Requirement already satisfied: six>=1.9.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (1.16.0)
Requirement already satisfied: rsa<5,>=3.1.4 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (4.7.2)
Requirement already satisfied: httplib2<1.0.0,>=0.19.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-packa
ges (from google-api-python-client) (0.31.0)
Requirement already satisfied: google-auth-httplib2<1.0.0,>=0.2.0 in c:\users\eddie chow\appdata\roaming\python\python39
\site-packages (from google-api-python-client) (0.2.0)
```

(Optional) Data Collection - Reading Weather Data from Google Sheet

Python Coding

2. Open jupyter notebook
3. Type following code

```
# Import Library
import os
import csv
from google.oauth2.service_account import Credentials
from googleapiclient.discovery import build

SCOPES = ['https://www.googleapis.com/auth/spreadsheets']
SERVICES_ACCOUNT_FILE = "graceful-disk-484104-a2-152ebd9c692e.json"

credentials = Credentials.from_service_account_file(SERVICES_ACCOUNT_FILE, scopes=SCOPES)
service = build('sheets', 'v4', credentials=credentials)

sheet_id = '1BVdJkKlZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0'
range_name = "'Sheet1'!A1:F2"  # Specify the range with the sheet name
```

(Optional) Data Collection - Reading Weather Data from Google Sheet

```
# Retrieve the data
result = service.spreadsheets().values().get(spreadsheetId=sheet_id, range=range_name).execute()
values = result.get('values', [])
print(values)
```

```
[['Date', 'Coordinates', 'Temperature', 'Atmosphere Temperature', 'Air Humdity', 'Cloudiness'], ['2026-05-18T04:00:00.000Z', '', '{"day":25.64,"min":24.87,"max":26.2,"night":25.36,"eve":25.89,"morn":24.87}', '1010', '86', '47']]
```

```
if not values:
        print('No data found.')
else:
        # Save to CSV file
        with open('weatherforecastdata.csv', 'w', newline='', encoding='utf-8') as csv_file:
                writer = csv.writer(csv_file)
                writer.writerows(values)  # Write all the rows

        print('Data has been saved to weatherforecastdata.csv')
```

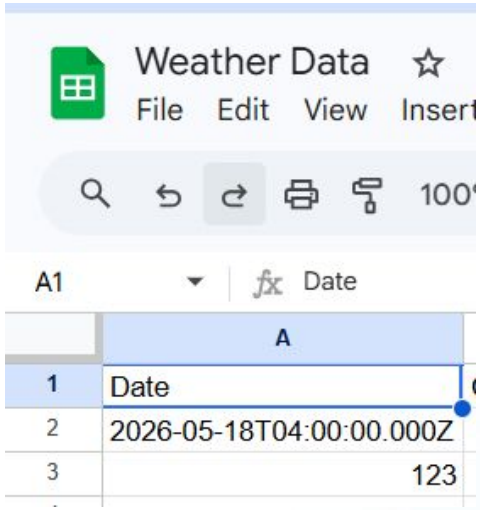
Google Sheet Content Update

```
    # Update the range to just the starting cell for appending
     result = service.spreadsheets().values().update(
         spreadsheetId=sheet_id,
         range=range_name,  # Change range here
         valueInputOption=value_input_option,
         body=body
     ).execute()

     print(f"{result.get('updatedCells')} cells updated.")
     return result

sheet_id = '1mqrxumbU1Si5aGmkg2jpffu7zn9uWxFfvXX1Wnuwkas'
range_name = "Sheet1!A3"
value_input_option = 'USER_ENTERED'
values = [['123']]

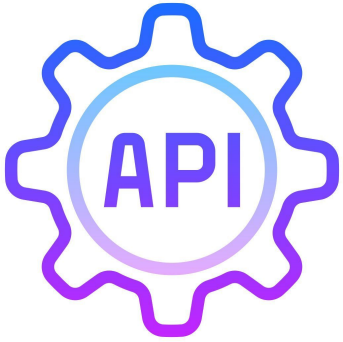
update_values(sheet_id, range_name, value_input_option, values)
```



Structured Data Collection Method

Main Data collection are get from Google Sheets, API and Web Scraping







kaggle

https://www.kaggle.com/

開放數據平台
DATA.GOV.HK

https://data.gov.hk/en/

Data Collection - Web Crawler

Definition

- a web crawler is a computer program that browses the World Wide Web in a methodical, automated manner. (Wikipedia)

Utilities:

- Gather pages from the Web.
- Support a search engine, perform data mining and so on.

Object:

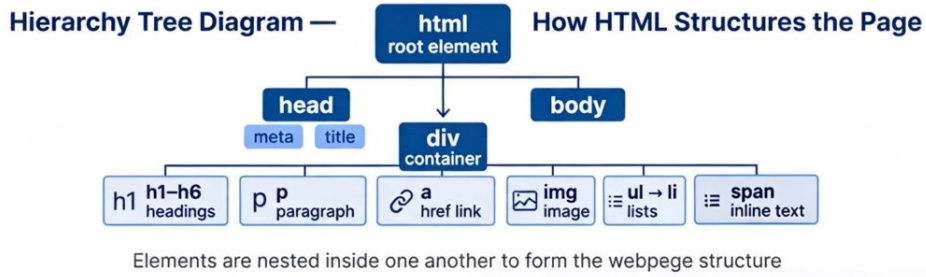
- Text, video, image and so on.
- Link structure

Some scraping knowledge

1. HTTP: the communication protocol
2. HTML: the language in which web pages are defined
3. JS: javascript (code executing in the browser)
4. CSS: style sheets, how web pages are styled. Important, but does not contain data.
5. JPG, PNG, BMP: images, usually not interesting
6. CSV / TXT / JSON: data, interesting !!!

Webpage Structure & DOM Hierarchy

HTML DOM Hierarchy — Structure & Key Elements



Elements are nested inside one another to form the webpege structure

CODE EXAMPLE

```
1  <(DOCTUPE html>
2  <html>
3  <head>
4  <meta chraset="UTF-8" />
5  <title>Page Title</title>
6  </head>
7  <body>
8  <div class="container")
9   <h1>Welcome to HTML</h1)
10  <p"This is a <span>paragaph> of text..p>
11  <a href="https://example.com")Learn More<a>
12  <img src="image.png" alt="Sample Image" />
13  <ul>
14  <li>First list item</li>
15  <li>Second list item</li>
16   </ ul>
17    </div>
18  </body>
19  </html>
```

Key: Nesting elements creates the structure — head for metadata, body for visible content

RENDERED WEBPAGE

https://example.com

Welcome to HTML

This is a ' **'paragaph' of text.**
of ıemure
Learn More →

Sample Image

- First list item
- Second list item

html = root | head = metadata | body = visible content

div groups elements into sections for layout

Common elements: headings, paragraph, link, image, list, span

Detailed HTML Tags Explained

Quick reference: common HTML tags with purpose, example, and usage tips — for learning & training

Tag	Purpose	Example	Usage Tip
	Block container for layout	Content	Groups content for layout; not visible itself
<p>	Paragraph text	<p"This is a paragraph."p>	Represents a paragraph; adds natural spacing before and after
<a href>	Hyperlink with attribute href	<a href="https://example.com"> Visit=/a>	href is the destination URL; use title for tooloip
<button>	Clickable button	<button type="submit"> Submit</button>	Triggers an action; use type="button" or "submit"
<img src>	Embeds an image	<img src="logo.png" alt="logo">	src is the image path; alt is required for accessibility
class / id	Attributes for styling & JavaScript	<div class="card" id="main"=/div>	class is reusable for multiple elements; *id* is unique to one element

Code Snippet Example

```
<!DOCTYPE html>
<html lang="en">
` <body>
· <div class="container" id="main">
· <h2>Welcome="h2>
· <pThis is a <a href="https://example.com"
title="Go to site">..p>
· <button type="button">Clink Me="button">
· <img src="photo.jpg" alt="Sample photo" }>
· </div>
· </body>
  </html>
```

Tip: Use class for styling multiple elements. Use *id* for one unique element. Always include *alt* on <img> for accessibility.

Best Practice: Keep HTML semantic • Test links • Use accessible alt text • Validate your HTML

Slide 1 • HTML Tags Reference • Training

Visual DOM Nesting & Box Model

HTML CODE

```
1 '
  <div id="container">
2 '
   <div class="card">
3 '
    <p"This is a paragraph with a
    <a href="https://example.com"}link</a..p>
4 '
    </div>
5 '
    </div>
```

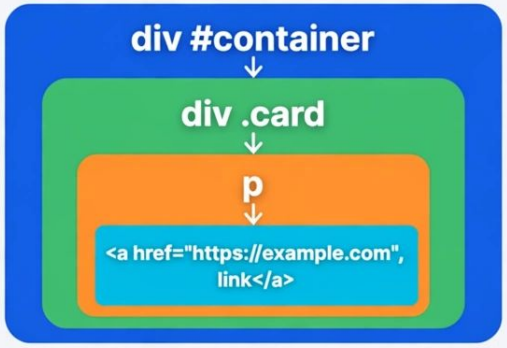
Key Elements:

- — container element for grouping other elements
- <p> — paragraph element for holding text content
- <a href> — link element that navigates to a URL

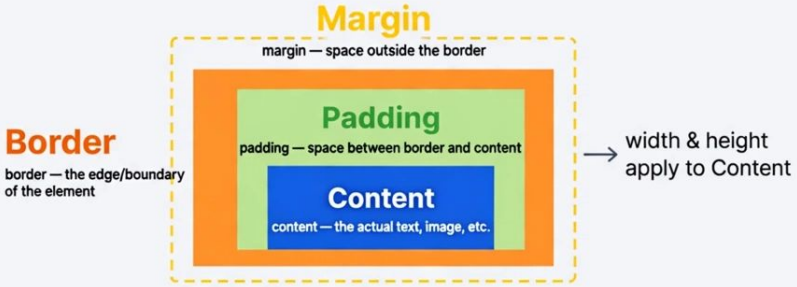
Tip: Elements can be nested inside other elements to create hierarchy

VISUAL LAYOUT & DOM STRUCTURE

DOM Tree (Nested)



CSS Box Model Illustration



How it works: The browser parses the HTML to build the DOM tree → Each element is rendered using the CSS Box Model → Margin, Border, and Padding control the spacing and layout around the content

Obtaining Data – Web Scraping

1. Open Browser and enter https://www.hk01.com/
2. Right Click and Select "Inspect"



3. In the "Elements" tab, the Content are in html structure with

Obtaining Data – Web Scraping

Web Scraping

- Using programs to get data from online
- Often much faster than manually copying data!
- Transfer the data into a form that is compatible with your code

Obtaining Data – Web Scraping

Why scraping the web?

- Vast source of information; can combine with multiple datasets
- Companies have not provided APIs
- Automate tasks
- Keep up with sites / real-time data
- Fun!

Obtaining Data – Web Scraping

Robots.txt

- Specified by web site owner
- Gives instructions to web robots (e.g., your code)
- Located at the top-level directory of the web server
  - E.g., http://google.com/robots.txt

```
User-agent: *
User-agent: Yandex
Disallow: /search
Allow: /search/about
Allow: /search/howsearchworks
Disallow: /sdch
Disallow: /groups
Disallow: /index.html?
Disallow: /?
Disallow: /goto?
Allow: /?hl=
Disallow: /?hl=*&
Allow: /?hl=*&gws_rd=ssl$
Disallow: /?hl=*&*&gws_rd=ssl
Allow: /?gws_rd=ssl$
Allow: /?pt1=true$
Disallow: /imgres
Disallow: /u/
Disallow: /setprefs
Disallow: /m?
```

Obtaining Data – Web Scraping

HTML

- Tags are denoted by angled brackets
- Almost all tags are in pairs e.g., Hello
- Some tags do not have a closing tag e.g.,

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>Ttle</title>
  </head>
  <body>
    <h1>Body Title</h1>
    <p>Body Content</p>
  </body>
</html>
```

Obtaining Data – Web Scraping

HTML

- <html>, indicates the start of an html page
- <body>, contains the items on the actual webpage (text, links, images, etc)
- <p>, the paragraph tag. Can contain text and links
- <a>, the link tag. Contains a link url, and possibly a description of the link
- <input>, a form input tag. Used for text boxes, and other user input
- <form>, a form start tag, to indicate the start of a form
- <img>, an image tag containing the link to an image

Obtaining Data – Web Scraping

How to Web scrape:

1. Get the webpage content
    - Requests (Python library) gets a webpage for you
2. Parse the webpage content
    - (e.g., find all the text or all the links on a page)
    - BeautifulSoup (Python library) helps you parse the webpage.
    - Documentation: https://beautiful-soup-4.readthedocs.io/en/latest/

The Big Picture Recap

Data Sources	Files, APIs, Webpages (via Requests)
Data Parsing	Regular Expressions, Beautiful Soup
Data Structures/Storage	Traditional lists/dictionaries, PANDAS
Models	Linear Regression, Logistic Regression, kNN, etc

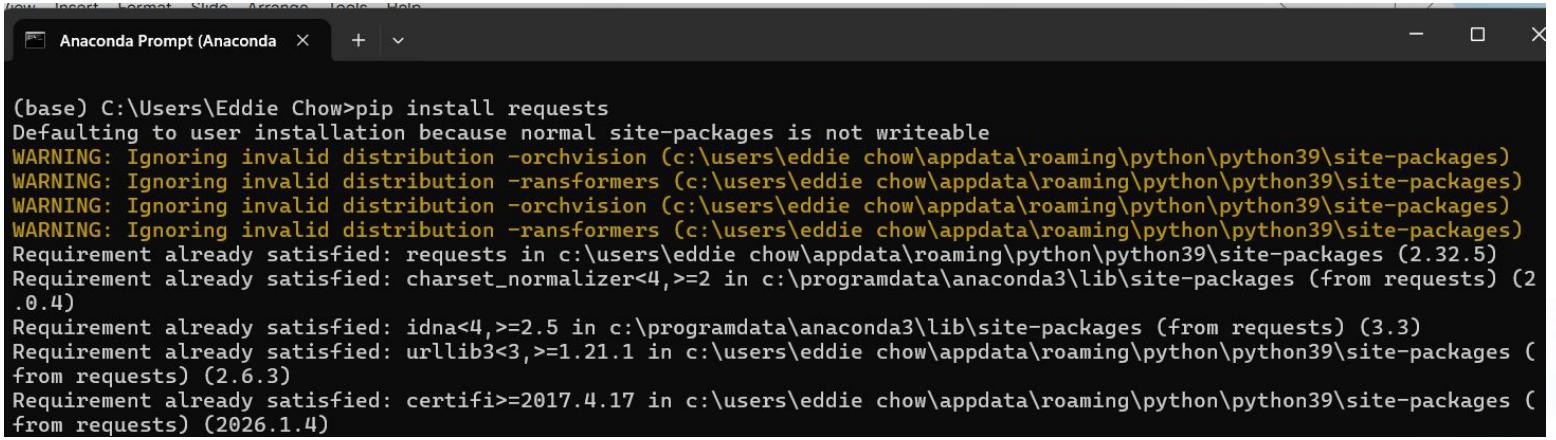
BeautifulSoup only concerns webpage data

Obtaining Data – Web Scraping

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.

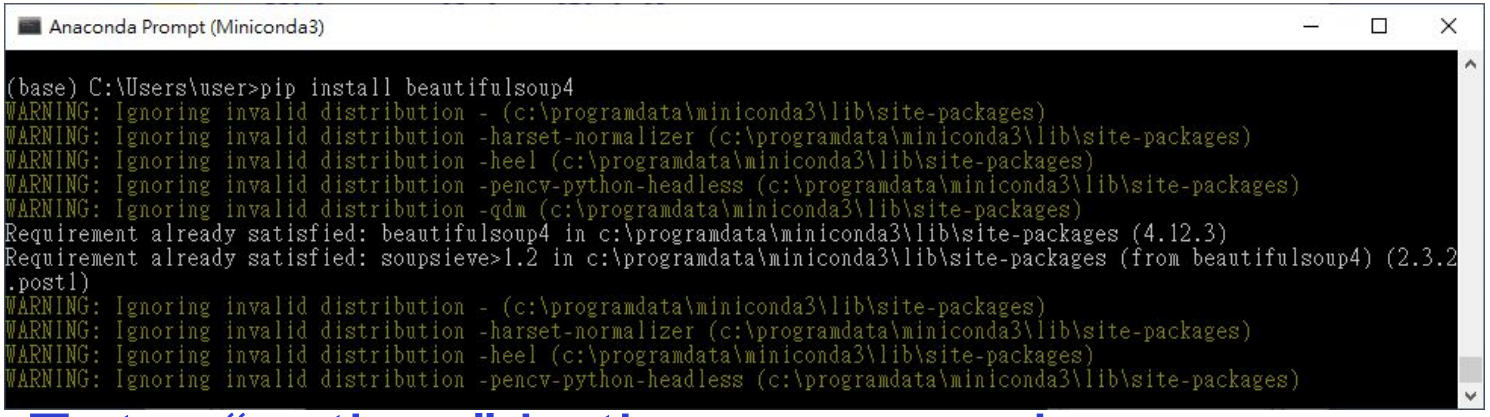


2. Install python package requests by entering “pip install requests”

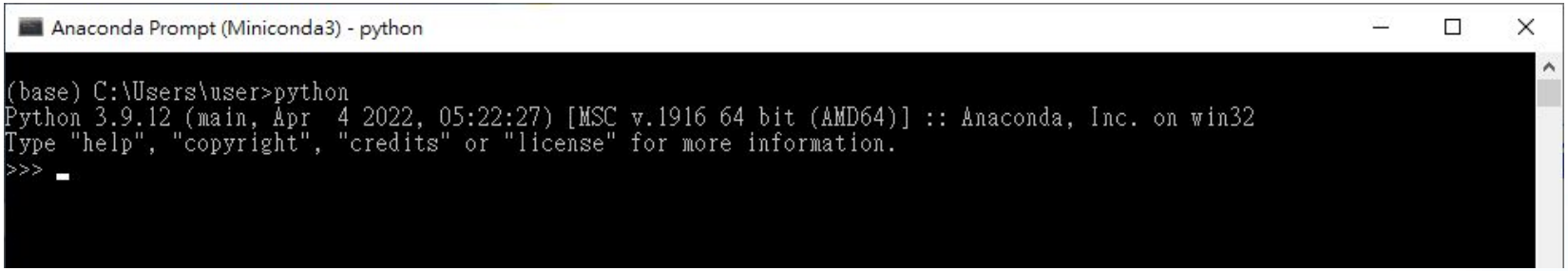


Obtaining Data – Web Scraping

3. Install python package beautifulsoup4 by entering “pip install beautifulsoup4”



4. Enter “python” in the command



Obtaining Data – Web Scraping

1. Get the webpage content

Requests (Python library) gets a webpage for you

```
import requests

url = "https://www.nytimes.com/international/"

page = requests.get(url)
page.status_code
page.content
```

Obtaining Data – Web Scraping

1. Get the webpage content

Requests (Python library) gets a webpage for you

```
import requests

url = "https://www.nytimes.com/international/"

page = requests.get(url)
page.status_code
page.content
```

Gets the status from the webpage request. 200 means success. 404 means page not found.

Obtaining Data – Web Scraping

1. Get the webpage content

Requests (Python library) gets a webpage for you

```
import requests

url = "https://www.nytimes.com/international/"

page = requests.get(url)
page.status_code
page.content
```

Returns the content of the response, in bytes.

Obtaining Data – Web Scraping

2. Parse the webpage content

BeautifulSoup (Python library) helps you parse a webpage

```
from bs4 import BeautifulSoup

soup = BeautifulSoup(page.content, "html.parser")
soup.title
soup.title.text
```

Obtaining Data – Web Scraping

2. Parse the webpage content

BeautifulSoup (Python library) helps you parse a webpage

```
from bs4 import BeautifulSoup

soup = BeautifulSoup(page.content, "html.parser")

soup.title

soup.title.text
```

Returns the full context, including the title tag. e.g.,

```
<title data-rh="true">The New York Times - Breaking News</title>
```

Obtaining Data – Web Scraping

2. Parse the webpage content

BeautifulSoup (Python library) helps you parse a webpage

```
soup = BeautifulSoup(page.content, "html.parser")
soup.title
soup.title.text
```

Returns the text part of the title tag. e.g.,

```
The New York Times - Breaking News
```

Obtaining Data – Web Scraping

BeautifulSoup

- Helps make messy HTML digestible
- Provides functions for quickly accessing certain sections of HTML content

Exampl

```
import bs4
## get bs4 object
soup = bs4.BeautifulSoup(source)
## all a tags
soup.findAll('a')
## first a
soup.find('a')
## get all links in the page
link_list = [l.get('href') for l in soup.findAll('a')]
```

Obtaining Data – Web Scraping

HTML is a tree

- You don't have to access the HTML as a tree, though;
- Can immediately search for tags/content of interest (a la previous slide)

Exampl

```
tree = bs4.BeautifulSoup(source)

## get html root node
root_node = tree.html

## get head from root using contents
head = root_node.contents[0]

## get body from root
body = root_node.contents[1]

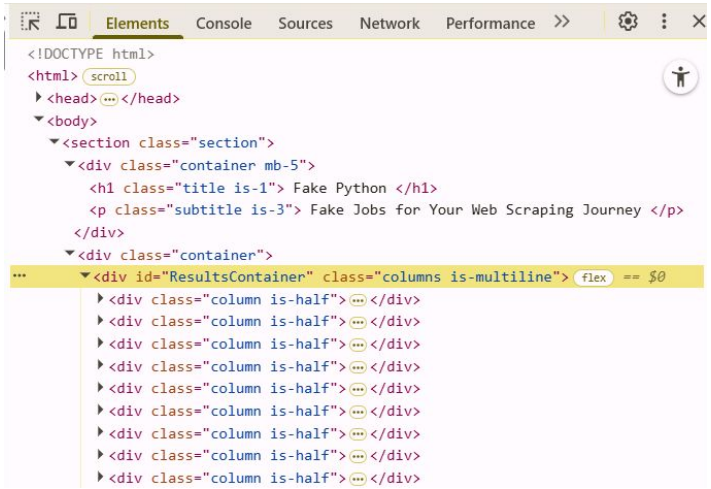
## could directly access body
tree.body
```

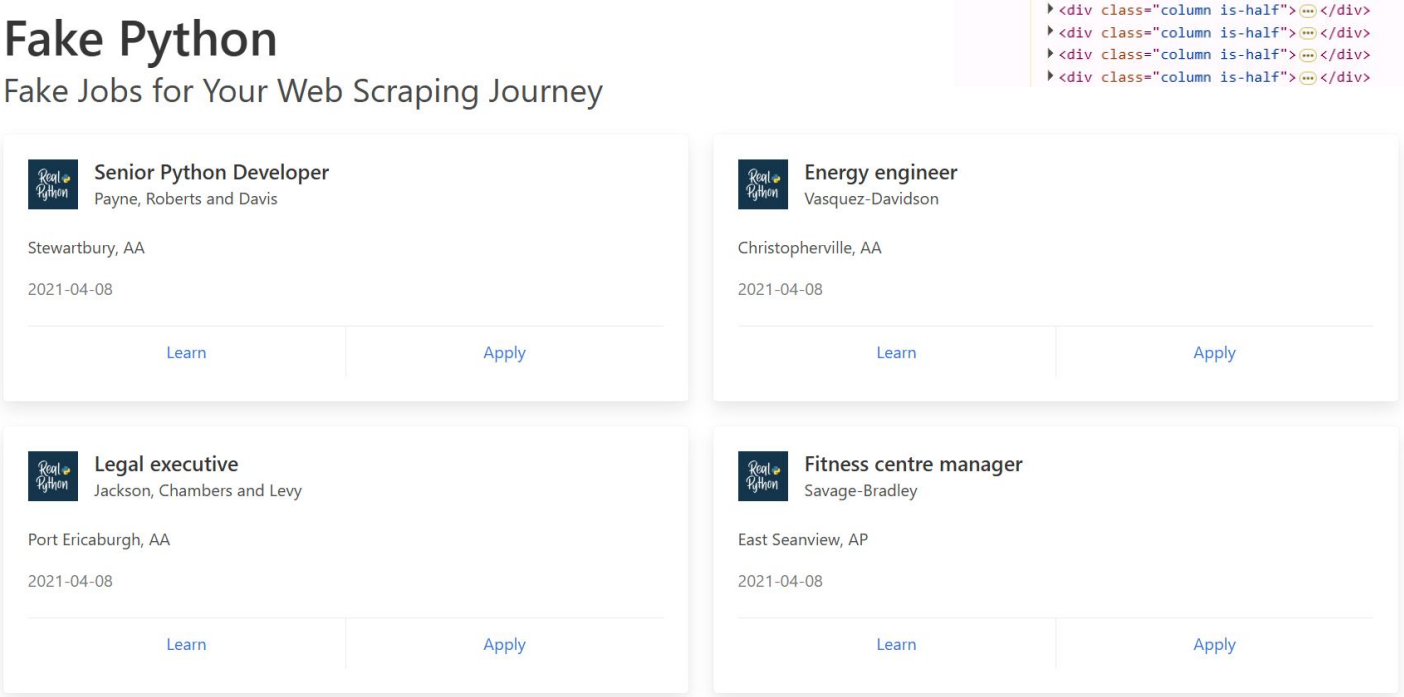
Obtaining Data – Web Scraping

```
import requests
from bs4 import BeautifulSoup

URL = "https://realpython.github.io/fake-jobs/"
page = requests.get(URL)

soup = BeautifulSoup(page.content, "html.parser")
```





Obtaining Data – Web Scraping

URL = "https://realpython.github.io/fake-jobs/"
page = requests.get(URL)

soup = BeautifulSoup(page.content, "html.parser")

results = soup.find(id="ResultsContainer")
print(results.prettify())

```
>>> print(results.prettify())
<div class="columns is-multiline" id="ResultsContainer">
 <div class="column is-half">
  <div class="card">
   <div class="card-content">
    <div class="media">
     <div class="media-left">
      <figure class="image is-48x48">
       <img alt="Real Python Logo" src="https://files.realpython.com/media/real-python-logo-thumbnail.7f0db70c2ed2.jpg?_
_no_cf_polish=1"/>
      </figure>
     </div>
     <div class="media-content">
      <h2 class="title is-5">
       Senior Python Developer
      </h2>
      <h3 class="subtitle is-6 company">
       Payne, Roberts and Davis
      </h3>
     </div>
    </div>
```

Obtaining Data – Web Scraping

Find Elements by HTML Class Name

You've seen that every job posting is wrapped in a `<div>` element with the class `card-content`. Now you can work with your new object called `results` and select only the job postings in it. These are, after all, the parts of the HTML that you're interested in! You can pick out all job cards in a single line of code:

Python

```
>>> job_cards = results.find_all("div", class_="card-content")
```

Here, you call `.find_all()` on `results`, which is a `BeautifulSoup` object. It returns an iterable containing all the HTML for all the job listings displayed on that page.

Take a look at all of them:

Python

```
>>> for job_card in job_cards:
...     print(job_card, end="\n" * 2)
...
```

```
<div class="card-content">
<div class="media">
<div class="media-left">
<figure class="image is-48x48">
<img alt="Real Python Logo" src="https://files.realpython.com/media/real-python-logo-
</figure>
</div>
<div class="media-content">
<h2 class="title is-5">Senior Python Developer</h2>
<h3 class="subtitle is-6 company">Payne, Roberts and Davis</h3>
</div>
</div>
<div class="content">
<p class="location">
        Stewartbury, AA
      </p>
<p class="is-small has-text-grey">
<time datetime="2021-04-08">2021-04-08</time>
```

Extract title Content

```
>>> for job_card in job_cards:
...     title_element = job_card.find("h2", class_="title")
...     print(title_element)
...
<h2 class="title is-5">Senior Python Developer</h2>
<h2 class="title is-5">Energy engineer</h2>
<h2 class="title is-5">Legal executive</h2>
<h2 class="title is-5">Fitness centre manager</h2>
<h2 class="title is-5">Product manager</h2>
<h2 class="title is-5">Medical technical officer</h2>
<h2 class="title is-5">Physiological scientist</h2>
<h2 class="title is-5">Textile designer</h2>
<h2 class="title is-5">Television floor manager</h2>
<h2 class="title is-5">Waste management officer</h2>
<h2 class="title is-5">Software Engineer (Python)</h2>
<h2 class="title is-5">Interpreter</h2>
<h2 class="title is-5">Architect</h2>
<h2 class="title is-5">Meteorologist</h2>
```

Extract Content of title, company, location

Python

```
>>> for job_card in job_cards:
...     title_element = job_card.find("h2", class_="title")
...     company_element = job_card.find("h3", class_="company")
...     location_element = job_card.find("p", class_="location")
...     print(title_element)
...     print(company_element)
...     print(location_element)
...     print()
...
<h2 class="title is-5">Senior Python Developer</h2>
<h3 class="subtitle is-6 company">Payne, Roberts and Davis</h3>
<p class="location">
        Stewartbury, AA
      </p>

<!-- ... -->
```

Full Article : https://realpython.com/beautiful-soup-web-scraper-python/

Obtaining Data – Web Scraping

1. Open Browser and enter https://www.hk01.com/
2. Right Click and Select "Inspect"



3. In the "Elements" tab, Select the Content that you want to extract

Obtaining Data – Web Scraping

4. Right Click and click “Edit As HTML”
Copy part of the code until “/div>



Obtaining Data – Web Scraping

5. Open Perplexity at https://www.perplexity.ai/ or Gemini at https://gemini.google.com/app

6. Enter prompt “write python beatifulsoap4 to extract data from website url https://www.hk01.com/ with html code {htmlcode}”

Where {htmlcode} = “ <button data-testid="hamburger-menu-button" class="box-border w-6 px-0 py-3 m-0 transition-all duration-[0.6s] ease-[cubic-bezier(0.19,1,0.22,1)] [transform:translateZ(0)] border-none outline-none rounded-none cursor-pointer shadow-none bg-transparent focus:shadow-none focus:outline-none" aria-label="mobile-menu"><span class="relative block h-px rounded-[1px] transition-all duration-[0.6s] ease-[cubic-bezier(0.19,1,0.22,1)] w-[70%] left-[15%] bg-black/[0.78] before:w-full after:w-full before:absolute after:absolute before:left-0 after:left-0 before:content-[' '] after:content-[' '] before:h-px after:h-px before:rounded-[1px] after:rounded-[1px] before:transition-all before:duration-[0.6s] before:ease-[cubic-bezier(0.19,1,0.22,1)] after:transition-all after:duration-[0.6s] after:ease-[cubic-bezier(0.19,1,0.22,1)] before:bg-black/[0.78] after:bg-black/[0.78] before:-top-1.5 before:rotate-0 after:top-1.5 after:rotate-0"></button> ”

7. Copy the code and paste in jupyter notebook

Obtaining Data – Web Scraping

8. In original perplexity prompt, type the following prompt "Add code that save it into csv file"

	A
1	text,url,class
2	,/,
3	港聞,/zone/1/%E6%B8%AF%E8%81%9E,
4	兩會焦點,/issue/10355/%E5%85%A9%E6%9C%832025-%E5%9C%8B%E9%9A%9B%E5%B1%80%
5	娛樂,/zone/2/%E5%A8%9B%E6%A8%82,
6	最平酒店,https://clk.omgt3.com/?AID=2200313&PID=55029&uid=content&uid2=channel_旅遊_editor
7	國際,/zone/4/%E5%9C%8B%E9%9A%9B,
8	即時,/latest,
9	熱榜,/hot,
10	生活,/zone/8/%E7%94%9F%E6%B4%BB,
11	科技,/zone/11/%E7%A7%91%E6%8A%80%E7%8E%A9%E7%89%A9,
12	中國,/zone/5/%E4%B8%AD%E5%9C%8B,
13	體育,/zone/3/%E9%AB%94%E8%82%B2,
14	01深圳,/zone/25/01%E6%B7%B1%E5%9C%B3,
15	經濟,/zone/14/%E7%B6%93%E6%BF%9F,
16	觀點,/zone/12/%E8%A7%80%E9%BB%9E,
17	健康,/zone/24/%E5%81%A5%E5%BA%B7,
18	好食玩飛,/zone/19/%E5%A5%BD%E9%A3%9F%E7%8E%A9%E9%A3%9B,
19	女生,/zone/6/%E5%A5%B3%E7%94%9F,
20	熱話,/zone/7/%E7%86%B1%E8%A9%B1,
21	藝文格物,/zone/9/%E8%97%9D%E6%96%87%E6%A0%BC%E7%89%A9,
22	社區,/zone/10/%E7%A4%BE%E5%8D%80,
23	教育,/zone/23/%E6%95%99%E8%82%B2,
24	简,/,"font-light, pointer-events-none, !text-light-n5"
25	
26	
27	Mobile Menu Details:
28	Exists: True

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THANK YOU

